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Survey paper

Histopathology image analysis for colorectal cancer using deep learning techniques: A comprehensive survey

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ABSTRACT

Histopathology images are considered the benchmark modality for the definitive diagnosis of colorectal cancer owing to their ability to extract rich phenotypic information. Nowadays, histopathology images are acquired and digitalized using Whole Slide Image scanners. This shift enables the integration of machine learning and deep learning into the pathological workflow and opens a new avenue for computational pathology. A typical workflow of computational pathology for colorectal cancer can be formulated as a three-phase pipeline: pre-processing, analysis, and post-processing. However, critical examination of the existing studies has primarily centered on the analysis phase due to the widespread use of curated datasets. Consequently, pre-processing and post-processing stages are often overlooked. This methodological bias leads to a fragmented research pipeline, limiting the generalizability required to handle real-world clinical practice. The survey showcases a three-phase pipeline that unifies approaches across pre-processing, image analysis, and post-processing methods. In the first stage, an overview of methods for effectively preparing histopathological images through pre-processing is presented. The second phase surveys recent developments in computational models for tissue classification, segmentation, detection, and biomarker prediction. Finally, this review presents techniques used in colorectal cancer histopathology to enhance network output by post-processing. In addition, existing open issues and future research directions are discussed for innovative researchers. By consolidating the findings of several studies, this review underscores the pressing need for a unified approach that effectively bridges the gap between current research and clinical acceptance.

1. Introduction

Colorectal cancer (CRC), a malignancy originating in the colon or rectum, is a major contributor to cancer-related morbidity and mortality globally. According to the latest GLOBOCAN 2022 statistics, it ranks as the third most common cancer in terms of incidence, with over 1.926 million new cases, and the second leading cause of cancer-related mortality, responsible for over 903,000 deaths worldwide (Bray et al., 2024). The progression of CRC often begins with precancerous polyps, which develop due to genetic mutations (Biller and Schrag, 2021; Dekker et al., 2019). Since CRC is highly curable when detected at an early stage, early and accurate diagnosis is critical for improving patient survival rates (Guitton et al., 2023). Unfortunately, CRC is often diagnosed at an advanced stage, resulting in a poor prognosis and worsened survival outcomes (Khan and Lengyel, 2023). The clinical workup for CRC involves a combination of methods, including physical examination, imaging methods such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), and Ultrasound (US), along with colonoscopy, histopathology, and biomarker tests. Although imaging modalities provide crucial diagnostic information, they often yield non-definitive diagnoses. Moreover, tumor biomarker analysis provides essential insights for the early prediction and prognosis of CRC, but it is resource-intensive. In contrast, histopathology, the microscopic examination of tissues and cells, is considered the gold standard for definitive diagnosis, as it allows pathologists to see the extent of cancer at the cellular level (Nature Biomedical Engineering Editorial, 2017; Xu et al., 2016).

Conventional pathologists meticulously prepare and examine stained tissue sections on glass slides under optical microscopes (Al-Thelaya et al., 2023). This manual examination is the standard way to diagnose CRC and offers advantages such as cost-effectiveness and widespread availability (Baxi et al., 2022). Nevertheless, it is labor-intensive and time-consuming (Jehanzaib et al., 2025). A more fundamental limitation of this approach is its subjective nature of interpretation, which can contribute to inter-observer variability. This variation may arise from disparities in the diagnostic protocols and the diverse experience levels of individual pathologists (Fig. 1), thereby affecting decision-making

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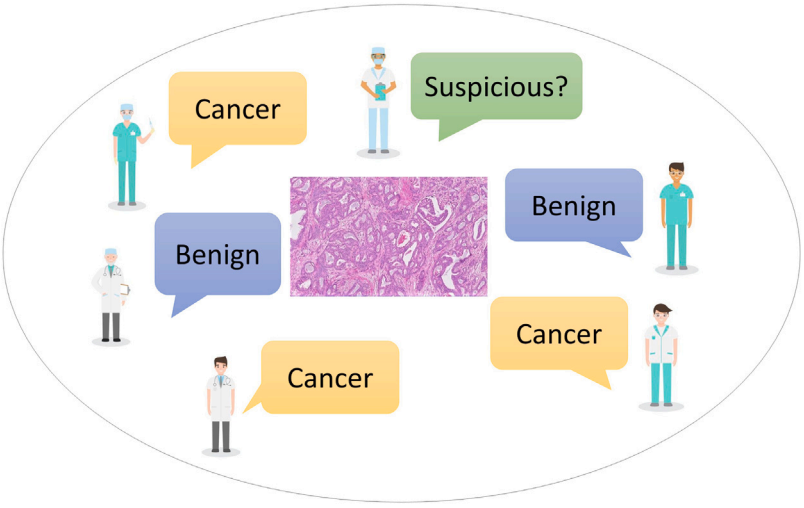


Fig. 1. Example of inter-observer variability showing differing pathologist opinions for the same CRC histopathology image.

and potentially leading to misdiagnoses (Brunnström et al., 2017). The advancement of technology ushered in a new era of pathology, known as Digital Pathology (DP), which provides the foundation for the digitization, storage, and efficient management of pathology slides. This field initially relied on static images captured by fixed digital cameras mounted on microscopes (Park et al., 2012; Farahani et al., 2015). The major breakthrough in DP came with the introduction of whole-slide scanners in the late 1990s. It enabled the conversion of the entire tissue onto a glass slide into high-resolution digital images known as Whole Slide Images (WSIs). These WSIs are multi-gigabyte images that contain complete and intricate pathology features essential for accurate diagnosis (Archana and Jeevaraj, 2024). Pathologists can now comprehensively examine tissue in a single view, optimizing and standardizing the diagnostic pathology workflow. Moreover, it also facilitates easy data sharing among experts and improves diagnostic reproducibility (Hamida et al., 2021; Atallah et al., 2022). However, while DP addressed geographical barriers, it nevertheless entails heavy manual work and suffers from image irregularities and limited traceability (Das et al., 2020b).

The digitization of pathology and the abundance of large, open-access datasets, such as the TCGA Pan-Cancer Atlas, provided the necessary data foundation for the advancement of Computational Pathology (CPath) (Tomczak et al., 2015). CPath is a growing field that couples DP with Artificial Intelligence (AI) algorithms to automate and streamline the analysis of WSIs. Fig. 2 shows the workflows of the conventional approach, DP, and CPath. The primary goal of CPath is to build intelligent digital diagnostic workflows that serve as assistive computer-aided diagnostic (CAD) systems in clinical pathology. These workflows cover the full spectrum of CPath, ranging from pre-processing and segmentation to tumor detection, tissue classification, prognostic prediction, and post-processing (Smith et al., 2021). The advancement of AI, specifically machine learning (ML) and deep learning (DL), presents a promising avenue for the effective execution of CPath's workflows. Early CRC studies in CPath focused on extracting handcrafted features from manually defined regions of interest (ROIs) and applying conventional techniques for automated diagnostic analysis of pathology images (Gurcan et al., 2009). The application of conventional diagnostic techniques includes CRC tissue classification (using Support Vector Machine, K-Nearest Neighbors, and Logistic Regression) and pathological segmentation (using thresholding and region-growing methods). However, these techniques required extensive feature engineering (Archana and Jeevaraj, 2024; Srinidhi et al., 2022).

A major breakthrough in AI occurred with the success of DL techniques around 2012. Over the last few decades, DL methods, particularly Convolutional Neural Networks (CNNs), have achieved impressive results across several computer vision tasks, such as image classification, object detection, and image segmentation. This success has propelled DL methods across the biomedical domain, ranging from signal processing (Yu et al., 2020, 2019, 2024a) to image analysis (Suganyadevi et al., 2022; Mansoor et al., 2026), with major breakthroughs in diagnostic accuracy. For CRC histopathology, DL approaches have been widely applied for both diagnostic and prognostic analysis. Key pioneering diagnostic studies include tissue classification, gland and nuclei segmentation, and tumor detection, whereas in prognostic modeling, DL methods have contributed to predicting disease progression and survival outcomes. While CNNs have achieved excellent results in CRC histopathology diagnostic and prognostic analysis by learning local cellular features, they struggle to capture long-range dependencies due to their limited receptive fields (Tan et al., 2023; Yengec-Tasdemir et al., 2024). This fundamental limitation often restricted the use of CNNs, especially for CRC prognostic modeling. Advanced Transformer architectures (Liu et al., 2025; Gao et al., 2024) captured long-range dependencies and enhanced contextual understanding through self-attention mechanisms. While supervised learning is the method of choice for most studies, the field of CRC histopathology has also embraced strategies such as Transfer Learning (TL), Self-Supervised Learning (SSL), Weakly Supervised Learning (WSL), Few-Shot Learning (FSL), and Continual Learning (CL) to maximize learning efficiency from limited labeled data.

The ongoing development of AI methods, particularly DL, is improving diagnostic and prognostic outcomes in CRC histopathology (Wu et al., 2022; Mall et al., 2023). However, it is worth noting that the performance of these models is highly dependent on the quality and complexity of the raw histopathology images (Wang et al., 2024; Komura and Ishikawa, 2019). Unlike radiological images, these digital images differ significantly in size and structural complexity. Moreover, the staining procedure, though inherent to the visualization process, results in inter-laboratory stain variability due to a lack of standardized protocols (Hoque et al., 2023). Thus, dedicated image pre-processing steps are required to standardize the input. Beyond this, the massive size and computational constraints of modern architectures mandate the use of patch-based analysis strategies. This creates the pressing need for dedicated post-processing techniques to aggregate patch-level outputs and refine final predictions. Nevertheless, heavy dependence on benchmark datasets has somewhat reduced the emphasis on dedicated pre- and post-processing strategies, making the current domain analysis centric. A comprehensive understanding of CRC analysis and willingness to achieve clinical acceptability and trust, therefore, requires a unified framework that systematically integrates all three phases—Pre-processing, Analytical, and Post-processing.

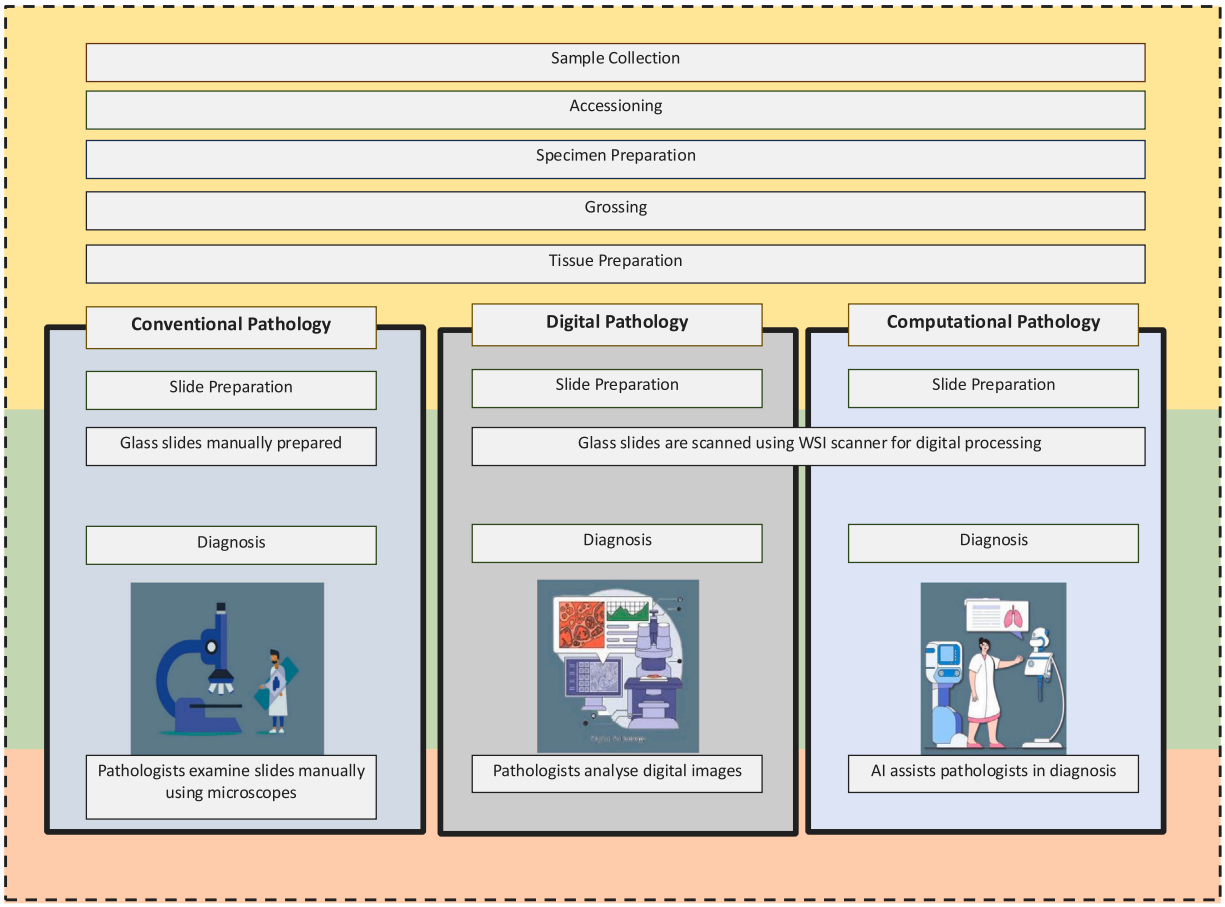


Fig. 2. Comparative workflow of conventional, digital, and computational pathology in terms of diagnostic style, starting with microscope-based examination (conventional), digital screen review (DP), and AI-assisted analysis (CPath).

1.1. Related work and research gaps

Although a unified framework is crucial, fully automated CRC diagnostic systems are still facing challenges due to inherent limitations in histopathology images. These inherent limitations, such as heterogeneity, structural complexity, and large size, hinder the application of AI models in CRC histopathology analysis. Thus, it necessitates different pre- and post-processing approaches. This requires a critical investigation into the existing literature to identify why a comprehensive, end-to-end workflow analysis is missing. A review of existing literature shows that previous studies have focused merely on a few such techniques. To provide a concise overview of the limitations of existing systematic reviews, we summarize their scope, focus, and methodology in Table 1. In addition to the existing review listed in Table 1, several comprehensive reviews (Baxi et al., 2022; Li et al., 2022b; Abdel-Nabi et al., 2023; Cooper et al., 2023; Kuntz et al., 2021; Rai, 2024; Park et al., 2022; Ahmedt-Aristizabal et al., 2022) have focused on CAD systems across multiple cancer types. In contrast, some authors (Alboaneen et al., 2023; Tharwat et al., 2022; Tamang and Kim, 2021; Kavitha et al., 2022; Pacal et al., 2020) have analyzed CRC in their review papers but have worked across various modalities, such as histopathology, endoscopy, and colonoscopy.

We have thoroughly scanned eighteen existing reviews on CRC histopathology analysis to identify trends and gaps in the literature. Out of these eighteen, five studies (Jing et al., 2023; Davri et al., 2022; Wu et al., 2023; Echle et al., 2021; Thakur et al., 2020) focused on classification, segmentation, detection, and prediction in CRC histopathology. Nevertheless, the remaining 13 studies (Baxi et al., 2022; Li et al., 2022b; Abdel-Nabi et al., 2023; Cooper et al., 2023; Kuntz et al., 2021; Rai, 2024; Park et al., 2022; Ahmedt-Aristizabal et al., 2022; Alboaneen et al., 2023; Tharwat et al., 2022; Tamang and Kim, 2021; Kavitha et al., 2022; Pacal et al., 2020) may limit the in-depth understanding of histopathology-based CRC analysis. Consequently, these broad reviews present only a superficial overview of CRC-specific methodologies and inherent histopathology challenges. This limitation is evident in the bar graph in Fig. 3. This analysis confirms a critical gap in the existing literature: the focus remains fragmented, either by analytical task (Table 1), cancer type, or imaging modality (Fig. 3). Moreover, to the best of our knowledge, no existing comprehensive review offers a unified, end-to-end framework that systematically integrates the crucial Pre-processing, Analytical, and Post-processing stages necessary for developing clinically robust CRC diagnostic systems.

1.2. Search strategy and study selection

The systematic search is carried out using scholarly databases such as PubMed, Scopus, ScienceDirect, and Google Scholar, covering publications from 2015 to 2025. The query keyword is: ((“colorectal” OR “CRC” OR “colon” OR “bowel” OR “intestine”) AND (“histopathology” OR “WSI” OR “whole slide image”) AND (“classification” OR “detection” OR “segmentation” OR “prediction” OR “preprocessing” OR “post-processing”) AND (“deep learning” OR “machine learning” OR “CNN” OR “MIL” OR “transformer”)).

The entire selection process (PRISMA) is shown in Fig. 4. Initially, a total of 230 articles are selected. Following the systematic exclusion of duplicate and non-relevant studies, a total of 156 articles were selected for detailed review. The irrelevant publications are excluded based on the following criteria: (a) studies not related to CRC (b) studies focusing

Table 1
Comparative analysis of existing systematic reviews for CRC histopathology.

References	Cancer scope	Pre-processing coverage	Post-processing coverage	Key insight	Limitation
Jing et al. (2023)	CRC	Limited	Not	Comprehensive analysis of ML/DL classification, segmentation and detection for Intestinal Histopathology datasets.	– Lacks analysis of the full end-to-end workflow. – Focus is primarily on analytical tasks.
Davri et al. (2022)	CRC	Not	Not	Comprehensive trend analysis of deep learning for CRC classification, grading, and Microsatellite Instability (MSI) prediction.	– Complete omission of pre- and post-processing techniques. – Focus only on core analysis of CRC histopathology.
Wu et al. (2023)	CRC	Limited	Not	Analysis of pathomics and deep learning for CRC tissue/MSI status.	– Lacks discussion of complete pre- and post processing pipeline. – Limited scope of analytical tasks covered.
Echle et al. (2021)	CRC	Not	Not	DL achieves state-of-the-art (SOTA) performance for MSI status prediction directly from WSIs.	– Limited analytical scope (only 20 studies). – Lacks generalized coverage of CRC tasks.
Thakur et al. (2020)	CRC	Not	Not	Classification of algorithms based on features including tumor, gland, prognosis, and tumor microenvironment (TME).	– Lack coverage of newer technologies. – Focus is primarily on specific tasks.

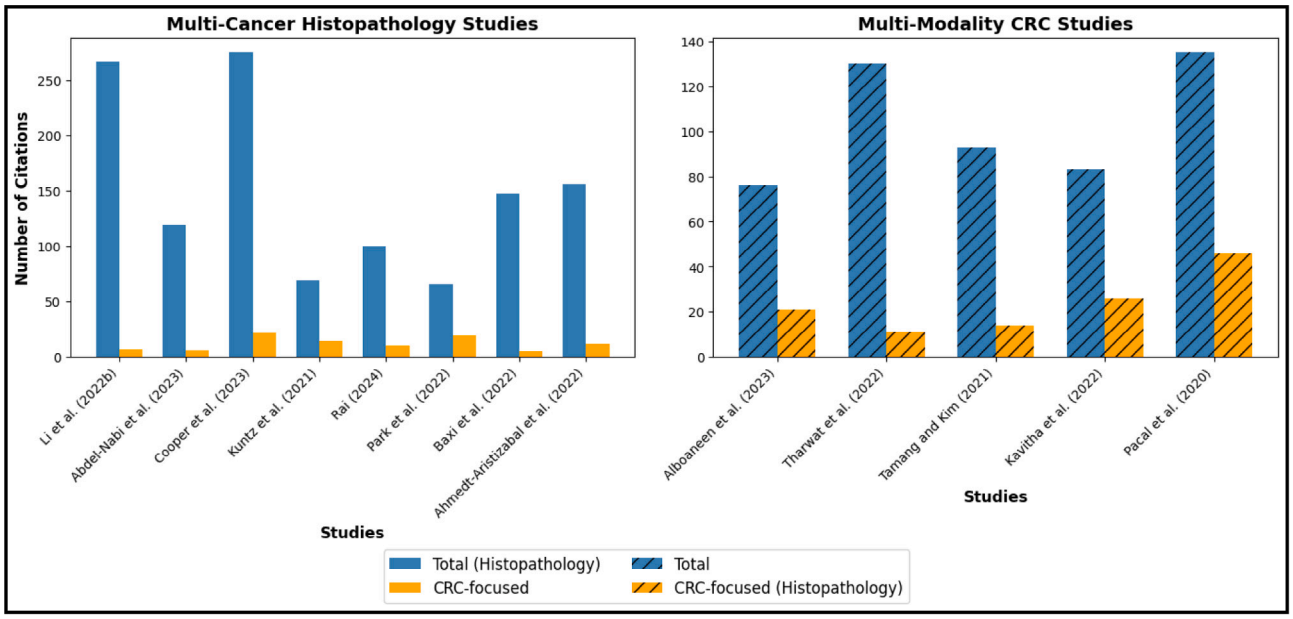


Fig. 3. Overview of literature citations across recent reviews: (Left) shows a comparison between the number of CRC-focused references and total references in multi-cancer histopathology studies. (Right) shows a comparison between the number of histopathology-focused references and total citations in multi-modality CRC studies.

on modalities other than histopathology images, (c) studies published prior to 2015. A final set of 156 papers covering studies related to pre-processing, classification, detection, segmentation, prediction and post-processing, is analyzed in this review.

1.3. Contributions and structure of the review

Building on the summary of existing reviews (Table 1 and Fig. 3), our work offers the following key contributions:

1. This review is the first to systematically present CRC analysis as an end-to-end workflow, integrating Pre-processing, Analytical, and Post-processing stages to cover the complete CRC histopathology pipeline (summarized in Fig. 7).
2. We provide a systematic and in-depth analysis of the under-analyzed Pre-processing components and Post-processing methods specifically for CRC histopathology.
3. We analyze the application of both ML and DL methods in the Analytical phase for CRC histopathology diagnostic (classification, segmentation, and detection) and prognostic (survival prediction, MSI status) tasks.

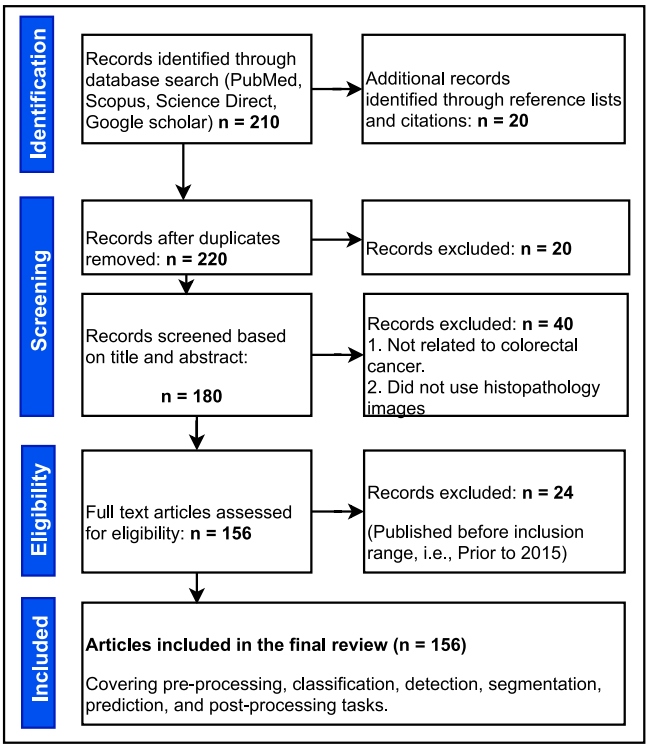


Fig. 4. PRISMA-based study selection process for CRC histopathology analysis.

This review summarizes a total of one hundred fifty six papers on colorectal histopathology analysis published from 2015 to 2025. The organization of the review is inspired by the CPath pipeline, presenting the analysis of CRC histopathology in a three-phase pipeline: Pre-processing, Analysis, and Post-processing. To achieve this, the structure of the paper is as follows: Section 2 provides an overview of the computational foundations for CRC histopathology analysis, which includes histopathology image modalities, publicly available CRC datasets, overview of ML and DL methods. Section 3 outlines widely used pre-processing techniques in CRC histopathology analysis. Section 4 delves into existing study on image analysis that utilizes ML and DL methods. Section 5 outlines standard post-processing methods in existing histopathology CRC literature. Section 6 discusses the current research landscape in CRC CPath diagnosis. Section 7 emphasizes existing open issues and potential future research directions. Finally, Section 8 presents the conclusion of the survey. The overall workflow of the review is illustrated in Fig. 6.

2. Computational foundations

This section introduces the fundamental components of computational CRC histopathology analysis. It covers commonly used histopathology image types, benchmark datasets, conventional feature extraction techniques, and classification methodologies, ranging from traditional ML approaches to advanced DL frameworks widely adopted in the literature for CRC tissue analysis.

2.1. Histopathology image modalities

Histopathology image preparation is a complex process encompassing several stages, from tissue sectioning to final slide generation. Among these stages, the tissue staining process is particularly crucial. Staining enables the differentiation of various cellular components and structural contexts within the tissue, enabling different structures to be visually distinguished. Different staining methods are employed in slide preparation, each designed to highlight specific cellular or tissue features for diagnostic interpretation. Therefore, histopathology images can be categorized based on the specific staining protocols employed. Two commonly used CRC histopathology stain types, H&E and IHC, are illustrated in Fig. 5.

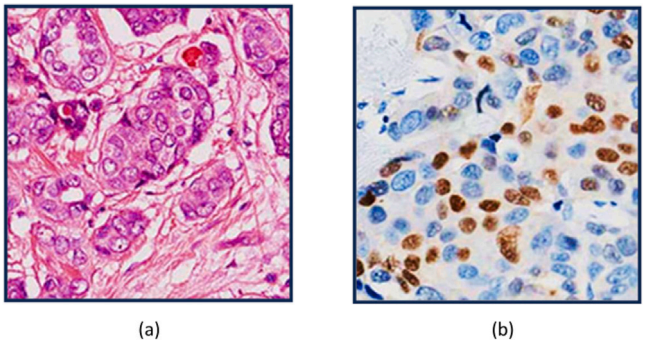


Fig. 5. Examples of CRC histopathology images: (a) H&E-stained, (b) IHC-stained.

- **Hematoxylin and Eosin (H&E):** In histopathology, H&E staining has been a well-established method for over 50 years, in which Hematoxylin stains nuclei blue while Eosin stains the cytoplasm and connective tissue pink. Due to widespread availability and a trove of data and publications, the H&E method remains the method of choice among histopathologists (Fox, 2000).
- **Immunohistochemistry (IHC):** It is another popular staining technique widely used for molecular analysis. Unlike H&E, which is suitable for cellular morphology analysis, IHC uses antibodies to detect specific protein expressions and molecular markers which are crucial for prognostic evaluation (Rizzardi et al., 2012).

2.2. Publicly available histopathology datasets for CRC

Large and high-quality datasets are foundational assets for any data-driven analysis. Although the advent of DP has improved data storage and management, digitizing histopathology datasets has required the involvement of multiple entities, ranging from high-cost equipment to individual or a team of pathologists for accurate annotations, making this process costly and time-consuming. The public release of these datasets requires additional ethical approvals to protect patient privacy. The advancement of modern architectures further demands substantial amounts of data for effective training, which can impede the rapid development of automated analysis pipelines. Nevertheless, open repositories such as the TCGA offer publicly available CRC WSIs for research and development. Now, medical institutions and researchers are collaborating to curate and release highly annotated CRC datasets in the public domain. Many of these datasets offer detailed annotations, such as image- or patch-level labels for classification and pixel-level annotations for segmentation, and are widely used for automated CRC analysis. The specific details of these datasets are outlined in Table 2 to ensure better clarity and accessibility for reference.

- **CoN-SeP** (Graham et al., 2019c): Colorectal Adenocarcinoma Nuclear Segmentation and Phenotypes (CoNSeP) dataset consists of 41 digital slides of colorectal tissue and has approximately 24K manually annotated nuclei along with corresponding labels across seven distinct cell types. As CoNSeP provides instance-level masks for both nucleus segmentation and classification, it is ideal for analyzing both tasks.

Table 2
Summary of available public histopathology datasets for the analysis of colorectal cancer.

Dataset	Class	Images	Annotation	Task	Description	Link	Limitations
Warwick-QU (GLaS) (Sirinukunwattana et al., 2015b)	Two	165	Pixel-level gland annotation	Gland segmentation	All 165 H&E-stained slides were scanned using a MIRAX MIDI Slide Scanner at a pixel resolution equivalent to 20x magnification	https://warwick.ac.uk/fac/sci/dcs/research/tia/glascontest/	Released as cropped image regions without WSI context, Small dataset size, Limited tissue diversity
CRCHistoPhenotypes (Sirinukunwattana et al., 2016)	Four	100	Point-level nucleus centroids	Nuclei detection and classification	The coordinates of the centroids were used for nucleus annotation in .mat format	https://warwick.ac.uk/fac/sci/dcs/research/tia/data/crchistolabelednucleihe	Limited patient diversity due to WSIs from only 9 patients, Released as fixed-size patches without WSI context, Partial Labeling
Kather 5000 images (Kather et al., 2016)	Eight	5K	Patch-level annotation	Tissue classification	It serve as a benchmark for texture-based methods	https://zenodo.org/records/53169	Released as fixed-size 150×150 px patches without WSI context, Small dataset size
CoN-SeP (Graham et al., 2019c)	Seven	41	Instance-level annotation	Nuclei segmentation and classification	Many significant overlaps exist among nuclei with indistinct boundaries, and several artifacts like ink exist in the dataset	https://warwick.ac.uk/fac/sci/dcs/research/tia/data/	High nuclear overlap and indistinct boundaries, Presence of artifacts, Limited diversity due to samples from 16 patients.
NCT-CRC-HE-100K (Kather et al., 2019a)	Nine	100k	Patch-level annotation	Tissue classification	All patches were standardized by color normalization using Macenko's method	https://zenodo.org/records/1214456	Released as non-overlapping patches without WSI context, No patient-level linkage, Class Imbalance
LC25000 (Borkowski et al., 2019)	Two	10,000 colon tissues	Patch-level annotation	Tissue classification	This dataset comprises 25,000 histology patches of lung and colon cancer images, out of which 10,000 are colon tissue images	https://academictorrents.com/details/7a638ed187a6180fd6e464b3666a6ea0499af4af	Released only as pre-cropped patches without WSI context, Heavily Augmented
CRAG (Graham et al., 2019a)	Two	213	Pixel-level gland annotation	Gland segmentation	Includes gland annotations overlaid on the original H&E images	https://warwick.ac.uk/fac/sci/dcs-/research/tia/data/mildnet/	Released as cropped tiles without WSI context, Limited tissue diversity
CRC-TP (Javed et al., 2020)	Seven	280k patches	Region-level tissue annotation	Tissue classification	All patches were extracted from 20 H&E stained WSIs	https://warwick.ac.uk/fac/cross_fac/tia/data/crchistolabelednucleihe/	Released as fixed-size 150×150 px patches without WSI context, Class imbalance, Tissue overlap within patches
Lizard (Graham et al., 2021)	Six	291 WSIs	Pixel-level nuclei annotation	Instance segmentation and classification	This dataset comprises labeled nuclei in H&E-stained colon tissue images from six different datasets	https://warwick.ac.uk/fac/cross_fac/tia/data/lizard/	Imbalanced nucleus-class distribution, Heterogeneous staining and scanner variations due to multi-centric sources
DigestPath (Da et al., 2022)	Two	872 tissue sub-images from 476 WSIs	Pixel- and region-level tumor annotation	Cell detection and tissue segmentation	Multi-center digestive pathology dataset with colorectal tissue subset	https://digestpath2019.grand-challenge.org/	Released as region-specific images, Multi-center dataset with heterogeneous staining and limited case diversity
CCTGS (Arslan et al., 2025b)	Five	103 WSIs	Pixel-wise segmentation masks	Tumor Grade Segmentation	First dataset with segmentation masks for all three tumor grades (G1, G2, G3) and Normal Mucosa	https://eakbas.github.io/metu-cctgs/	Single-Center Data, No patient follow-up or survival data available

- **CRCHistoPhenotypes** (Sirinukunwattana et al., 2016): This data set comprises 29,756 manually annotated nuclei, each marked at its center, extracted from 100 H&E-stained colorectal adenocarcinoma histology images. Among these nuclei, 22,444 have corresponding class labels across four cell types. The CRCHistoPhenotype serves as an ideal source for both the detection and classification of nuclei.
- **Warwick-QU (GLaS)** (Sirinukunwattana et al., 2015b): This data set was released as part of the Gland Segmentation Challenge organized by MICCAI 2015. GlaS comprises 165 H&E-stained colorectal histology slides with manually annotated glandular boundaries for both benign and malignant tissue and serves as a benchmark for evaluating gland segmentation algorithms.
- **CRC-TP** (Javed et al., 2020): This dataset includes 280K patches from 20 region-level-annotated digital H&E-stained CRC slides. Each patch in this is non-overlapping and fixed at 150×150 pixels. Importantly, each patch was initially normalized using the Macenko method and is associated with a class label corresponding to one of seven distinct tissue types. All multi-label patches

(multiple tissue phenotypes) were intentionally discarded during dataset preparation. For CRC patch-level classification, CRC-TP serves as a valuable benchmark dataset.

- **Kather 5000 images** (Kather et al., 2016): The Kather dataset comprises 5K patches of size 150 × 150 px from 10 anonymized H&E stained CRC tissue slides. It contains 8 distinct tissue classes and includes texture characteristics across different scales. The primary aim was to serve as a public benchmark for texture based classification methods in CRC histopathology.
- **NCT-CRC-HE-100K** (Kather et al., 2019a): This dataset contains 100K H&E-stained non-overlapping image patches of size 224 × 224 pixels extracted from 86 distinct CRC patients. Each patch in this dataset was initially normalized by the Macenko stain normalization method and is associated with a class label corresponding to one of nine distinct tissue types. Similarly, a non-normalized version of this dataset with the same WSIs but different patches, known as NCT-CRC-HE-100K-NONORM, was also released publicly. The NCT-CRC-HE-100K dataset serves as a valuable benchmark for CRC histopathology patch classification.
- **CRAG** (Graham et al., 2019a): The Colorectal Adenocarcinoma Gland (CRAG) dataset comprises a total of 213 histology patches extracted from 38 H&E-stained colorectal WSIs. Each patch has a size of 1512 × 1512 pixels and is provided with manually annotated instance-level gland segmentation masks.
- **LC25000** (Borkowski et al., 2019): The LC25000 dataset comprises 25K H&E-stained lung and colon tissue images. It originally contained 750 lung tissue images with three class labels and 500 colon tissue images with two class labels. However, with augmentation, this dataset comprises a total of 25K images, with an equal number of samples in each class.
- **Lizard** (Graham et al., 2021): The Lizard is the largest known colorectal histology dataset for instance nucleus segmentation and classification. It provided an annotation set of nearly 50,000 nuclei extracted from 291 H&E-stained images collected from six established sources (GlaS, CRAG, CoNSeP, DigestPath, CRCHistoPhenotypes, and TCGA) of colon tissue.
- **DigestPath** (Da et al., 2022): The DigestPath dataset, developed in collaboration with multiple Chinese medical institutions, comprises two subsets devoted to gastrointestinal pathology: (a) a Signet Ring Cell Detection dataset that contains H&E-stained histopathology slides scanned at $40X$ magnification, and (b) a Colonoscopy Tissue Segmentation and Classification dataset that includes H&E-stained slides scanned at $20X$ magnification for lesion analysis.
- **CCTGS** (Arslan et al., 2025b): The Colorectal Cancer Tumor Grade Segmentation (CCTGS) dataset contains pixel-wise segmentation masks for 103 H&E-stained colon WSI. It is a multi-class dataset that comprises segmentation masks for both tumor grades and Normal Mucosa. Moreover, it is the newest publicly available resource and serves as a benchmark for evaluating multi-class semantic segmentation in CRC tumor grading.

2.3. Conventional ML methods

The use of ML methods in CPath began around 2010. During this period, ML approaches focused on extracting handcrafted features from manually defined ROIs and applying traditional classification algorithms for analysis. The entire process of converting raw image data into a diagnostic decision involves two major components: Feature extraction and Classification method. The early phase of feature extraction depends heavily on handcrafted features generated by predefined algorithms (Öztürk and Akdemir, 2018a). Over time, the scope of feature extraction in ML has expanded beyond handcrafted features to include automatically learned representations. The subsequent section outlines the ML methodology widely used in histopathology in terms of how visual information is extracted and used to categorize.

2.3.1. Feature extraction

The entire ML methodology hinges on a feature extraction module because the more discriminative features they extract, the more accurate the predictions are. Feature extraction is a technique that transforms raw information into quantifiable descriptors that allow computational decision-making. This segment embarks on a panoramic exploration of feature extraction techniques used with ML methods.

- **Handcrafted Features (HCFs):** Handcrafted features (HCFs) are explicitly designed descriptors derived using predefined algorithms and domain knowledge. In the realm of histopathology images, HCFs are specifically designed to determine texture, morphological, color, and gradient-based descriptors that capture the visual characteristics of tissue and cellular architecture. While these features are highly interpretable due to their intentional design, they are often sensitive to variability, particularly in complex histopathological images.
- **Deep Features (DFs):** In contrast to HCFs, which require manual design and domain expertise, deep features (DFs) are complex and hierarchical representations of data learned automatically by DL models such as CNNs. The concept of transfer learning (TL) refers to the reuse of model weights trained on one task for other related tasks and has transformed the way features are extracted. Pre-trained models such as ResNet, GoogleNet, and VGGNet are CNN architectures trained on large image datasets, most commonly ImageNet, and are often used to extract feature representations directly from complex histopathology images. Consequently, DFs are now extensively used as input vectors for conventional machine learning classifiers, including SVMs, KNN, bagging, and RF, across a variety of modern CPath frameworks. However, these features are often complex, abstract, and high-dimensional, thereby requiring substantial computational resources and memory for processing.

2.3.2. Conventional ML classification methods

A conventional ML classifier is a computational model or mathematical function that maps input data to predefined categories. This section outlines conventional ML classifiers that use either HCFs or DFs to learn discriminative patterns and categorize data into one of the appropriate classes. These classifiers are used at the core analytical stage in CPath pipelines, where the extracted features serve as input vectors for predictive modeling.

- **K-Nearest Neighbor (KNN)** KNN is a straightforward yet powerful classification technique in machine learning. However, with some modification, it can also be used to solve the regression problem. It is a non-parametric supervised learning technique, often described as a lazy learner, since it does not require explicit learning during the training process (Talo, 2019). It is widely used for applications where prior knowledge about the data distribution is limited. The core idea of KNN is to assign a class label to a query sample based on the classes of its nearest neighbors in the feature domain. It determines the K nearest samples using distance metrics such as Euclidean or Manhattan distance and predicts the class through majority voting. The value of K is a hyperparameter in KNN, and it should preferably be an odd value to avoid ties. The simplicity of its metric-based learning makes KNN popular for a variety of complex image classification tasks. It is often used as a baseline classifier for patch-level tissue classification, where the similarity between local feature patterns derived from either HCF or DF serves as a key discriminative factor.
- **Support Vector Machine (SVM)** SVM is a supervised learning approach commonly used for classification tasks (Marneni and Vemula, 2023). The core idea behind the SVM classifier is to find

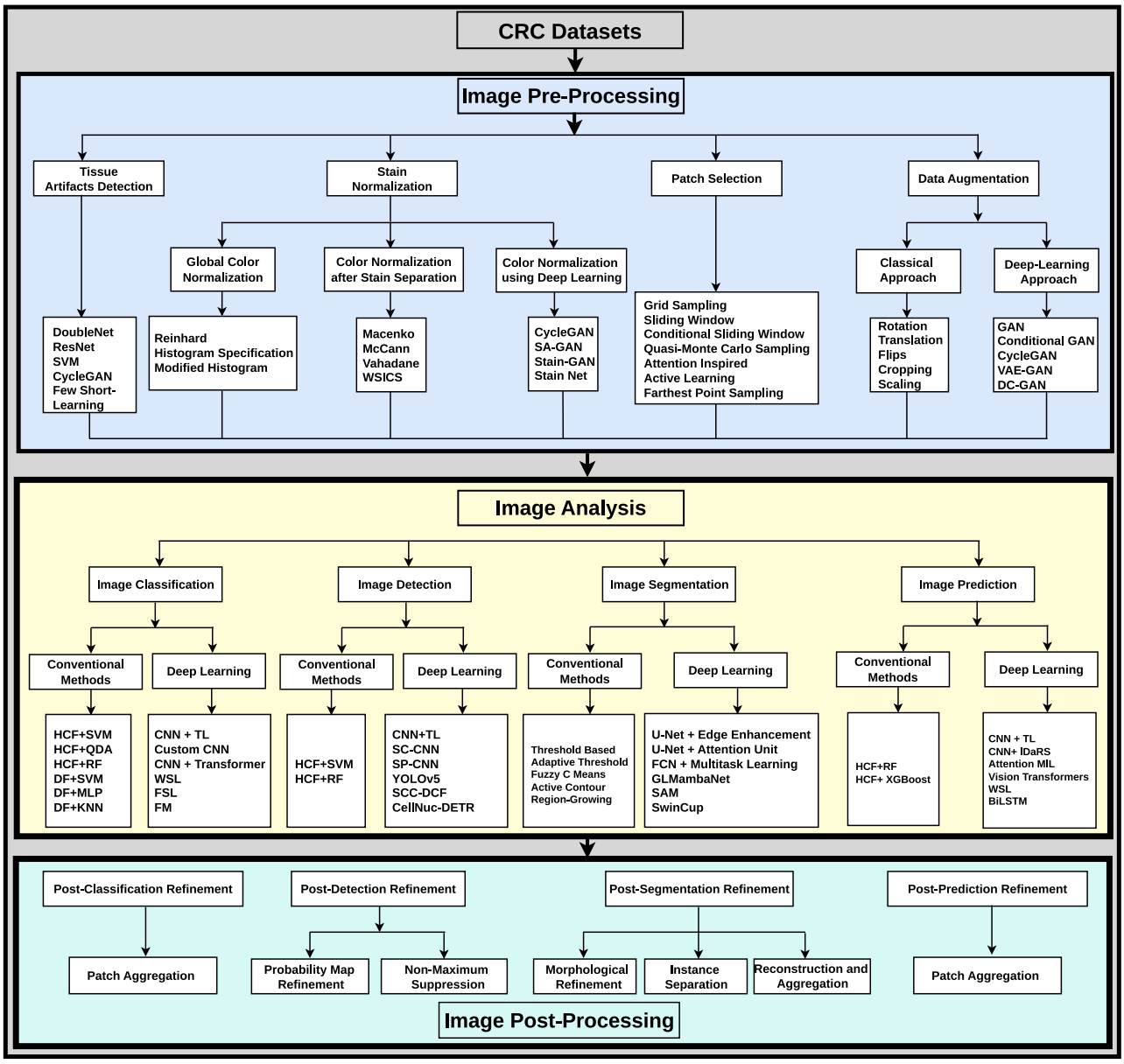


Fig. 6. A flow diagram of conventional and deep learning-based methods for colorectal cancer analysis using histopathology images, with the following aspects: (a) CRC datasets, (b) image pre-processing, (c) image analysis across classification, detection, segmentation, and prediction tasks, and (d) image post-processing.

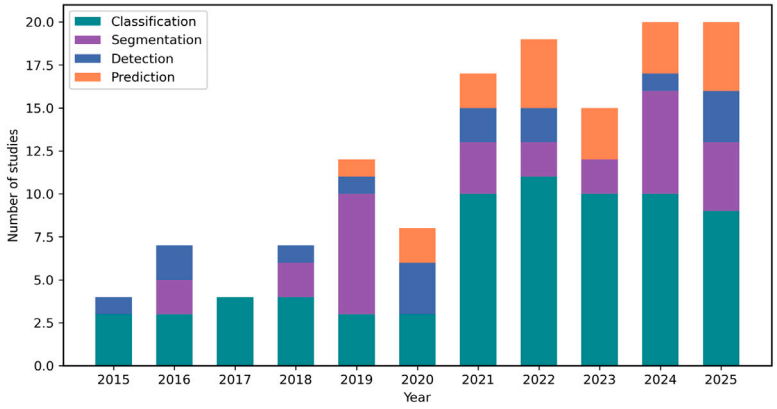


Fig. 7. Year-wise distribution of CRC histopathology analysis across classification, detection, segmentation, and prediction tasks (Section 4).

an efficient computational method by which it learns a margin-maximizing hyperplane that linearly separates different classes in feature space. This hyperplane maximizes the distance from the bordering data points of each class, which are known as support vectors. This approach can help to improve generalization accuracy of the SVM model. While SVM is effective for linearly separable datasets, it can also handle non-linear datasets by using robust kernel methods such as radial basis function (RBF), polynomial, or sigmoid kernels. Handling non-linearity allows SVM to tackle complex tissue classification challenges, making it a widely used ML technique in histopathology for feature-based image classification.

- **Logistic Regression (LR)** LR is a family of supervised ML techniques commonly used to solve classification problems. The goal of the LR model is to find a decision boundary that effectively separates the two classes in the feature space. The idea behind the LR technique is based on the concept of odds, defined as the ratio of probability of success to probability of failure. The parameters for LR models are the feature weights. The linear combination of these features is passed through the non-linear logit function for further classification. LR uses the logit function, often referred to as the sigmoid function, which converts a probability score into binary values for classification. Depending on the problem statement, the shape of the decision boundary is influenced by assumptions made by different ML algorithms. Despite using a non-linear activation function, LR still assumes a linear decision boundary in the feature space. Therefore, while LR behaves elegantly with linearly separable classes, suitable modifications to its cost function and the use of discriminative feature representations can also make it effective for tissue classification tasks in histopathology (Subasi, 2020; Khairunnahar et al., 2019).
- **Random Forest (RF)** RF is a powerful ML technique used for classification. It uses an ensemble learning paradigm that integrates and trains multiple decision trees (DTs) in parallel on various subsets of the training data using the bootstrapping technique, followed by aggregation (Zhao, 2013). The terms bootstrapping and aggregation are collectively called bagging. The primary goal of the bagging method is to mitigate overfitting by reducing the variance of the RF classifier. As bagging consists of two techniques, the goal of bootstrapping is to make sure that each distinct DT in RF is unique. This randomness enables RF to handle high-dimensional data effectively and minimizes overfitting. RF uses a majority vote technique to predict the outcome, thus improving generalization. The interpretability, scalability, and strong generalization capability of RF make it a reliable model for tissue classification, segmentation, and diagnostic prediction. Moreover, RF can easily process HCFs and high-dimensional features such as DFs and shows robust performance across various histopathology datasets.
- **Quadratic Discriminant Analysis (QDA)** QDA is an ML technique commonly used to solve classification problems. The central goal of the QDA is to recognize the quadratic hyperspaces that separate the data instances in the feature space. In QDA, the covariance matrix is created for each class $k \in \{1, \ldots, k\}$ rather than using an identical covariance matrix Σ_k. Thus, QDA allows the decision boundary to be quadratic rather than linear. This enables QDA to model more complex, non-linear decision boundaries between classes. In addition, QDA would be the better choice when the nature of the dataset is heteroscedastic (Wu and Hao, 2022). As a result, QDA has shown its excellent performance in complex histopathology tissue classification when integrated with appropriate feature extraction and selection techniques (Chen et al., 2024).
- **Light Gradient Boosting Machine (LightGBM)** LightGBM is an ensemble learning model built on the Gradient Boosting Decision Tree framework and is commonly used to solve classification problems. LightGBT was introduced by Ke et al. (2017), researchers from Microsoft. The core idea of LightGBM is to effectively handle high-dimensional features that other conventional GBMs struggle with. For this, it uses the Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) approaches, as well as histogram-based sampling. Handling high dimensionality and faster and smoother training made this useful for CRC histopathology image classification tasks.
- **Multilayer Perceptron (MLP)** MLP is a feed-forward neural network consisting of an input layer, one or more hidden layers, and an output layer. The interconnected layers of perceptrons, particularly hidden layers, enable the MLP to learn and capture complex non-linear feature relationships from data. Backpropagation is another integral component of an MLP that is used during training to optimize the network weights. Unlike traditional ML classifiers, MLPs are non-linear classifiers that do not rely heavily on manual feature engineering and can effectively work with both HCF and DFs. The automatic modeling of complex relationships enables MLPs to be used effectively for complex tissue classification tasks in histopathology.

2.4. DL frameworks and feature evolution

The advent of DL, particularly after the 2012 ImageNet competition, marked the beginning of a golden age for CPath. The primary objective of its emergence was to address the feature-engineering bottleneck and the domain-shift vulnerability inherent in ML pipelines. DL models revolutionized the way features were extracted by making this process automatic and data-driven. This automation of feature learning not only reduced the reliance on manual intervention but also accelerated the development of CPath applications (Yu et al., 2024b). DL models now learn and capture complex and heterogeneous features directly from raw histopathology images, making DL the dominant analytical method in CPath. Nowadays, DL methods are applied to a variety of CPath analytical tasks, ranging from tissue classification and segmentation to the detection and prediction of biomarkers. Moreover, it has achieved SOTA performance in a variety of CPath analysis tasks. The field is still evolving, and the subsequent section outlines popular DL methods widely adopted for histopathology image analysis.

2.4.1. End-to-end feature learning and conventional architectures

The field of DL has become immensely popular with the introduction of Convolutional Neural Networks (CNNs). The major components of any CNN include convolutional layers, pooling layers, and fully connected layers. The former extracts local spatial features, pooling layers subsequently reduce spatial dimensions, and fully connected layers perform the final classification. The automatic feature extraction and end-to-end feature learning capabilities of CNNs make them highly popular, and as a result, they have achieved SOTA performance across diverse histopathological tasks, including tissue classification (Ozturk and Akdemir, 2019), segmentation (Ozturk and Akdemir, 2019), and biomarker prediction (Fu et al., 2020). Notwithstanding this, CNNs are designed for fixed-size patch-level inputs and operate with a limited receptive field. This limits their adaptability, especially when long-range dependencies are required, such as in predicting disease progression and capturing complex tissue-level variations in CRC. The subsequent subsection outlines foundational CNN architectures widely adopted in CRC histopathology for diagnosis and prognosis.

- **VGG-Net** VGG-Net (Visual Geometry Group Network) is a composition of a convolutional neural network developed by Simonyan and Zisserman (Simonyan and Zisserman, 2014) in 2014. It was designed to explore the consequences of network depth on the accuracy of CNNs for large-scale image recognition. Experiments with VGGNet demonstrated that increasing depth with smaller convolutional filters improves performance. Moreover, its smaller, repeated convolutional filters allow the network to learn

complex, multilevel features from the image. As a result, VGG-Net was widely adopted in early studies of CRC histopathology classification, either as a robust feature extractor or backbone. VGG16 and VGG19 are the commonly used variant models of the VGG-Net family. However, its higher number of parameters hinders its wider applicability in CRC histopathology.

- **Res-Net** He et al. (2016) developed a model in 2015 called Residual Network (ResNet). In the meantime, vanishing gradients had become a major issue for training deeper networks due to learning stagnation. ResNet transformed DL by introducing skip connections, also known as residual connections, to mitigate the vanishing gradient. Skip connections connect the input from a previous layer directly to a subsequent layer, encouraging the training of deeper networks. This impressively enhanced feature learning and representation capacity of the networks without degrading performance. As a result, the adaptability of ResNet has increased tremendously in a variety of biomedical image analysis tasks, including CPath. Today, ResNet and its variants are widely used as both feature extractors and backbones for various CRC histopathology image analysis, achieving impressive results by directly extracting stable, discriminative, and less redundant features from complex histopathology images.
- **Dense-Net** DenseNet is defined as a Densely CCNN (Connected Convolutional Neural Network), suggested by Huang et al. (2017) in 2016, aiming to enhance the concept of residual network connection in the deeper neural network. DenseNet extends the idea of residual connections in ResNet by introducing dense connectivity (dense blocks), in which each layer receives inputs from all preceding layers. As a result, DenseNet improves information flow in connected neural networks using reusability with fewer parameters and reduced overfitting. The feature reusability and parameter efficiency of DenseNet or its variants (DenseNet-169, and DenseNet-121) make it widely adaptable for histopathology image analysis, where it can act either as a feature extractor to generate fine-grained morphological and cellular features or as a backbone to support end-to-end learning (Riasatian et al., 2021).
- **Inception-Net** InceptionNet, also referred to as the GoogleNet, was developed by a group of Google researchers in 2014 and introduced in the paper "Going Deeper with Convolutions" by Szegedy et al. (2015). The purpose behind introducing this model was to optimize the computational cost during training of the deep networks and mitigate the vanishing gradient problem. The major contribution is the Inception module, which performs parallel convolutions with different kernel sizes and enables multi-scale feature extraction within a single layer. To optimize computational cost, it applied factorization of larger convolutions into smaller ones, and reduced the vanishing gradient problem by using auxiliary classifiers. It outperforms VGG and AlexNet in several image classification tasks. Due to its computational efficiency and multi-scale feature-learning capabilities, it is widely used as a backbone or feature extractor for CRC histopathology classification tasks. In many instances, this model is embedded with the ResNet architecture, Inception-ResNet, to enhance image classification accuracy.
- **U-Net** The advent of the U-Net has been a great success for the field of biomedical imaging. It is a CNN-based DL architecture introduced by Ronneberger et al. (2015) for pixel-level image segmentation. As the name suggests, it follows a U-shaped structure comprising an encoder, a decoder, and skip connections. The encoder learns the spatial characteristics of an image during downsampling, while the decoder performs upsampling and convolution to reconstruct spatial details and generate the final segmentation map. Skip connections are designed to preserve the spatial information lost during downsampling. U-Net is effective even with a small number of training cases. U-Net and its modern variants, such as hybrid U-Nets where the encoder is replaced with powerful CNNs (ResNet, DenseNet, EfficientNet, or Swin Transformers), as well as attention U-Nets, are widely used for various CRC histopathology segmentation tasks, such as gland and nuclei segmentation.

2.4.2. Advanced architectures and novel learning strategies

The field of DL is growing rapidly, with modern architectures and learning strategies being developed and adopted in the histopathology image analysis. This field has rapidly evolved to overcome shortcomings of CNNs, such as their inability to model global context and to account for WSIs' implicit constraints. This section outlines advanced architectures and novel learning strategies recently adopted in CRC histopathology to address these shortcomings.

- **Transformers** Transformers (Vaswani et al., 2017) are an advanced DL architecture that uses self-attention mechanisms to identify long-range correlations in image data. As a result, they have been widely adopted in CRC histopathology, where histopathology images are first treated as sequences of patches and split into non-overlapping, fixed-size regions, unlike CNNs, which are based on hierarchical feature representations and receptive fields. A key advantage of Transformer-based architectures is the ability to model entire spatial relationships within an image. Consequently, Transformer-based models are increasingly adopted in CRC, as they can capture not only cellular structures in histopathology images but also relationships between tissue components, achieving SOTA results in CRC diagnosis and, specifically, prognosis. However, Transformers are computationally expensive due to the self-attention module, and their complexity scales quadratically and increases exponentially with image resolution.
- **Weakly Supervised Learning (WSL)** WSL proves to be a data-efficient learning strategy that allows models to be trained with only coarse-grained (image-level) annotations (Nisha and Mishra, 2026). In the realm of CRC histopathology image analysis, where comprehensive pixel- or patch-level annotations are difficult to obtain and highly time-consuming, WSL enables the effective training of DL models using only slide-level labels (Kuang et al., 2024). MIL is a widely used WSL framework for CRC image analysis. The main principle of the MIL method is to conceptualize the WSI as a bag containing multiple patches and, in general, the bag is defined as positive if at least one of its patches is labeled positive. Other MIL aggregation methods, such as attention-based pooling and transformer-based aggregators, are also used in CRC histopathology analysis. Beyond CRC histopathology diagnostic tasks, MIL itself provides a major advantage by enabling the prediction of high-level clinical and molecular attributes using only slide-level labels.
- **Few Shot Learning (FSL)** Few-shot learning is a novel strategy that enables DL models to learn new categories effectively when only a few labeled examples are available (Pachetti and Colantonio, 2024). FSL proves a data-efficient learning approach specifically for the CRC histopathology domain, where rare subtypes, uncommon morphological patterns, and scarcity of annotated samples often limit the creation of large-scale, balanced datasets. The rationale behind FSL is to generalize the knowledge acquired from a large base dataset and then quickly adapt this knowledge to classify newer categories using minimal data. Among the literature we reviewed, metric-based learning (Prototypical Networks) and meta-learning approaches (Model-Agnostic Meta-Learning (MAML)) are commonly applied for CRC histopathology domain.
- **Self Supervised Learning (SSL)** SSL is a powerful learning strategy that proves to be a data-efficient approach for training DL models in no-supervision settings (Zhang et al., 2025). SSL, unlike supervised learning, compels DL models to learn robust feature representations from tissue images by creating pretext tasks from

an ample unlabeled WSI dataset. This becomes of utmost importance in CRC histopathology, where collecting annotated data is costly and time-consuming. The primary goal of SSL is to train a powerful encoder (or backbone) that can learn high-level representations from histology images itself. This process results in a powerful feature encoder that can then be efficiently finetuned on a variety of downstream tasks such as classification or genetic biomarker prediction. SSL techniques such as contrastive learning and masked-image modeling are widely used in CRC histopathology image analysis. After the success of large models such as ViTs, SSL becomes instrumental for the development of pathological Foundation Models (FMs) such as CTransPath and HIPT, often trained on diverse, large-scale, pan-cancer histopathology datasets that capture a universal feature set across different tumor types and organs. Both Vision FMs and Vision–Language FMs, such as PLIP and CONCH, are adopted for a wide variety of pathology tasks, ranging from tissue classification and detection to predicting genetic biomarkers in CRC histopathology images, with either small fine-tuning or zero-shot settings.

3. Image pre-processing

The exponential growth of digitalization has established ML and DL as the preferred choices for CRC histopathology analysis. However, this technological shift has simultaneously introduced challenges, such as massive storage requirements and a lack of standardization, which hinder the seamless adoption of these techniques. Since the performance of both methods is contingent upon the quality of raw WSIs, pre-processing has become an indispensable prerequisite (Khan et al., 2014). Nevertheless, the purpose of pre-processing differs as ML uses it to facilitate manual feature engineering while DL relies on it to ensure standardized input for automated learning.

To this end, modern CAD systems attempt to manage the computational and memory demands of WSIs for real-time use. Yet, manual intervention and diverse scanning hardware can introduce artifacts and staining variability into automated CRC pipelines. Since quality control is often limited to subjective visual assessments, these variations across staining batches and laboratories are frequently overlooked. Beyond these technical variations, the scarcity of annotated CRC data further exacerbates the computational pipeline. Ensuring a robust pre-processing workflow is therefore vital, as any discrepancies at this stage will propagate and compromise the diagnostic and prognostic pipeline. In light of this, based on the literature reviewed so far, this paper classifies CRC pre-processing techniques into four essential categories: tissue artifact detection, color or stain normalization, patch selection, and data augmentation.

3.1. Tissue artifact detection methods

Artifacts including folds, debris, blur, and contaminants emerge during histological preparation and digitization. These anomalies, exacerbated by hardware limitations like scanner resolution and illumination, can disrupt the scanning system during optimal focus selection and negatively affect downstream feature mining (Wu et al., 2015). Modern whole-slide scanners produce digital images that capture intricate phenotypic details at gigapixel resolution. In most cases, a histology slide of size 20 mm X15 mm can produce a WSI of size 80,000 X60,000 pixels (Khened et al., 2021). Given that a large portion of these images consists of non-informative background, identifying and removing this portion of the WSIs is crucial to optimizing computational efficiency. Nevertheless, segmenting tissues from the background is not sufficient, as the foreground invariably contains artifacts such as bubbles, pen marks, and tissue folds. In such cases, methods like Otsu thresholding (Otsu, 1978) often fail, as they only separate foreground from background based on grayscale intensity. These methods sometimes wrongly classify artifacts as tissues and propagate this misclassification into subsequent analysis (Schreiber et al., 2024).

Artifacts not identified during slide preparation are a formidable challenge, mostly when overlapping with cancerous regions (Duenweg et al., 2023). Yet, histopathologists are highly experienced and well-equipped to handle such challenges, but if these artifacts remain unaddressed, they result in lower image quality and impede automated analysis (Kothari et al., 2013). Wright et al. (2020) also observed this and reported that excluding 12 slides with quality issues from the CRC QUASAR trial dataset improved RF model accuracy by 3%. However, this approach required manual exclusion by pathologists. To address this, several automated frameworks have been developed to handle these artifacts and improve the accuracy of CRC image analysis.

A manual quality control approach for addressing artifacts affects the degree of automation. Ke et al. (2023) proposed an automated pre-processing framework using AR-Classifier and AR-CycleGAN models for detecting and restoring artifacts in colon histopathology images. This framework addresses artifacts ranging from tissue folds and dye stains to out-of-focus regions through a three-step pipeline involving artifact classification, detection, and restoration. Similarly, Hossain et al. (2023) used the DoubleUNet model for multi-organ artifact segmentation, including the colon. This model effectively segmented tissue artifacts and exhibited a strong agreement with pathologists' severity evaluations. Foucart et al. (2018) developed a modified deep residual network to detect and segment artifacts, including tissue folds, blur, tears, and ink stains. Data-efficient approaches are used and explored in this domain. For instance, Shaikh et al. (2022) introduced an FSL framework to identify and exclude regions with blur and tissue crush artifacts from subsequent analysis. Additionally, Shakhawat et al. (2020) proposed a quality assessment technique using an SVM for artifact detection in low-resolution WSIs and achieved 99% accuracy. It also evaluated the quality of the remaining slide in terms of noise and sharpness.

3.2. Stain normalization methods

Stain normalization is a common technique in CPath to standardize color appearances and minimize variances in WSIs caused by diverse scanners and staining protocols (Hoque et al., 2023; Gurcan et al., 2009). Staining is used during the image acquisition to highlight intra- and extra-cellular structures, but the process is sensitive to the source image quality. Artifacts such as dust or spots can impede accurate color distribution (Anghel et al., 2019). This loss could be prevented by removing artifacts prior to normalization (Moyes, 2021).

In histopathology, hematoxylin and eosin (H&E) are the primary dyes used to stain tissue slides (de Haan et al., 2021). Hematoxylin stains components like nuclei blue or purple, whereas eosin stains eosinophilic structures pinkish (Macenko et al., 2009). Nevertheless, the lack of standardization in staining protocols creates a formidable challenge of variability across different protocols and even different batches (Zheng et al., 2019). This can adversely affect the performance of CRC diagnostic and prognostic systems. Consequently, both statistical and DL algorithms are now widely adopted to normalize stain variability in WSIs. Stain normalization techniques are commonly classified into three types based on the normalization approach: Global Color Normalization, Color Normalization after Stain Separation, and Color Transfer using Deep Learning-based Techniques. This classification is illustrated in Fig. 8.

3.2.1. Global color normalization

The global normalization technique uses methods like Principal Component Analysis (PCA) to decouple color and intensity information in various spaces such as RGB and l$\alpha\beta$. It is particularly suitable for histopathological images due to the auto-correlation and spatial dependencies among pixel values. The most common method is Reinhard color normalization. It was developed by Reinhard et al. (2001) for

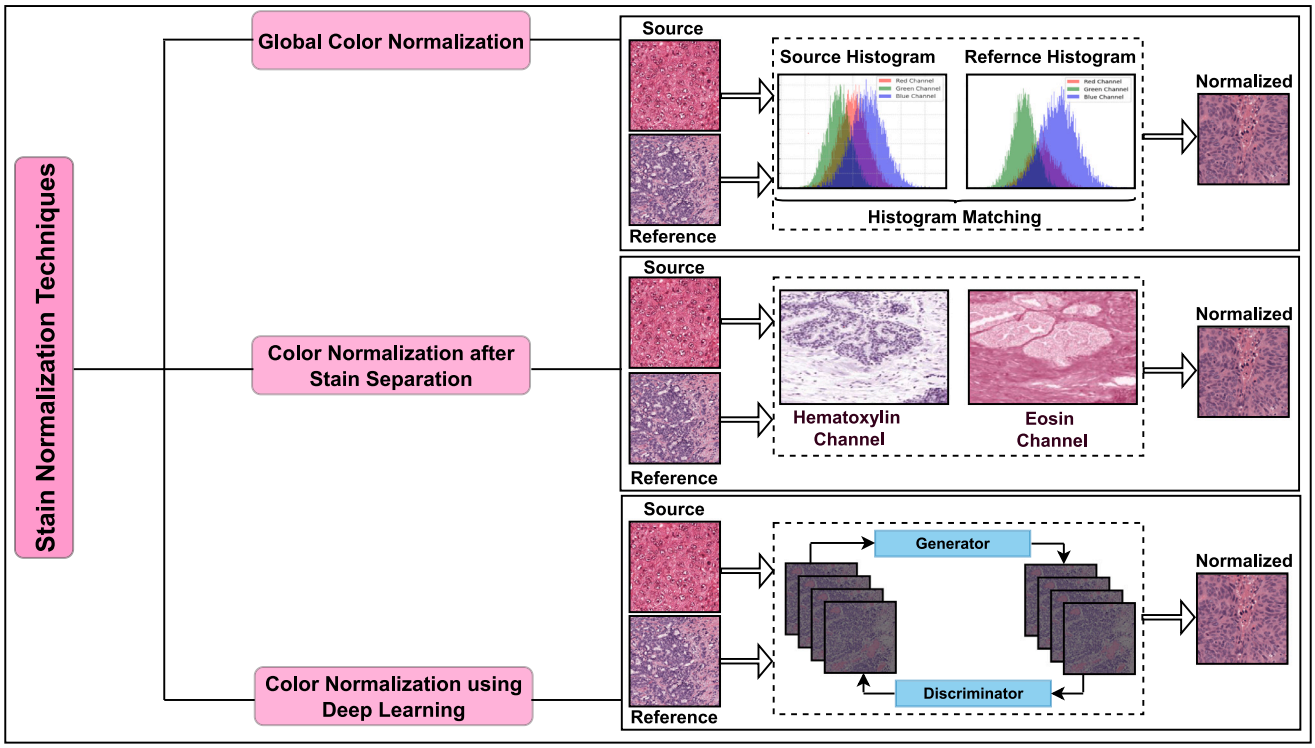


Fig. 8. Three common strategies for stain normalization: The topmost diagram illustrates the global color normalization approach, where global features such as contrast, brightness, and RGB histograms from the reference image guide normalization of the source image; the middle diagram depicts color normalization after stain separation, where individual stain components (hematoxylin and eosin) are isolated to modify the original image based on the overall color distribution using techniques like color deconvolution; the bottom diagram illustrates stain normalization using deep learning approaches, where the stain appearance of the source image is transformed to match that of the reference image using techniques such as GANs.

natural images. Over time, it has been widely adopted for histopathology images. This technique transfers the color properties of a reference (target) image to a source image by matching their mean and variance using a linear transformation in the $l\alpha\beta$ perceptual color space. Some modifications to the Reinhard method have also been explored. For instance, Roy et al. (2021) addressed limitations of the Reinhard algorithm, such as data loss and computational complexity, by proposing a modified version specifically for H&E stained CRC images. This approach satisfied all four hypotheses of color normalization and achieved high accuracy without data loss. Another popular method is Histogram Specification (HE) (Coltuc et al., 2006), originally designed for real images, is now a widely used color normalization method for histopathology slides. This method computes the source image histogram and matches it to a target histogram to map color and intensity values. However, it may sometimes introduce artifacts in the normalized slides (Zanjani et al., 2018a).

3.2.2. Color normalization after stain separation

Histopathology slides are stained with distinct dyes to differentiate components like nuclei and cytoplasm for microscopic examination. The stain separation technique employs color deconvolution to classify pixels into individual stain channels, such as Hematoxylin (H only) and Eosin (E only). This approach explains the correlation between light absorption and the concentration of an absorbing substance according to the Beer–Lambert equation. Finally, a standardization-driven strategy is used to decompose and normalize these light-absorbing stains individually (Ruifrok et al., 2001).

Macenko et al. (2009) introduced a widely used method for stain separation in histology images. This method assumed that the color of every pixel was a linear combination of two stain vectors. This method automatically recognized different stain vectors and applied stain deconvolution to separate the stains into distinguishable components. Later, McCann et al. (2014) used the same strategy, except that H-only images exhibited higher contrast than E-only images. Vahadane et al. (2016) made further improvements and developed the Structure-Preserving Color Normalization (SPCN) algorithm. The method used sparse regularized NMF (SNMF) to separate different stains while preserving tissue structure. In another study, Bejnordi et al. (2015) introduced the Whole-Slide Image Color Standardizer (WSICS), an automatic stain-specific normalization method that utilized chromatic and density distributions within the Hue-Saturation-Density (HSD) color model.

3.2.3. Color transfer using deep learning-based techniques

Color deconvolution improved stain estimation, but it sometimes failed to capture spatial relationships among tissues, particularly when diverse dyes were used (Bejnordi et al., 2015; Zanjani et al., 2018b). Consequently, DL models, specifically Generative Adversarial Networks (GANs) and their variants, have become the method of choice for color normalization in WSIs. GANs are a class of unsupervised learning techniques that comprise two neural networks: a generator and a discriminator, which use adversarial training to generate synthetic data that resembles real samples. Recently, GANs and their variations have gained enormous success in image enhancement and color normalization tasks. Recently, models like CycleGAN (Cycle-Consistent Adversarial Networks) (Zhu et al., 2017) have become popular in histopathology for color normalization tasks.

de Bel et al. (2021) used a CycleGAN network with residual learning to address stain variability in colon and kidney tissue segmentation tasks. This network was trained on single-source data, evaluated on multiple sources, and handled both H&E- and PAS-stained slides. The proposed model increased the overall Dice coefficient by 4% over regular stain normalization and 9% over non-normalized data for colon tissue segmentation. Similarly, Smit et al. (2023) employed the same CycleGAN approach for stain normalization to develop semi- and fully automated techniques for TSR scoring in colon cancer slides. It

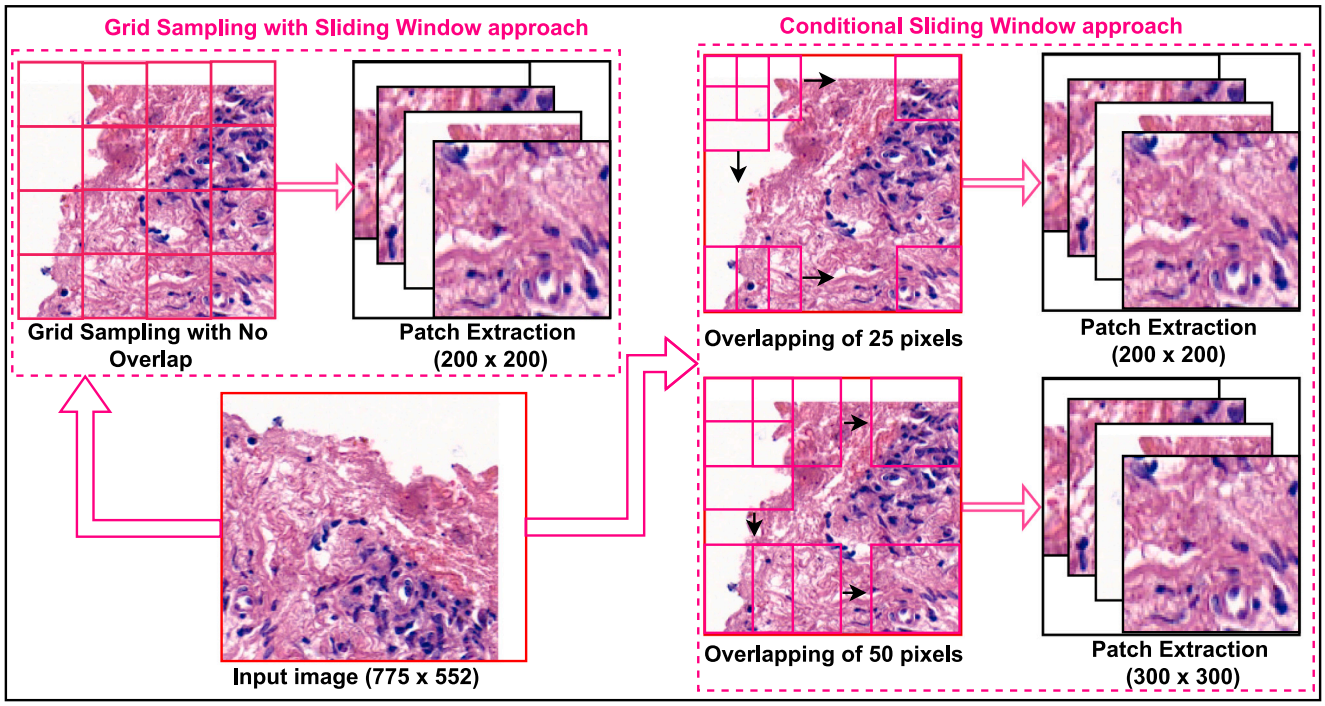


Fig. 9. Patch selection strategies: (Left) Grid sampling with no overlap: the WSI is initially divided into grid-like structures, followed by a sliding window approach to extract all patches of size (200 × 200) for model training. (Right) Conditional Sliding Window approach: the upper WSI uses a 25-pixel overlap to extract (200 × 200) patches, while the lower WSI applies a 50-pixel overlap to extract larger patches of size (300 × 300) to support multi-scale patch extraction from the WSI.

achieved Spearman correlations of 0.88 and an intraclass correlation coefficient (ICC) of 0.78 for the visual and semi-automated methods.

Other GAN variants have also been explored for CRC histopathology. For instance, Kausar et al. (2021) proposed SA-GAN, a GAN-based architecture with one generator and two discriminators for stain acclimation, which assimilated color distributions of the entire dataset instead of a single reference image. Its efficiency was validated on multiple datasets, including CRC Glas. Furthermore, Yengec-Tasdemir et al. (2023) used both Stain-Net (Kang et al., 2021) and Stain-GAN (Shaban et al., 2019) to mitigate domain shift in CRC histology where the Stain-GAN approach outperformed other existing methods in tissue classification tasks.

3.3. Patch selection methods

Advances in computational power and imaging hardware have transformed CPath workflows. Modern scanners produce WSIs comprising millions of pixels, resulting in image sizes of up to 6 GB (GB) (Romero Lauro et al., 2013). Despite the success of DL techniques like CNNs, training models on such high-resolution images is computationally impractical (Hou et al., 2016). With today's computational capacity, processing an entire WSI typically presents two major challenges:

1. Image downscaling is required, which may cause a loss of small-scale visual information.
2. Models could only learn a subset of the discriminative features in the image, leading to data inefficiencies.

To address these challenges, researchers adopted patch-based methods, training models on high-resolution patches rather than entire slides. This approach allowed the models to encode essential discriminative features effectively (Hou et al., 2016; Das et al., 2020a). However, selecting relevant patches from the WSI slide for training is crucial. Numerous studies have explored automated patch selection methods for CRC histopathology analysis.

Grid sampling with a sliding window approach is the most common technique for splitting WSIs into smaller patches for model development. This method splits the WSI into a grid-like structure, where a sliding window extracts all patches. Grid sampling provides uniformity during patch selection and improves computational efficiency, but it may include patches of non-informative tissue regions. This can add unnecessary processing overhead for automated tissue analysis models (Öztürk and Akdemir, 2018b). To optimize this, Haryanto et al. (2021) proposed a Conditional Sliding Window (CSW) algorithm to select valid patches from CRC WSIs. This approach improved the traditional sliding window by considering image dimensions such as width, height, and channels for appropriate patch selection. Fig. 9 shows the working procedure of grid sampling with a sliding window technique and the conditional sliding window technique.

Farthest Point Sampling (FPS) (Eldar et al., 1997) is another sampling method that offers an alternative selection of patches that maximizes spatial distance. This method identifies the center of each patch and repeatedly selects the next farthest patch to ensure uniform sampling across the WSI. Although FPS ensures wide spatial diversity, it fails to extract diagnostically relevant patches.

Attention-based sampling techniques use DL models to extract patches by focusing on diagnostic relevance rather than spatial diversity. This mechanism assigns higher attention weights to patches representing important histological features and directs the model toward diagnostically relevant areas. For instance, Broad et al. (2022) designed an attention-inspired algorithm based on a quasi-random sampling method for CRC WSIs. This method initially used sparse randomized grid patterns and then iteratively applied dense sampling to strategically select image patches from informative regions.

Active Learning (AL) based patch selection is another popular approach in histopathology used for both manual annotation reduction and model training optimization. This strategy identifies important patches with low model confidence through an iterative process. For instance, Shen and Ke (2021) proposed an active-learning-based automated patch-localization model that selected the most representative patches within the ROI from CRC histology images. Experiments

showed that this approach reduced the training set size by 38% and achieved 92.70% accuracy in CRC tumor classification.

3.4. Data augmentation methods

Limited training data and class imbalance remained major bottlenecks in CPath. Both ML and DL techniques depended on large datasets. However, DL models require substantial data to avoid overfitting and achieve effective performance (Shorten and Khoshgoftaar, 2019). Unfortunately, the field of CRC histopathology lacks access to this big data, resulting in the problem of model overfitting. To address this, data augmentation played a crucial role by creating new training samples through transformations of existing data. This artificial expansion reduced overfitting and improved model generalization (Berman et al., 2021). Data augmentation offered two primary approaches to achieve this: classical transformations and generative methods.

Classical techniques utilized basic image transformations to expand the training dataset and improve model performance. Common classical techniques include rotations, reflection, flips, cropping, scaling, and hue adjustments. Some of these techniques are widely applied to histopathology images. For instance, random reflection assisted models to learn invariant features regardless of tissue orientation (Berman et al., 2021). Similarly, random rotation allowed models to capture tissue structure variations from diverse angles. Nevertheless, these methods transform existing samples instead of the creation of entirely new data.

To achieve greater diversity, recent studies extensively adapted GAN-based models to generate synthetic images. Unlike classical transformations, a GAN-based augmentation model can create entirely new, realistic histology patches by learning the underlying distribution of the original dataset. Various studies have demonstrated that GANs like CycleGAN produce high-fidelity images that mitigate data scarcity and class imbalance in CRC datasets. For instance, Wei et al. (2019) addressed class imbalance in CRC polyp datasets using a CycleGAN-based image translation method. The model generated synthetic images for rare polyp classes while it preserved the original tissue structure. It improved AUC by 10% for the detection of sessile serrated adenomas.

Paired image translation, using models such as Conditional GANs (cGANs), is a widely adopted method that uses labeled image pairs to generate specific tissue categories. For instance, Deshpande et al. (2022) developed a SAFRON model based on cGAN to generate giant colorectal tissue slides with ground truth annotation. The framework used tissue slides from the DigestPath and CRAG datasets for training, where SAFRON not only achieved a high Dice index for gland segmentation but also preserved tissue morphological features. Li et al. (2022a) designed a multi-scale cGAN framework to produce high-resolution CRC histopathology images. Using a pyramid of Markovian discriminators, the framework captured high-frequency features and achieved an FID of 0.0059 on the GlaS dataset. Similarly, Krause et al. (2021) developed a histology CGAN model to generate realistic synthetic MSI and non-MSI colon images and observed that models trained on synthetic data matched the performance of real data with an AUROC of 0.743. Finally, Sujatha et al. (2024) compared Variational Autoencoder GAN (VAE-GAN) and Deep Convolutional GAN (DC-GAN) variants for augmenting the EBHI-seg colorectal cancer dataset. The validation results showed that the VAE-GAN model exhibited minimal discriminator loss. Table 3 summarizes various image pre-processing methods used in CRC analysis using histopathology images along with their respective outcomes in brief.

4. Image analysis

Image analysis involves interpreting histopathology images to classify cancer subtypes, detect specific cellular structures, segment tumor regions, and predict molecular biomarkers using ML and DL methods. This section outlines the involvement of ML and DL architectures across relevant tasks.

4.1. Image classification

Image classification is a well-known computer vision task in which the aim is to categorize the input image into the appropriate class. In the context of CRC histopathology classification, studies have adopted two popular approaches. The first uses traditional ML techniques, where classification is performed using HCFs or DFs extracted from pre-trained models, followed by ML classifiers. The second approach uses DL frameworks to classify CRC tissue, enabling automatic extraction of discriminative features. Building on these approaches, this section summarizes the methods and contributions of various studies that use ML and DL techniques for classifying CRC using tissue images.

4.1.1. Conventional classification methods

Conventional ML approaches form the historical foundation of automated CPath in a two-stage process: feature engineering and classification. However, these approaches are heavily dependent on HCF engineering when modeling discriminative classifiers. Nevertheless, they are adopted for CRC patch-level classification tasks. An extensive corpus of literature focuses on reformulating ML classifiers, primarily SVMs, using optimal HCF extraction and selection. For instance, Rathore et al. (2015) created an early hybrid feature set integrating geometric, morphological, SIFT, and GLCM features and achieved high accuracy with an SVM classifier in colon tissue classification. Similarly, Naiyar et al. (2015) used an SVM ensemble after extracting HCFs like EFD and LSLD from colon tissue images and achieved 96.65% accuracy. In addition, Kather et al. (2016) introduced an 8-class benchmark dataset of 5000 CRC tissue images and, using multiple texture descriptors, found that the RBF-SVM was the most accurate conventional classifier for multi-class prediction. Awan et al. (2017) developed the Best Alignment Metric BAM that quantifies tumor grade and glandular shape and used an RBF-SVM for classification based on glandular morphology, achieving 97% accuracy. Sahran et al. (2018) introduced a novel feature selection technique, SVM-RFE(AC), to select non-redundant features from colon and prostate tissue images and found that this feature selection technique is suitable for both single and ensemble SVM classifiers.

Some studies focused on integrating robust classifiers and optimizing ML pipelines. Kallipolitis and Maglogiannis (2019) used Bag-of-Visual-Words (BoVW) to create visual tokens after extracting SURF features on the Warwick-QU dataset and trained an RF classifier to achieve higher accuracy. Dabass et al. (2020) proposed a Colormap Enhanced Image Sharpening technique combined with an SVM classifier and found that it improved image quality and the identification of critical components. Grosu et al. (2021) trained a bootstrap resampling-based RF classifier to distinguish between benign and premalignant polyps by integrating CT colonography data and achieved an AUC of 91%. Trivizakis et al. (2021) trained multiple RBF SVMs and MLPs using a wavelet-based multi-scale feature extraction approach for classifying CRC tissue images, achieving up to 84% classification accuracy even with ANOVA feature selection. Alqudah and Alqudah (2022) extracted GLCM-based texture features across three color space channels and found that QDA achieved the highest accuracy among multiple ML models trained for a multi-class classification task. Among ensemble methods, Gradient Boosting frameworks showed superior performance. Hage Chehade et al. (2022) trained and compared six ML-based classifiers on lung and colon images and found that XGBoost achieved the highest accuracy. Similarly, Tripathi et al. (2023) achieved higher accuracy with CatBoost after training multiple ML models on nine-class CRC tissue. In a recent study, Chhillar and Singh (2024) showed that LightGBM trained on a combined feature set of color histograms and Haralick features was not just interpretable but also achieved 99.92% accuracy.

HCFs provide good interpretability but are time-consuming, sensitive to staining variations, and require domain expertise. As a result, approaches such as combining DF with an ML classifier have also been

Table 3
Summary of the pre-processing strategies used in colorectal cancer analysis.

Task	Reference	Dataset	Methods	Outcome
Artifact detection	Wright et al. (2020)	CRC QUASAR trial	Pathologists manually verified image quality issues like tissue folds, tears, and bubbles, then trained a RF model	3% improvement on accuracy of classification and calculation of TSR in CRC trial (after excluding images that contained quality issues)
	Hossain et al. (2023)	26 WSIs of multiple organs, including the colon	Tissue fold and air bubble segmentation using DoubleUNet	Improvement of 9% in accuracy for tissue artifacts segmentation over state-of-the-art methods
	Foucart et al. (2018)	–	Artifact detection and segmentation using modified ResNet with fuzzy-target scheme using a Gaussian filter	Optimally finds the largest artifact-free tissue area
	Shaikh et al. (2022)	WSIs of colon tissue and several organs	Identification of various tissue artifacts such as tissue blur, fold, and pen mark using few-shot learning approach	Trained using small samples and outperforms the standard transfer learning approach
	Shakhawat et al. (2020)	26 WSIs of multiple organs, including the colon	Tissue artifacts such as blur, fold and bubble detection using handcrafted features and SVM	Accuracy in artifact detection is 0.99
	Ke et al. (2023)	Colon cancer WSIs for evaluating AR framework and NCT-CRCHE-100K for downstream analysis	Tissue artifacts such as out-of-focus areas, folds, ink marks, blur, and others are detected and restored using the AR-Classifier and AR-CycleGAN	Structural Similarity (SSIM) in presence of de-artifact and re-artifact processing is 0.9597
Stain normalization	Reinhard et al. (2001)	Natural images	Color transfer using mean and variance in the perceptual color space ($l\alpha\beta$ space)	–
	Roy et al. (2021)	CRC histology	Modified the Reinhard algorithm by incorporating more hypotheses	Pearson correlation coefficient 0.99
	Macenko et al. (2009)	12 slides with different staining variations	Singular value decomposition (SVD)-based geodesic method	–
	Vahadane et al. (2016)	Colon, bladder, stomach, and prostate tissues from TCGA	Stain separation using sparse regularized Non-negative Matrix Factorization (SNMF)	–
	Bejnordi et al. (2015)	155 WSIs	HSD color transform, 2D chromatic distribution, and inverse HSD transform	–
	de Bel et al. (2021)	Colon and kidney tissue	Stain normalization using CycleGAN with residual learning technique	Improvement of Dice coefficients by 4% over regular stain normalization methods in colon tissue segmentation task
	Smit et al. (2023)	75 colon WSIs	Stain normalization using CycleGAN	Spearman rank correlation of 0.88 for TSR score
	Kausar et al. (2021)	GlaS	Stain acclimation generative adversarial network (SA-GAN)	SA-GAN outperformed the existing SOTA methods in terms of transferring stain colors between datasets
	Yengec-Tasdemir et al. (2023)	470 CRC histology slides	Address domain shift problem using stain-GAN and stain-Net	Stain-GAN method outperformed other methods
Patch Selection	Haryanto et al. (2021)	GlaS	Conditional sliding window (CSW) based sampling	Model trained using samples generated by the CSW algorithm outperformed the model trained on the original data and achieved a sensitivity of more than 80% in classification task
	Broad et al. (2022)	CRC QUASAR trial	Attention-inspired algorithm based on a quasi-random sampling method	This approach was 3 to 6.3 times faster than tile-by-tile WSI analysis
	Shen and Ke (2021)	CRC WSIs from TCGA	Active learning-based automated patch-localization	Reduced the training data size by 38%

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Table 3 (*continued*).

Task	Reference	Dataset	Methods	Outcome
Data Augmentation	Deshpande et al. (2022)	CRAG	SAFRON framework based on conditional GAN	CRC gland segmentation task performance was improved when trained with synthetic data generated by SAFRON and achieved a Dice index score of 0.97
	Wei et al. (2019)	427 colorectal polyps WSIs	CycleGAN	AUROC is improved by 10% for detecting sessile serrated adenomas in CRC
	Li et al. (2022a)	GlaS	Multi-scale conditional GAN	F1 Score is 0.886 for SSL-segmentation on GlaS dataset
	Sujatha et al. (2024)	EBHI-seg colorectal histology	VAE-GAN and DC-GAN	DC-GAN surpassed VAE-GAN in terms of generating realistic synthetic colorectal histology images
	Krause et al. (2021)	CRC WSIs from TCGA	Conditional GAN	Improved AUROC in MSI detection when trained using hybrid data (real+ synthetic) as well as it exhibited that the generated images retain genetic alteration information

– Indicates that the information is unavailable.

explored for CRC tissue classification. Fan et al. (2021) showed this using Alexnet DFs with an SVM and achieved 99.44% accuracy on colon and lung tissue images. DFs are complex and not interpretable; however, DFs combined with an ML approach are still in progress in CRC histopathology classification. A recent study by Junaid et al. (2025) made this approach interpretable by applying the chi-squared test to features extracted with MobileNetV2 and DenseNet121 and achieved 97.96% accuracy with SVM. Sharkas and Attallah (2024) proposed Color-CADx with three CNNs for DFs, applied DCT and ANOVA for fused-feature selection, achieving the best results with SVM. Ohata et al. (2021) used 18 pre-trained CNN architectures as feature extractors and integrated them with multiple ML classifiers for CRC tissue classification. After evaluating various model combinations, DenseNet169 with SVM achieved the highest accuracy of 92.08%. Babu et al. (2021) compared three pre-trained CNNs with Bayesian-optimized SVM to identify the best-performing colon tissue classification model and found that Inception-V3 with SVM achieved the highest accuracy of 99%. Tripathi and Singh (2020) fine-tuned multiple pre-trained CNNs and implemented a cascaded ensemble approach using HCFs, DFs, and their fusion. They found that ensemble DFs from multiple pre-trained models improved MLP performance but combining HCFs with DFs showed no gain in colon nuclei classification. Table 4 summarizes the ML methods, results, and key strengths used in CRC tissue classification.

4.1.2. Deep learning based classification methods

CNNs are widely used DL techniques that allow for automated feature extraction and serve as the foundation for image classification, segmentation, and detection tasks. Over the last decade, CNNs have also become popular in histopathology image analysis tasks. Over time, several enhancements have been proposed by researchers to produce more accurate CNN models for the classification of CRC using histopathology images. This section presents recent studies related to colorectal cancer classification using histopathology images.

The initial success of DL in CRC histopathology image classification relied on CNN architectures (ResNet, VGG, DenseNet), which often used TL to adapt pre-trained models from natural image datasets to the histology domain. Korbar et al. (2017) trained a ResNet-152-based model for the classification of five colon polyp types in histology slides. This method achieved the best performance among tested architectures with an overall accuracy of 93.0%. Anju and Vimala (2023) finetuned the VGG-16 model for classifying CRC tissue into nine classes, and this finetuned VGG-16 clearly outperformed GoogLeNet, DenseNet, and AlexNet. Kather et al. (2019a) finetuned multiple popular CNN models (VGG19, SqueezeNet, ResNet50, and GoogLeNet) to classify CRC tissue into nine distinct classes. The VGG19 network achieved the best accuracy in the independent set. Raju and Rao (2022) finetuned ResNet50 and ResNet152 models to the last ten layers for end-to-end classification of CRC histopathology images. The Adaptive ResNet152 achieved the highest accuracy of 98.38% and outperformed ResNet50. Pre-trained CNNs often serve as feature extractors in TL workflows. Irawati et al. (2022) compared seven pre-trained CNN architectures using TL to classify colon tissue images into two classes and found ResNet101, ResNet152, and VGG19 to be the best performers. Alhanaf et al. (2022) used ten pre-trained CNN models and evaluated them using two approaches: normal classification and k-fold cross-validation. Darknet53 achieved 96.6% accuracy in 5-fold cross-validation, while ResNet50 achieved 98.7% accuracy in normal classification.

Some studies focus on feature optimization and ensemble methods for classifying CRC histopathology images to improve the robustness of the learning model. For instance, Hamida et al. (2021) evaluated multiple DL methods using pre-trained, from-scratch, and fine-tuned approaches. They found that a fine-tuned ResNet could achieve the best accuracy on benchmark CRC datasets. In the context of feature fusion, Alotaibi et al. (2025) implemented the Osprey Optimization Algorithm (OOA) to optimize features from multiple pre-trained models and achieved 99.39% accuracy using the Deep Belief Network (DBN). For ensemble methods, Albashish (2022) used majority voting and product rules, while Ghosh et al. (2021) employed a non-linear weighted fusion strategy to combine predictions from diverse CNN architectures. Furthermore, Ke et al. (2025b) found that a stacking-based ensemble strategy outperforms single models and compared the model's focus on different tissue regions using Grad-CAM heatmaps.

TL, although it provides an essential foundation for deeper networks with limited data, the domain shift between natural images (ImageNet) and H&E-stained histopathology often results in the loss of fine-grained morphological details. Studies are designing their own custom, shallow CNNs for CRC classification. Kumar et al. (2023) designed a shallow CNN model, CRCNN-Net, with only 3.76 million parameters for classifying multiple CRC tissues. It outperformed four popular pre-trained architectures (DenseNet121, InceptionResNetV2, Xception, and VGG16) in comparative analysis. Basha et al. (2018) designed RCCNet, a simpler architecture with just 7 layers and 1.51M parameters for colon-nuclei classification and achieved 80.61% accuracy. Yildirim and Cinar (2022) chose a deeper architecture and developed MA_ColonNET, a 45-layer CNN designed for binary classification of colon tissue images. It achieved 99.75% accuracy on 10K colon tissue images. Yadav et al. (2022) examined the impact of different activation functions (ELU, ReLU, PReLU, LReLU) on CRC classification using a shallow

Table 4
An overview and findings of studies employing machine learning techniques for colorectal cancer classification.

Author	Year	Datasets	Feature extraction/Feature selection	Classifier	Results	Key contributions	Limitations
Junaid et al. (2025)	2025	EBHI-Seg	DFs (MobileNetV2, DenseNet121)/ chi-squared test	SVM, KNN, DT	ACC 97.96%	Enhanced interpretability using chi-squared selection on deep features	Lacks external validation
Chhillar and Singh (2024)	2024	LC25000	Texture and Color/	LightGBM	ACC 99.92%	Interpretable feature engineering based approach	Evaluated on only one dataset, reducing generalizability
Sharkas and Attallah (2024)	2024	NCT-CRC-HE-100K, Kather-5000 images	DFs (AlexNet, ResNet50, DenseNet201)/ ANOVA	Cubic SVM, Ensemble Discriminant	ACC 99.3%	Multi-CNN feature fusion with DCT-based spectral encoding and ANOVA-selected features for efficient CRC classification	The study does not examine the individual contribution of each CNN
Tripathi et al. (2023)	2023	NCT-CRC-HE-100K	Fractal-based texture features using Differential Box Counting/	KNN, DT, RF, CatBoost, NB	ACC (CatBoost) 90.67%	SMOTE technique is used for generating new samples to balance the classes within the dataset	Some AUCs are very poor because of class imbalance
Hage Chehade et al. (2022)	2022	LC25000	Texture and Geometric / RFE	LightGBM, XGBoost, SVM, RF, LDA, MLP	ACC (XGBoost) 99%	Image enhancement with Unsharp Masking; SHAP for model explainability	SHAP alone may not fully capture ML models' decision-making process
Alqudah and Alqudah (2022)	2022	Kather-5000 images	3D Haralick /	QDA, SVM, KNN (K=5), CDT, and ANN	ACC 97.32%	Required less time in feature extraction and classification	Lack of validation
Fan et al. (2021)	2021	LC25000	DFs using AlexNet/	SVM	ACC 99.44%	Verified using multi-class breast datasets	Lower validation accuracy
Babu et al. (2021)	2021	ImediaTreat, GlaS	DFs using Inception-V3, AlexNet, and VGG-16/	Bayesian-optimized SVM	ACC 99% (Inception-V3 + SVM)	Compared deep features from three CNNs and identified Inception-V3 with Bayesian SVM as the most effective	Lacks cross-dataset generalization
Trivizakis et al. (2021)	2021	Kather-5000 images	Texture/ ANOVA	RBF-SVM	ACC 84%	Optimization of wavelet parameters	High computational complexity
Ohata et al. (2021)	2021	Kather-5000 images	DFs from 18 pre-trained CNNs/	NB, KNN, RF, MLP, and SVM	ACC 92.08% (DenseNet169 + RBF-SVM)	The study performed an extensive comparison of CNN-based feature extractors and ML classifiers to select the optimal combinations	The work is limited to a single-dataset comparison
Grosu et al. (2021)	2021	107 colorectal polyps	Shape, Histogram and GLCM/	RF	AUC 91%	Combines histopathology with CT colonoscopy	Did not validate their method against other classifiers

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CNN and found that ELU activations performed best. Similarly, Tiwari (2018) and Ciompi et al. (2017) examined the impact of different color models (RGB, YUV, LAB, and HSV) and color normalization methods (WSICS Bejnordi et al., 2015 and Macenko Macenko et al., 2009) in CRC tissue classification using shallow CNNs and found that the HSV color model achieved the highest accuracy, while stain standardization improved CRC classification accuracy by up to 20%.

Improvements in CNNs can also be an approach for a variety of studies to enhance their robustness in CRC classification. Wang et al. (2017) designed Bilinear-CNN (BCNN) for H&E-stained CRC

Table 4 (*continued*).

Author	Year	Datasets	Feature extraction/Feature selection	Classifier	Results	Key contributions	Limitations
Tripathi and Singh (2020)	2020	CRC-HistoPhenotypes	DFs (AlexNet, VGGNet) and HCFs (Texture, Color)/	MLP	AUC 99.81% (Deep Ensemble)	Used Adaptive Synthetic Sampling to address class imbalance during training across both HCF and DF branches	Unaddressed artifacts and background noise weakened the HCFs
Kallipolitis and Maglogiannis (2019)	2019	Warwick-QU	SURF and Haralick/	MLP, LR, NB, RF	ACC 100% (RF)	Applied Inverse Document Frequency technique to incorporate the importance of each visual token	Did not investigate the relationship between visual tokens and the extraction of meaningful information
Dabass et al. (2020)	2019	Warwick-QU	SURF/	SVM	ACC 94%	Combines image enhancement techniques with pathologists' feedback	Not evaluated on other datasets
Sahran et al. (2018)	2018	Standard colorectal H&E dataset	Color Transformation and Haralick/ SVM-RFE-AC	SVM (RBF)	ACC 95.8%	Addresses overfitting with ensemble learning	No normalization for varying intensity
Awan et al. (2017)	2017	38 H&E-stained slides of colorectal adenocarcinoma	Shape features using BAM/	SVM	ACC 97%	Novel approach for distinguishing tumor regions	Some misclassifications due to overlap between low- and high-grade tumors
Kather et al. (2016)	2016	Kather-5000 images	Texture features using higher- and lower-order histogram, GLCM, LBP, Gabor filters/	RBF-SVM	ACC 87.4%	Released a benchmark CRC multi-class dataset which comprises 5000 microscopic images	N/F
Rathore et al. (2015)	2015	174 colon histology slides	Morphological and geometric features using K-Means algorithm, Texture features using GLCM and SIFT /	SVM	ACC 99.18%	Applied different pre-processing techniques based on features and requirements	The dataset used is quite small
Naiyar et al. (2015)	2015	174 colon histology slides	Elliptic Fourier descriptors (EFD), Morphological features (MF), and Least Square Lumen Distance (LSLD)/	Ensemble SVM	ACC 96.65%	Features are computationally inexpensive in terms of extraction time	Dataset is relatively small

ACC: Accuracy; AUC: Area Under Curve; N/F: Not Found.

tissue classification, where the H and E stain channels were processed separately and fused using a bilinear vector. It achieved 92.6% accuracy and outperformed several conventional CNNs in multi-class CRC tissue classification. Baidar Bakht et al. (2023) introduced DeepBLS, which integrates DF extractors and the Broad Learning System (BLS) for tissue classification. BLS is a flat network that uses incremental learning to perform feature mapping and classification. DeepBLS outperformed existing methods and achieved F1 score improvements of 2% and 6% on benchmark CRC datasets. Khazaee Fadafen and Rezaee (2023) proposed dResNet, a modified ResNet-101 with dilated convolutions and an attention module, and integrated it with DeepSVM for CRC tissue classification. This combination achieved 99.76% accuracy on the NCT-CRC-HE-100K dataset. Izzaty et al. (2022) combined attention squeeze-and-excitation (SE) layers with EfficientNet-B0 and trained it using transfer learning for multi-class CRC classification. It achieved 97% accuracy and outperformed baseline models. Dabass et al. (2022) integrated attention-based learning and multi-level convolutional modules into a CNN framework to capture multi-scale spatial features for classifying cancer grades and tissue structures in color-normalized colon cancer histology slides. It also used three XAI techniques Grad-CAM, Occlusion Sensitivity mapping, and LIME to improve model explainability. Ke et al. (2025a) used a Multi-Head Self-Attention (MHSA) technique for multimodal feature fusion that adaptively integrates both discriminative as well as complementary features from multiple domain-specific feature extractors to improve CRC classification. Liang et al. (2020) applied the shearlet transform to extract multi-scale features and developed the MFF-CNN framework for CRC classification. MFF-CNN processed a three-channel input: the original histopathology image, the magnitude coefficients, and the phase coefficients extracted using shearlet decomposition. By combining these,

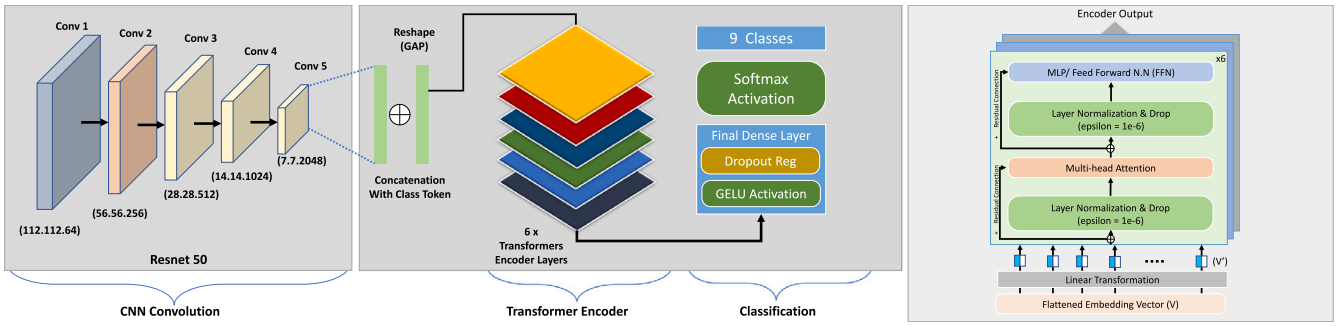


Fig. 10. Architecture of hybrid CNN–Transformer (TransNetV) for CRC histopathology classification (Tanveer et al., 2024).

it achieved a classification accuracy of 96%. Kumar et al. (2024) enhanced CRC tissue images using Tikhonov-based unsharp masking and developed a lightweight three-headed network called Multiheaded CNN (MHCNN) for tissue classification. It achieved a multi-class classification accuracy of 96.62%. Haj-Hassan et al. (2017) used multispectral histology images and applied the active contours technique to segment regions of interest (ROIs) for training a CNN-based classification model. It achieved 99.17% accuracy in a three-class CRC classification task. Kim et al. (2023) proposed a slide-level classification system where ResNet-18 was fine-tuned for patch-level classification of CRC lesions. Patch-level predictions from WSIs were then aggregated using a decision tree for final classification. Sirinukunwattana et al. (2016) proposed a Neighboring Ensemble Predictor (NEP) within a unified CNN framework for nuclei classification in CRC tissue and achieved an AUC of 0.917.

Different learning strategies have been used in CRC histopathology image classification, ranging from human-like methods such as curriculum learning and active learning to more advanced approaches. For instance, Wei et al. (2021) developed a curriculum-learning-based approach using ResNet-18 and trained it on images in multiple steps. Image difficulty was determined according to the annotator protocol (easy images first, then hard images). The proposed approach attained an AUC of 88.2% in CRC polyp classification. Similarly, Raczkowska et al. (2019) proposed ARA-CNN, an active learning-based Bayesian CNN that not only learns discriminative features during training but also measures uncertainty per patch to detect mislabeled samples for re-annotation. It achieved 92.44% accuracy in an eight-class CRC classification task. Peng et al. (2019) developed a multitask DL framework for simultaneous classification and image retrieval of CRC tissue images. It used Triplet loss, Cauchy hashing loss, and Cross-Entropy as multi-task objectives, and attained a best classification accuracy of 95% with multitask ResNet-18.

Though CNNs have achieved SOTA results across various histopathology image classification tasks, CNNs struggle to model long-range dependencies. Architectures such as ConvNeXt and transformer are used in the CRC literature to capture global contextual information and improve classification accuracy. For instance, Yengec-Tasdemir et al. (2023) used two ConvNeXt architectures (ConvNeXt-Tiny and ConvNeXt-Base) to capture both local and long-range dependencies and designed an ensemble framework to classify histopathology images into two classes. They achieved 91.1% on the external CRC dataset and outperformed all other single CNNs. Tanveer et al. (2024) combined a CNN with a Transformer encoder, TransNetV, to improve global context reasoning on image patches for CRC histopathology classification. This hybrid framework, as illustrated in Fig. 10, showed better performance on benchmark CRC datasets and achieved up to 98.9% accuracy. Fu et al. (2025) combined patch embeddings from ResNet-18 with local features obtained by Multi-Scale Grouping (MSG) to improve the TransMIL feature encoding for CRC histopathology classification. This hybrid approach not only achieved 93.7% accuracy on the TCGA colon dataset but also outperformed existing MIL and transformer-based methods. Zeid et al. (2021) compared two transformer variants, Vision Transformer (ViT) and the Compact Convolutional Transformer (CCT), on an eight-class CRC dataset. The main finding of the study was that ViT captured global context better than CNNs but CCT performed better on gland structures and outperformed ViT overall. Transformers capture a better global context in histopathology images due to their self-attention modules, but it is computationally expensive. Choudhary et al. (2025) reduced its complexity by adding a BiLSTM to an MLP head of the ViT that used multi-scale features from the Capsule-Assisted Res2Net to capture global context within tissue patches. It achieved up to 4% higher accuracy and lower overfitting than baselines in CRC tissue classification. To enhance interpretability, Mir et al. (2025) elucidated ViT decision-making in CRC histopathology by applying attention-based visualization, LIME, and TransLRP. After extensive experiments on two CRC datasets, they found that TransLRP provided more accurate and clinically trustworthy explanations than the other methods. Yengec-Tasdemir et al. (2024) used supervised contrastive learning (SCL) with the Big Transfer (BiT) model to develop a CRC multi-class classification system. It achieved 87.1% accuracy and outperformed traditional deep CNN models.

Data-efficient approaches such as WSL, FSL, and SSL are adopted in CRC histopathology image classification to develop robust models that require inexact, limited, or no supervision, respectively. WSL allows training models using only weak supervision derived from coarse, slide-level diagnostic labels. For instance, Bilal et al. (2022) developed the WSL framework using iterative draw and rank sampling (IDaRS) to classify normal and neoplastic colon biopsies. They fine-tuned ResNet-34 using only slide-level labels and achieved AUROCs of 0.9895 and 0.9746 on the internal and TCGA cohorts. Tan et al. (2023) employed a WSL method to build a transformer-based MIL framework for CRC lymph node metastasis classification from histology slides. The proposed model required only slide-level binary labels and obtained an accuracy of 95.3%. Saeed et al. (2025) used a lightweight MobileNet model to develop a WSL-based MIL framework for CRC classification and improved the aggregation step by considering multiple top-ranked tiles. Experiments on CRC WSIs showed that this framework improves slide-level diagnostic accuracy and was less sensitive to noisy tile predictions. However, interpretability is a key limitation that affects transparency and clinical reliability. To address this, Ilse et al. (2018) combined WSL with an attention mechanism and introduced an MIL-based classification framework. The novelty of this framework was the incorporation of an attention-based fusion strategy that not only optimally selected informative patches based on attention weights for slide-level prediction but also provided insights into the contribution of each instance. Similarly, Iizuka et al. (2020) introduced a hybrid CNN-RNN architecture for CRC slide-level classification. The Inception-V3 first extracted patch embeddings and the RNN-based aggregator then captured the tissue's global structural context to achieve slide-level classification. Along with high diagnostic performance, the approach also provided interpretability through CNN-generated probability heatmaps. Few-shot learning techniques are a rapidly evolving set of methods for CRC tissue classification, as they enable the development of a classification framework that can learn with limited data. To

explore this, Sekhar et al. (2024) investigated five SOTA FSL methods (ProtoNet, MAML, DeepEMD, SimpleShot, and LaplacianShot) for CRC histopathology image classification and found that the non-episodic methods SimpleShot and LaplacianShot achieved better results across three CRC datasets. Hasan et al. (2025) enhanced the Prototypical Network by integrating powerful pathological foundation models with small adapters for task specific adaptability. Bias-Diminishing Cosine Similarity Prototypical Network (BDSPN) achieves the best performance across benchmark datasets. Li et al. (2024c) used contrastive and transfer learning to develop a classification framework that can work with limited samples in CRC histopathology. It used VGG-16 as the backbone network and fine-tuned it using a combination of contrastive loss and cross-entropy loss. This approach achieved 94.99% accuracy on the test set with just 15 labeled samples per class. Merabet et al. (2025) improved the explainability of the Prototypical Network using XAI techniques and developed a few-shot CRC classification model based on a ConvNeXt-Tiny encoder. This model achieved up to 90% cross-domain accuracy using just 10 shots.

Self-supervised learning addresses the limited annotated dataset problem in CRC histopathology by enabling the model to learn directly from massive unlabeled histology slides. As a result, pathological foundational models have become highly popular in CPath, where DL models are pretrained on massive unlabeled histopathology datasets and then generalized to CRC histopathology tasks with limited samples. Adopting this approach, Filiot et al. (2023) pre-trained a ViT architecture on large pan-cancer histology images using advanced SSL techniques in the iBOT (Masked Image Modeling) framework. This framework achieved SOTA performance on both WSI-level and patch-level CRC classification and outperformed the ImageNet pre-training-based architecture. Huang et al. (2023) developed the PLIP model by fine-tuning CLIP on the large OpenPath medical image–text dataset, which demonstrated SOTA performance in zero-shot classification on the Kather colon dataset for CRC tissue types and outperformed the baseline general models such as CLIP. Alfasly et al. (2024) compared two domain-specific pathology models, KimiaNet and DINO-SSL-Path, with the foundational models BiomedCLIP and PLIP using both the in-house CRC dataset and the DigestPath dataset, and found that the domain-specific models consistently outperformed the foundation models. Jin et al. (2022) utilized domain-specific knowledge, such as cell-level and stain-level information, and developed an SSL-based knowledge distillation framework, HistoSSL, using ViT and ResNet architectures. The framework showed superior performance on the NCT-CRC dataset and outperformed other existing methods.

Summary: The classification of CRC histopathology images predominantly involves both ML and DL paradigms, as evidenced by 70 literature reviewed in this section. After analyzing these studies, it is concluded that SVM (Awan et al., 2017), RF (Grosu et al., 2021), and LightGBM (Chhillar and Singh, 2024) are the most commonly used methods for the classification of CRC histopathology images. It is imperative to note that, among the 19 ML studies analyzed, approximately 60% use the SVM approach to classify colon cancer. This dominance is largely due to SVM's robust performance in high-dimensional spaces and its effective handling of non-linear data through kernel functions. Alternatively, DL methods, particularly CNNs, have effectively surpassed traditional approaches by automating feature extraction. We analyzed 51 studies across DL-based classification methods and ascertained that CNNs are predominant in the literature and achieve SOTA results across several studies (Yildirim and Cinar, 2022; Dabass et al., 2022; Ke et al., 2025a). However, CNNs fail to model long-range dependencies due to their limited receptive fields. Modeling global understanding of histopathology images is important to capture subtle and complex tumor characteristics. To address this limitation, recent studies have shifted toward architectures such as ConvNeXt and Transformers to capture global contextual information. For instance, Yengec-Tasdemir et al. (2023) used ConvNeXt to capture both local and long-range dependencies, clearly outperforming single CNN-based models. Similarly, hybrid frameworks such as TransNetV (Tanveer et al., 2024) and ResNet–Transformer (Fu et al., 2025) approaches have shown impressive results by combining patch embeddings with CNN-based local features. Vision Transformers and Compact Convolutional Transformers reduce reliance on CNNs and enable more effective modeling of long-range dependencies. Although effective, it is worth noting that training ViT-based models requires large datasets and substantial computational resources. Later integration of BiLSTM–ViT (Choudhary et al., 2025) has reduced this complexity while further improving classification accuracy.

In contrast to architectural development, this field is witnessing the emergence of data-efficient learning strategies. Paradigms such as FSL, WSL, and SSL handle data scarcity efficiently and achieve promising results across various studies (Saeed et al., 2025; Hasan et al., 2025; Li et al., 2024c; Filiot et al., 2023). FMs are popularly evolving the field, outperforming SOTA methods with zero-shot classification on the Kather colon dataset for CRC tissue (Huang et al., 2023). However, recent evaluations indicate that while FMs offer broad generalization, domain-specific models still outperform them on specialized CRC classification datasets (Alfasly et al., 2024). Table 5 summarizes the methods, results, and key strengths employed in DL-based CRC tissue classification.

4.2. Image detection

Disease Detection in CRC histopathology is more than just classifying patches into appropriate classes. Rather, it focuses on identifying the location of the tumor or abnormal region within the WSI. This spatial understanding is essential because it provides valuable insights for diagnosis and treatment planning by highlighting and quantifying the location and extent of tumor tissue. Disease detection in CRC histopathology includes tasks ranging from tumor localization to detecting abnormal micro-structures within the slide. Both ML and DL methods have been explored in disease detection. However, ML methods can effectively classify image tiles but lack spatial awareness and boundary precision. On the other hand, DL architectures are nowadays appreciated for providing precise spatial localization via region proposal and regression or classification. This section presents both ML-based and DL-based studies that have performed disease detection in CRC histopathology.

4.2.1. Conventional detection methods

The application of conventional methods for defining the task of disease detection in CRC histopathology was often unclear due to the interchangeable use of terms such as classification, localization and detection. The conventional methods for detecting CRC tumors often rely on patch-based classification, followed by the generation of a spatial map of cancerous tissue components across the WSI. For instance, Kather et al. (2016) first extracted HCFs using GLCM and LBP and performed patch-based classification with different ML classifiers, such as SVM, KNN, and DTs. They finally reconstructed a large-scale spatial map that highlights regions predicted as tumor. Unlike directly identifying abnormalities within WSI, this method provides only a categorical understanding of the tumor's location. Glandular structures are important histological components in colorectal tissue and the morphology of these glands is a critical cue for diagnosis and grading in CRC. To automate the identification of these vital structures, Sirinukunwattana et al. (2015a) proposed a superpixel based approach for detecting glandular structures in colon histology and generated a Glandular Probability Map (GPM). The method extracted texture and color features from each superpixel and applied an RF classifier to assign a glandular probability score. Table 6 summarizes the ML methods, results, and key strengths used in CRC tissue detection.

Table 5
An overview and findings of studies employing deep learning techniques for colorectal cancer classification.

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Merabet et al. (2025)	2025	LC25000, EBHI	Data Augmentation	Few-shot Prototypical Network with ConvNeXt-Tiny backbone	ACC 98.5% (LC25000)	Data-efficient and interpretable colon cancer histopathology analysis	Expert validation performed, but only on benchmark datasets
Fu et al. (2025)	2025	TCGA	Tissue Masking, Patch Extraction	Hybrid CNN–Transformer model with a Multi-Scale Grouping (MSG) module	ACC 93.7% (TCGA)	Transformer MIL model integrating FPS-sampled patch features with multi-scale regional features (MSG)	Computational complexity is not discussed in the paper
Alotaibi et al. (2025)	2025	Warwick-QU	Median Filter	Multi-model fusion with DBN classifier	ACC 99.39%	Feature fusion with MobileNet (lightweight), SqueezeNet (compact), and SE-ResNet (attention-based)	Lacks external validation
Ke et al. (2025a)	2025	EBHI, COAD, Chaoyang, NCT-CRC-HE-100K	Tissue Masking, Patch Extraction	Domain-specific transfer learning with multi-model fusion using Multi-Head Self-Attention (MHSA)	ACC 99.68% (EBHI)	Grad-CAM interpretability for validating MHSA attention	Computationally intensive due to multi-model domain-specific training
Ke et al. (2025b)	2025	EBHI, NCT-CRC-HE-100K	Stain Normalization, Data Augmentation	Transfer learning with multiple CNN backbones and ensemble learning	ACC 99.80% (NCT-CRC-HE-100K)	Tested multiple ensemble strategies to enhance CRC classification performance	Demand higher computational resources due to multiple CNN backbones and ensemble strategies
Choudhary et al. (2025)	2025	NCT-CRC-HE-100K, Kather-5000 images	Trimmed Pixel Density-based Median Filter (TPDMF)	Hybrid CNN–Transformer framework using CAR2N and an optimized Bidirectional Vision Transformer (OBVT)	ACC 97.11% (Kather-5000 images)	Integration of the Bi-LSTM technique into the Vision Transformer MLP block reduces model complexity	The model suffers from high inference time and a non-interpretable detection process
Saeed et al. (2025)	2025	1132 CRC WSIs	Tissue Masking, Patch Extraction	Weakly supervised Multiple Instance Learning (MIL) with rule based aggregation	ACC 93.05%	Improves traditional MIL by using multiple representative tiles instead of a single tile	Lack of external validation on independent datasets
Mir et al. (2025)	2025	LC25000, KBSMC	None	Explainable Vision Transformer (XViT) with ViT attention maps, LIME, and TransLRP	ACC 96.2% (LC25000)	Introduced a ViT explainability pipeline in CRC histopathology classification	High explanation complexity of TransLRP can reduce clinical applicability

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4.2.2. Deep learning based detection methods

DL approach has significantly advanced disease-detection methodology in CRC histopathology. In the early stages, DL models enhanced patch-based analysis by using CNNs to generate more accurate WSI heatmaps for tumor localization. For instance, Xu et al. (2020) used an InceptionV3 architecture to classify CRC patches and aggregated predictions to create spatial heatmaps via simple thresholding, highlighting tumor regions with 94.8% accuracy. However, this averaging-based thresholding approach proved sensitive to false positives. To address this limitation, Ho et al. (2022) introduced a detection framework using a Faster R-CNN model to directly identify and localize high-risk glandular structures in CRC histopathology. The model detected dysplastic and malignant regions with a sensitivity of 97.4%. However, to ensure high detection rates, the model accepted a lower specificity of 60.3%. Dharshini et al. (2025) addressed these trade-offs by combining

Table 5 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Alfasly et al. (2024)	2024	Mayo Clinic CRC WSIs, DigestPath	Tissue Masking, Patch Extraction	Comparative evaluation of FMs (CLIP, PLIP, BiomedCLIP, DinoV2) and domain-specific models (KimiaNet, DinoSSLPath) in histopathology classification and retrieval	ACC 81.15% (KimiaNet)	Demonstrates superiority of domain specific histopathology models over general vision–language foundation models for CRC analysis	Compares models only based on their pretrained feature embeddings
Filiot et al. (2023)	2024	TCGA, NCT-CRC-HE-7K	Tissue Masking, Patch Extraction	ViT-based iBOT self-supervised model	ACC 95.8% (NCT-CRC-HE-7K)	Large-scale MIM pretraining improves histology feature quality	High computational cost
Tanveer et al. (2024)	2024	NCT-CRC-HE-100K, Kather-5000 images	Classical Data Augmentation	Hybrid CNN–Transformer architecture using ResNet50 and Transformer Encoder (TransNetV)	ACC 98.96% (Kather-5000 images)	Strong generalization across multiple datasets	Struggles in regions with overlapping tissue patterns such as between MUS and STR
Sekhar et al. (2024)	2024	NCT-CRC-HE-100K, CRC-TP, LC25000	None	Few-shot classification using ProtoNet, MAML, SimpleShot, LaplacianShot, and DeepEMD	ACC 89.5% (5-way 10-shot)	Standard fine-tuning methods outperform meta-learning approaches in CRC histopathology	N/F
Hasan et al. (2025)	2024	NCT-CRC-HE-100K, LC25000, Kather-5000 images	None	Few-Shot Adaptation of Foundation Models using Lightweight Adapters	ACC 96.55% (NCT-CRC-HE-100K with 20 shots)	High classification accuracy in low-data regimes	Struggles with domain shift
Li et al. (2024c)	2024	EBHI	Classical Augmentation	Few-shot learning with transfer learning using pretrained VGG16 and combined contrastive learning	ACC 94.99% (2 way 15-shot)	High accuracy with 15-shot setting	Lack of external validation
Kumar et al. (2024)	2024	NCT-CRC-HE-100K	Tikhonov-based Unsharp Masking (TBUM)	Lightweight Multi-headed CNN (MHCNN)	ACC 96.48%	Claims the first application of TBUM in CRC histopathology classification and demonstrates improved feature representation	Despite using a multi-class benchmark dataset, the study is limited to binary classification
Yengec-Tasdemir et al. (2024)	2024	UniToPatho, Private CRC WSIs	Patch Extraction, Stain Normalization	Supervised Contrastive Learning (SuperCon) framework using Big Transfer (BiT-M)	ACC 70.3%	Improved discriminative representations using SuperCon for multiclass CRC polyp classification	High computational cost due to BiT and SuperCon framework

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Table 5 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Yengec-Tasdemir et al. (2023)	2023	UniToPatho, EBHI	Stain Normalization	Ensemble of ConvNeXts (Tiny and Base variants)	ACC 95% (EBHI)	Comprehensively evaluates multiple stain normalization techniques while focusing on model explainability through Grad-CAM	Ensemble relies on simple averaging
Huang et al. (2023)	2023	OpenPath paired image–text dataset	Downsampling	CLIP fine-tuned on OpenPath using contrastive image–text alignment (PLIP)	F1 Score 0.565 (NCT-CRC-HE-100K)	Domain-specific vision–language pretraining improves zero-shot generalization	Sensitive to caption noise, magnification, and prompt changes
Tan et al. (2023)	2023	CRC lymph node WSIs	Tissue Masking, Patch Extraction	Transformer-based MIL (DT-DSMIL) with deformable transformer and dual-stream MIL aggregation	ACC 95.3%	Model explainability using probability maps and attention map	Large memory requirement due to deformable transformer on sparse slides
Kumar et al. (2023)	2023	Kather-5000 images, NCT-CRC-HE-100K	Class Filtering	Lightweight CNN (CRCCN-Net) and compared with multiple baselines	ACC 96.26% (NCT-CRC-HE-100K)	Parameter-efficient CRCCN-Net architecture with just 3.76 M parameters	CRCCN-Net requires more epochs for stable convergence on smaller datasets
Khaz-aee Fadafen and Rezaee (2023)	2023	NCT-CRC-HE-100K, Kather-5000 images	HSV Color Space Conversion	Hybrid dilated ResNet with attention and DeepSVM for multi-class CRC classification	ACC 99.76% (NCT-CRC-HE-100K)	Reduced computational complexity through NCA-based feature selection	NCA-based feature selection showed limited impact on classification accuracy
Baidar Bakht et al. (2023)	2023	Kather-5000 images, NCT-CRC-HE-100K	None	Hybrid framework combining VGG-19 feature extraction with a Broad Learning System (DeepBLS)	F1 0.99 (NCT-CRC-HE-100K)	Achieved low-cost, incremental learning with BLS	Limited generalization across datasets due to distributional differences
Anju and Vimala (2023)	2023	NCT-CRC-HE-100K, VAL-HE-7K	Stain Normalization	Fine-tuned VGG16	ACC 96.8%	External validation	VGG16 slow training
Kim et al. (2023)	2023	889 CRC WSIs, TCGA-COAD	Tissue Masking, Patch Extraction	ResNet-18 for patch-level classification with rule-based hierarchical aggregation	ACC 95.5%	End-to-end WSI-level colorectal lesion classification using a CNN	Training data sourced from a single medical institution may limit generalizability
Bilal et al. (2022)	2022	TCGA, Internal clinical cohort	Tissue Masking, Patch Extraction	Weakly supervised WSI classification using iterative draw-and-rank tile sampling with ResNet-34 backbone	AUROC 0.989 (internal)	Introduced IDaRS to iteratively select informative tiles using slide-level labels only	Sensitive to class imbalance, and iterative tile selection may overlook sparse pathological patterns
Jin et al. (2022)	2023	NCT-CRC-HE-100K, NCT-CRC-VAL-HE-7K	Patch Extraction, Data Augmentation, Stain Decomposition	Self-supervised contrastive learning with knowledge distillation for CRC classification	ACC 96.18% (NCT-CRC-HE-100K)	Implement domain-specific self-supervised learning using multi-level histopathology features	CRC evaluation limited to benchmark datasets

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Table 5 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Dabass et al. (2022)	2022	NCT-HE-CRC-100K, Kather-5000 images, HosC WSIs	Patch Extraction, Data Augmentation, Stain Normalization	Custom CNN with Enhanced Convolutional Learning Modules (ECLM) and Attention Learning Module (ALM)	ACC 97.7% (NCT-HE-CRC-100K)	Employed Grad-CAM, Occlusion Sensitivity Mapping, and LIME to enhance model explainability	High computational cost and longer training time due to attention modules
Albashish (2022)	2022	Kather-5000 images, Stoean CRC patch	None	Transfer learning using multiple CNN backbones with ensemble learning	ACC 91.28% (Kather-5000)	Ensemble-based transfer learning improves robustness and accuracy under limited data	Sensitive to errors due to simple decision-level fusion
Yildirim and Cinar (2022)	2022	LC25000	None	Custom 45-layer CNN (MA_ColonNET)	ACC 99.75%	Proposed a lightweight CNN with high patch-level classification accuracy	Limited to patch-level binary classification on a single dataset
Raju and Rao (2022)	2022	Kather-5000 images	Data Augmentation	Adaptive ResNet152 based on a modified ResNet152 and compared against VGG16 and a modified ResNet50	ACC 98.38%	Showed that an adaptive ResNet152 architecture with balance data augmentation improves multiclass CRC histopathology classification accuracy	Focuses on accuracy but lacks generalizability due to single-dataset evaluation
Izzaty et al. (2022)	2022	Kather-5000 images	Data Augmentation	Transfer-learning-based EfficientNet-B0 with integrated Squeeze-and-Excitation (SE) attention modules	ACC 97%	Incorporated SE attention layers for feature recalibration in multiclass CRC histology classification	Lack of generalizability due to evaluation on a single public dataset
Yadav et al. (2022)	2022	Kather-5000 images	Data Augmentation	CNN-based multi-class CRC tissue classification with a systematic evaluation of activation functions	Precision 96% (ELU)	Systematically analyzed the impact of different activation functions in a CNN for multi-class CRC tissue classification	Lacks assessment of activation function behavior in modern DNN
Irawati et al. (2022)	2022	LC25000	Data Augmentations	Comparative study of multiple CNN architectures using Transfer Learning	ACC 1.00 (VGG19)	Benchmarked multiple SOTA CNNs for binary colon cancer classification	Lack of external validation
Alhanaf et al. (2022)	2022	139 CRC WSIs	None	Transfer learning using multiple pretrained CNNs	ACC 98.7%	Comparison of multiple pretrained CNNs using normal and k-fold settings	Fine-tuning details not clearly specified
Zeid et al. (2021)	2021	Kather-5000 images	Image Normalization (standardization)	Vision Transformer (ViT) and Compact Convolutional Transformer (CCT)	ACC 94.75% (CCT)	Introduces Vision Transformers to histopathology	Limited generalization due to single dataset evaluation

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Table 5 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Ghosh et al. (2021)	2021	Kather-5000-images, NCT-HE-100K-CRC	None	Ensemble Deep Neural Network	ACC 99.13% (merge)	Ensemble learning on merged datasets achieved SOTA performance for multi-class CRC tissue classification	High computational demand due to multiple architectures
Wei et al. (2021)	2021	3152 Colorectal polyp	Data Augmentation	Annotator-agreement-based curriculum learning with ResNet-18	AUC 88.2%	Proposed using annotator agreement to rank medical image difficulty for curriculum learning	Requires multiple annotators to define reliable difficulty levels
Hamida et al. (2021)	2021	AiCOLO, Kather-5000-images, NCT-HE-100K-CRC	Stain Normalization, Data Augmentation	Transfer-learning-based CNN classification with multiple backbone architectures	ACC 99.98% (merged)	Proved that Transfer Learning outperforms training from scratch in patch classification	Evaluating multiple CNN backbones increases computational overhead
Raczkowska et al. (2019)	2021	Kather-5000 images	None	Active Learning based Bayesian CNN architecture with residual connections (ARA-CNN)	ACC 92.44%	Variational dropout based entropy uncertainty speed up learning by 45%	High computational cost during inference due to multiple variational dropout samples
Liang et al. (2020)	2020	16K colon tissue images	Shearlet Transform	Multi-scale Feature Fusion CNN (MFF-CNN)	ACC 96%	Effective multi-scale fusion of WSIs and shearlet features	Additional computation overhead due to shearlet decomposition
Iizuka et al. (2020)	2020	Colon WSIs	Tissue Masking, Patch Extraction, Data Augmentation	Inception-v3 CNN for tile classification with WSI aggregation using max-pooling and RNN	AUC 99%	Demonstrates high generalization across institutions and TCGA datasets	Computationally intensive due to RNN-based aggregation
Kather et al. (2019a)	2019	NCT-CRC-HE-100K, NCT-CRC-HE-VAL-7K	Stain Normalization, Data Augmentation	Transfer learning-based fine-tuning of CNNs	ACC 98.7% (VGG19)	Introduced a CNN-derived deep stroma score as an independent predictor of patient survival	Manually extraction of tumor regions
Peng et al. (2019)	2019	NCT-CRC-100K	None	Multi-task framework for CRC classification with kNN-based retrieval	ACC 99.4%	Improves interpretability of CRC classification using kNN-based retrieval	Computationally complex multi-loss training
Ilse et al. (2018)	2018	CRCHisto Phenotypes	None	Attention-based Multiple Instance Learning (MIL) with gated attention for CRC nuclei classification	ACC 90.4%	Introduces trainable attention-based MIL pooling for identifying diagnostically relevant patches	Large computational requirement due to attention weight computation for each instance
Basha et al. (2018)	2018	CRCHistoPhe-notypes	None	CNN with 7 trainable layers (RCCNet)	ACC 80.61%	A lightweight CNN optimized for low-resolution histology patches	Lack of external validation

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Table 5 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Tiwari (2018)	2018	Kather-5000 images	Color Space Conversion (HSV, LAB, etc.)	CNN with color models	ACC 95.8%	Color model analysis	Color-model focused
Haj-Hassan et al. (2017)	2017	Multispectral CRC	Patch Extraction, Tissue Masking	Active contours with custom CNN	ACC 99.17%	Demonstrated the effectiveness of tissue segmentation on classification accuracy	Need of pathologists in order to select ROI.
Wang et al. (2017)	2017	Kather-5000 images	Stain Decomposition (H&E)	Bilinear CNN (BCNN) applied on separate H and E stain components with bilinear pooling	ACC 92.6%	Stain-aware feature fusion for CRC histopathology classification	Large feature dimensionality
Korbar et al. (2017)	2017	697 WSIs	Data Augmentation	Modified ResNet architecture with 152 layers with projection mappings	ACC 93.0%	First automated system to classify 5 common types of colorectal polyps on WSIs	Inter-class confusion between hyperplastic and sessile serrated polyps
Ciompi et al. (2017)	2017	Kather-5000 images, 74 Rectal Cancer WSIs	Patch Extraction, Stain Normalization	Custom ConvNet using two different stain normalization algorithms	ACC 75.55%	Proved stain normalization is essential for both training and testing to reduce inter-lab color variations	Loss of biological color features due to template-based color matching
Sirinukunwattana et al. (2016)	2016	CRCHistoLabeled, NucleiHE	Patch Extraction, Color Deconvolution	Softmax CNN with a novel Neighboring Ensemble Predictor (NEP)	AUC 0.917	NEP improves accuracy using local ensembling without explicit nuclei segmentation	Performance drop for minority classes due to class imbalance

ACC: Accuracy; AUC: Area Under Curve; N/F: Information Not Found.

Table 6
Summary of studies employing conventional approaches for colorectal cancer detection.

Author	Year	Datasets	Methodology	Results	Key contributions	Limitations
Kather et al. (2016)	2016	Kather-5000	Multi-tissue classification using texture features and an SVM classifier	ACC 98.6% (Tumor vs. Stroma separation)	Introduced the first large-scale benchmark for multi-tissue detection	Fixed tile size (150 × 150 px) limits detection precision to coarse boundaries
Sirinukunwattana et al. (2015a)	2015	Bilkent colon histology	Superpixel-based handcrafted features with RF	Object-level Dice 0.746	Proposed a GPM to detect glandular regions based on texture and color features	Computationally intensive when small superpixels are used to capture fine morphological details

an ensemble model with a dual attention mechanism for CRC tumor detection. This model used Distance Transform and Watershed methods to achieve better tumor margins and defined approximate lesion regions.

Beyond macro-level regional detection, identifying key biological structures, such as nuclei and glands, is fundamental for determining tumor grade and differentiation in CRC histopathology. In this context, Sirinukunwattana et al. (2016) developed a Spatially Constrained CNN (SC-CNN) to predict the probability of each pixel being the center of a nucleus in CRC tissues. By employing a spatially constrained regression mechanism, SC-CNN effectively detected nuclei even in regions with clustered or overlapping nuclei. Tofighi et al. (2019) further enhanced the SC-CNN and introduced the Shape Priors CNN (SP-CNN) and Tunable SP-CNN (TSP-CNN) by explicitly adding prior nucleus shape information as a regularizer during training. This shape-guided learning improved detection accuracy, even in dense and overlapping nuclei, compared to SC-CNN. Javed et al. (2020) further extend this analysis to cellular community detection using SC-CNN and TSP-CNN to build cellular graphs from detected nuclei and analyze cell-to-cell connections for tissue phenotyping. Li et al. (2020) introduced a Position of Interest (POI) network to learn context-aware features and added a cascade residual fusion block to improve nuclei detection in CRC histopathology

images. After evaluation, the POI network achieved an F1 score of 0.837 and clearly outperformed existing approaches, including SC-CNN. However, the study by Höfener et al. (2018) found that using a simple FCN-based Probability Map (PMap) approach with proper configuration and post-processing yielded SOTA results for nuclei detection in histopathology images. In conclusion, they suggested that using appropriate post-processing had a greater impact on detection accuracy than increasing architectural complexity. Similarly, single-stage detector approaches have also been explored. For instance, Pons et al. (2025) used YOLOv5 for nuclei detection and introduced an adaptive workflow in which an LR classifier analyzed image features to automatically route difficult cases to a specialized model. This two-stage pipeline improved nuclei detection accuracy and added interpretability by identifying features associated with poor model performance.

Though CNNs achieved good results in nuclei detection, they required heavy offline training. To address this, Javed et al. (2021) introduced the Spatially Constrained Context-aware hierarchical Deep Correlation Filters (SCC-DCF) framework as a hybrid approach for nuclei detection in CRC histology. It used VGG-19 as a feature extractor and trained a DCF in such a way that shallow layers provided better nuclei localization and deep layers encoded more semantic information about nuclei. This framework achieved better results and outperformed 15 SOTA models in both accuracy and speed. In the next study, Javed et al. (2022) suggested that single features could not handle dense, overlapped, and cluttered nuclei in the nuclei detection task. Therefore, they introduced MP-DFF, a multi-level DCF-based feature-fusion method for nuclei detection in CRC histology. Its variant, MP-DFF-SCC, achieved an F1-score of 0.946 on the CRCHistoPhenotypes dataset.

Nuclei detection models often suffer from high false-positive rates due to heterogeneous and overlapping morphology. To address this, Xie et al. (2024) introduced the Dual-Branch Fully Convolutional Network (DB-FCN) for nuclei detection in CRC histopathology. This dual approach included a coarse detection branch followed by a fine detection branch for more precise detection. The coarse-to-fine approach successfully reduced false positive rates and outperformed existing detectors. Sun et al. (2021) focused on improving feature learning for nuclei detection and introduced the Similarity-based Region Proposal Network (SRPN). The key novelty of this approach was the integration of a specialized embedding layer with triplet loss into CNN-based detection architectures like Faster R-CNN and RetinaNet. Evaluation on two nuclei detection benchmarks demonstrated that SRPN not only learns discriminative features but also effectively reduces false positives. Despite reducing false positives, NMS-based post-processing remains a computational bottleneck. To address this, Pina and Dorca (Pina et al., 2025) used a Detection Transformer (DETR) and introduced CellNuc-DETR for simultaneous nuclei detection and classification. The key innovation of the CellNuc-DETR model was to formulate nuclei detection as a set-prediction problem rather than a pixel-level segmentation task, thereby eliminating the need for NMS-based post-processing. Upon evaluation, the proposed framework demonstrated impressive performance on two nuclei detection benchmarks. Table 7 outlines the DL algorithms, results, and key strengths used in CRC tissue detection.

Summary: We found a total of 16 studies on CRC disease detection, ranging from macro-structure detection to micro-level nuclei detection, of which 14 use DL-based approaches. This trend indicates that disease detection in CRC histopathology has shifted from simple classification to precise spatial localization of tumors and abnormal microstructures within tissue images. Conventional methods confined detection tasks to object localization using spatial maps and provided only categorical information (Kather et al., 2016). In contrast, DL methods provide more precise boundary information. For instance, using Faster R-CNN, Ho et al. (2022) directly identify and localize high-risk glandular structures with higher sensitivity. At the micro level, overlapping and cluttered nuclei are a formidable challenge. While earlier studies, such as SP-CNN (Tofighi et al., 2019) and POI-Net (Li et al., 2020), incorporated shape prior information to address this, Höfener et al. (2018) suggested that appropriate post-processing methods have a greater impact on detection accuracy than increasing architectural complexity. Pons et al. (2025) developed a hybrid framework using a single detector, YOLOv5x, and LR to route difficult cases during nuclei detection. False positive reduction in nuclei detection is a major issue. Coarse-to-fine detection (Xie et al., 2024) and improved feature learning using triplet loss (Sun et al., 2021) have reduced false-positive rates. Nevertheless, these approaches heavily depended on NMS-based post-processing. More recently, CellNuc-DETR (Pina et al., 2025) has further advanced the field by treating detection as a set-prediction task, thereby eliminating the need for NMS and anchor generation.

4.3. Image segmentation

Segmentation is a pivotal step in CPath that involves partitioning of histopathology images into meaningful segments or regions of interest (ROIs). In the context of CRC, this process is essential for delineating morphological structures, such as tumor regions, glands, nuclei, and stroma, from the background tissue (Ma et al., 2024). Unlike other medical imaging, segmentation in H&E stained histology slides has its own unique challenges due to stain variability, overlapping cellular structures, and the heterogeneity of tissue texture. Nevertheless, accurate segmentation is a prerequisite for quantitative analysis. Since manual segmentations are both time-consuming and subjective, automated methods have become essential for identifying and extracting key histological components. Conventional approaches were the method of choice in early research. However, modern application of such methods is often limited to coarse tissue separation prior to classification rather than to the precise segmentation of malignant glands or complex tumor regions. The remarkable changes in segmentation methodology occurred after the introduction of DL, particularly with the U-Net. The subsequent sections present applications of both conventional and DL-based approaches to CRC image segmentation.

4.3.1. Conventional segmentation methods

Prior to the DL era, conventional segmentation methods were the standard for extracting meaningful ROIs from CRC histology images. These methods typically rely on simple mathematical rules and are often applied to HCFs such as pixel intensity, color features, and texture gradients. Such techniques are computationally efficient and easy to implement. Nonetheless, they struggle with the high variability inherent in biological tissues.

Abundant classical image segmentation techniques have been applied to segment tumor regions from CRC tissue images. One such classical technique is thresholding, which exploits the intensity differences between the objects and the background. Techniques range from global thresholding to adaptive methods such as Otsu's algorithm (Otsu, 1978), which calculates optimal threshold values based on image histograms. In the context of CRC, these methods are often combined with color space transformations to enhance contrast. For example, Nawandhar et al. (2015) examined the Sobel, Canny, HSV color-space transformation, and vector gradient methods to detect cell nuclei in H&E-stained colon tissue images and found that converting tissue images to the HSV color space prior to applying Otsu's thresholding correctly marked nuclear locations compared to standard grayscale processing. But these simple algorithms are sensitive to complex morphological patterns, such as overlapping glands. To address this, Karthikeyan et al. (2024) used a hybrid approach by combining Otsu's thresholding with Watershed segmentation as a robust pre-processing step to isolate distinct objects and their boundaries from colon tissue images, before feeding them into a CNN classifier. Similarly, Banwari et al. (2018) used an adaptive thresholding method to segment glandular tissues from CRC histology images prior to classification. Staining heterogeneity is a known issue in histology images and also affects segmentation accuracy. To address this, Abdelsamea

Table 7
An overview and findings of studies employing deep learning techniques for colorectal cancer detection.

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Pina et al. (2025)	2025	PanNuke, CoNSeP, MoNuSeg, TCGA	Tissue Masking, Patch Extraction	DETR-based transformer for direct nuclei detection and classification	F1 Score 0.76 (CoNSeP)	Eliminate NFS and anchor generation	Requires large annotated datasets
Pons et al. (2025)	2025	CRCHistoPhenotypes	Data Augmentation	Adaptive workflow combining YOLOv5x for detection with a Logistic Regression classifier	F1 score 0.84 (CRCHistoPhenotypes)	Proposed an intelligent and hybrid workflow using a LR classifier that can trace difficult images	Its performance mainly depends on the LR's ability to trace images correctly
Dharshini et al. (2025)	2025	Kather-5000-images	Gaussian Blur, Watershed technique, Data Augmentation	Ensemble model with Dual Attention Mechanism	F1 score 0.94	Improved accuracy and interpretability using an ensemble approach with a dual attention	Dependence on a single dataset may restrict generalizability
Xie et al. (2024)	2024	CRCHistoPhenotypes, PanNuke, ccRCC	None	End-to-end Dual-Branch Fully Convolutional Network (DB-FCN)	F1 score 0.897 (CRCHistoPhenotypes)	Introduced a coarse-to-fine dual-branch method that reduces false positives	Performance degradation due to variations in staining methods
Ho et al. (2022)	2022	294 CRC WSIs	Tissue Masking, Patch Extraction	Faster R-CNN-based region detection followed by ML-based slide classification	Sensitivity 97.4% (dysplasia/malignancy detection)	Introduced a high-sensitivity AI framework for detecting dysplastic and malignant regions	High false-positive rate and limited dataset size reduces generalizability
Javed et al. (2022)	2022	CRCHistoPhenotypes, CoNSeP, PanNuke	Hematoxylin Channel Extraction(Color Deconvolution), Contextual Patches	Multi-level Pool DCFs Feature Fusion (MP-DFF)	F1 score 0.946 (CRCHistoPhenotypes)	First multi-level feature fusion algorithm for nucleus detection that combines low, mid and high-level features using Pool-early, Inter-pool, and High-level fusion	Computationally intensive
Javed et al. (2021)	2021	CRCHistoPhenotypes, CoNSeP, PanNuke	Hematoxylin Channel Extraction(Color Deconvolution), Contextual Patches Extraction	Spatially Constrained Context-aware Deep Correlation Filters using hierarchical VGG-19 deep features	F1 score 0.914 (CRCHistoPhenotypes)	Integrate spatial graph regularization and context-aware constraints into the DCF framework	Generalization performance reduces on completely novel dataset
Sun et al. (2021)	2021	MoNuSeg, DigestPath	Patch Extraction	Similarity-based Region Proposal Network (SRPN) based embedding layer with triplet loss into Faster R-CNN	F1-score 0.854 (MoNuSeg)	Introduced triplet loss to RPN for discriminative feature learning	Exhibits a trade-off between specificity and recall
Javed et al. (2020)	2020	CRCHistoPhenotypes, CRC-TP	Cell Detection and Classification	Cellular Community Detection using cell–cell graphs	F1 score 0.91 (CRC-TP)	Proposed the large CRC-TP dataset	Computationally expensive due to use of all-pairs shortest path in graph construction

(*continued on next page*)

Table 7 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Xu et al. (2020)	2020	307 CRC WSIs	Tissue Masking, ROI Extraction	Patch classification-based detection using Inception-V3	ACC 94.8% (cancer slides)	Proposed a robust DL screening model for tumor region identification in CRC WSIs	Performance sensitivity near the tumor boundary due to the patch classification approach
Li et al. (2020)	2020	CRCHistoPhenotypes	Data Augmentation	Staged framework using a POI network in addition with Cascade Residual Fusion Blocks for precise detection	F1 score 0.837 (CRCHistoPhenotypes)	Proposed POI network generates a proximity map to precisely locate nuclei centers that is further improved by Cascade Residual Fusion Blocks	Sequential two-stage pipeline may increases computational cost
Tofighi et al. (2019)	2019	Warwick-QU	Tissue Masking, Canny Edge Detection	Shape Priors CNN (SP-CNN) and Tunable SP-CNN (TSP-CNN) for nucleus detection	F1 score of 0.852 (using TSP-CNN)	Proposed Shape Priors DL networks (SP-CNN/TSP-CNN) that can reduce false positives through shape guidance	Shape similarity analysis is limited
Höfener et al. (2018)	2018	CRCHistoPhenotypes	Data Augmentation	Fully Convolutional Network (FCN) based Probability Map (PMap) approach with regression and post-processing	F1 score 0.816 (CRCHisto Phenotypes)	Proved that simple FCN based PMap models can match complex SOTA methods if configured properly	Highly sensitive to parameter tuning
Sirinukunwattana et al. (2016)	2016	CRCHistoLabeledNucleiHE	None	Spatially Constrained CNN (SC-CNN) for nucleus detection	F1 score of 0.802 (using SC-CNN M=2)	Proposed the SC-CNN for spatial regression of nucleus center likelihood that is locality sensitive and does not require segmentation	Computationally intensive for large WSIs

ACC: Accuracy.

et al. (2019) designed a cascade-learning framework using a Fuzzy Signed Pressure Force (FSPF) model based on fuzzy c-means clustering. This method effectively segmented epithelial layers in CRC images and was robust against noise and stain variations. Using a similar approach, Rathore and Iftikhar (2016), Rathore et al. (2019) utilized ellipse fitting to model glandular components in colon tissue. Rathore and Iftikhar (2016) combined K-means clustering and an optimized region-growing approach to segment epithelial regions, while Rathore et al. (2019) integrated geometric features with hierarchical clustering to achieve a comprehensive segmentation of glands, lumen, nuclei, and epithelial boundaries. Traditional segmentation methods are simple and straightforward to implement. However, they are commonly found to be useful only in basic classification tasks. Table 8 summarizes the methods, objectives, and results of traditional approaches used in CRC tissue segmentation.

4.3.2. Deep learning based segmentation method

The introduction of the U-Net architecture (Ronneberger et al., 2015), as illustrated in Fig. 11, has changed the landscape of medical image segmentation. Following the success of U-Net, researchers focused on enhancing its feature extraction capabilities to better distinguish tumor regions. For instance, Zhang et al. (2024a) proposed SC-Net, comprising a Feature Mapping Layer (FML) and a spiral multi-branch encoder with Multi-scale Feature Extraction Blocks (MFEBs). This spiral architecture captured heterogeneous semantic information across different receptive fields, resulting in the precise segmentation of complex colorectal tissues with a mean Dice score of 86.11%. Similarly, Yuan et al. (2024) proposed MMUNet using depthwise separable convolutions. The network also used External Attention to capture long-range dependencies and added morphological modules (EDM and EFM) to improve contour refinement and feature fusion. MMUNet achieved a maximum Dice score of 92.76% on the Glas dataset. Rastogi et al. (2022) optimized the classical U-Net by reducing the number of filters and designed a semantic segmentation model. The raw predicted masks were further refined using morphological operations, resulting in a gland segmentation Dice score of 0.82. CRC diagnosis critically depends on gland morphology. However, blurred boundaries in malignant cases make accurate gland instance segmentation difficult. Luo et al. (2024) designed BFMSE-Net that integrated the Boundary Fusion Module (BFM) for explicit edge refinement and the Multi-Scale Enhancement Module (MSEM) for multi-scale adaptation. BFMSE-Net accurately segmented densely clustered and edge-adhesive glands and also achieved SOTA performance. Kassani et al. (2022) and Sengupta et al. (2025) conducted comprehensive comparative benchmarking on

Table 8
An overview and findings of studies employing conventional techniques for colorectal cancer segmentation.

Author	Year	Datasets	Methodology	Results	Key contributions	Limitations
Karthikeyan et al. (2024)	2024	334 CRC histology images	Threshold-based salient segmentation method	ACC 91%	Segmentation assisted classification	Limited scope of segmentation
Abdelsamea et al. (2019)	2019	14 WSIs CRC, 15 Tissue Microarrays (TMA)	Fuzzy c-means algorithm and classification framework based on self-organizing maps	Dice 96.23%	Automated epithelium segmentation from colon WSIs and combines appearance and shape information	Sensitive to initialization and parameter tuning
Rathore et al. (2019)	2019	Private Colon Biopsy	Elliptic object-based segmentation with SVMs	ACC 88.17%	Introduced elliptic and near-elliptic epithelial cell modeling	Single institution dataset
Banwari et al. (2018)	2018	Warwick-QU	Automated adaptive threshold technique with SVM classifier	ACC 91.25%	CRC gland tissue segmentation and classification from histopathology images	Sensitive to noise and artifacts
Rathore and Iftikhar (2016)	2016	GlaS	K-means clustering with ellipse fitting	ACC 87.5%	Multi-scale radiomic feature framework	Less performance on extremely deformed glands
Nawandhar et al. (2015)	2015	Six different colon tissue images	Thresholding	HSV shows promising results	Evaluate and compare the effectiveness of multiple methods for segmentation of colon tissue	Sensitive to noise and artifacts

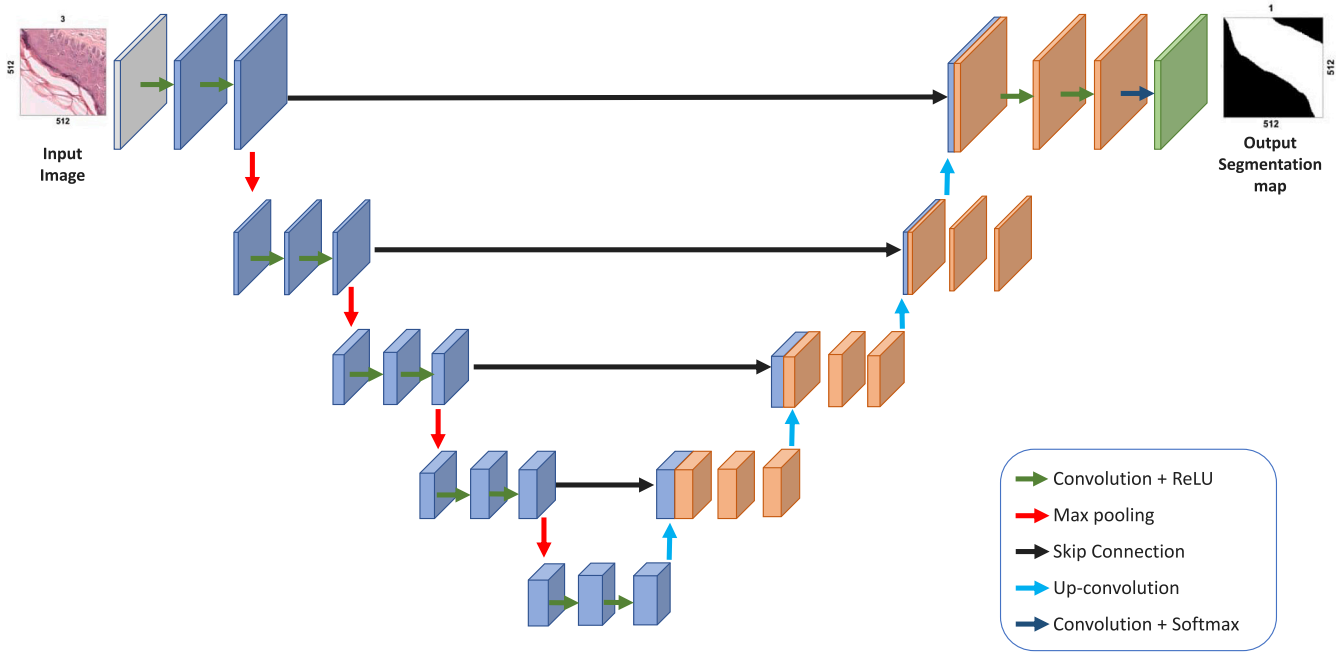


Fig. 11. U-Net architecture with an encoder–decoder structure and skip connections for CRC histopathology image segmentation.

semantic segmentation models. Study (Kassani et al., 2022) examined U-Net, FPN, and LinkNet with multiple CNN backbones and found that the DenseNet121–LinkNet setup outperformed all others with a Dice score of 0.8274. In contrast, study (Sengupta et al., 2025) showed that EfficientNet-B0 U-Net performed better than other U-Net variants and achieved the highest Dice score of 0.945.

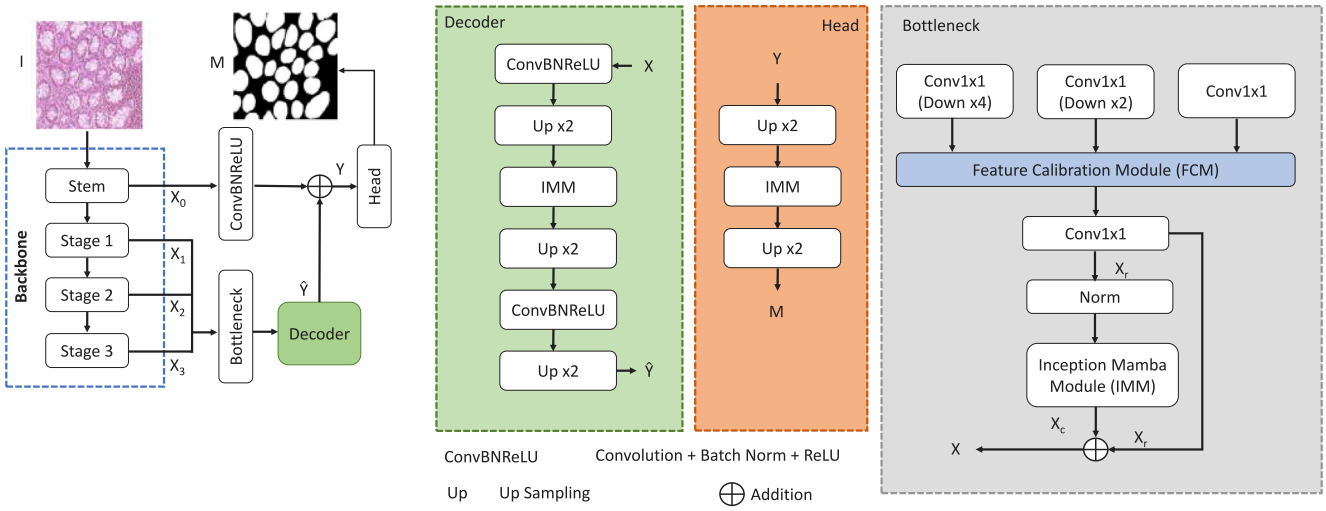


Fig. 12. Architecture of InceptionMamba for CRC tissue segmentation with Inception-Mamba module and a Feature Calibration Module (Kareem et al., 2025).

Touching or overlapping glands or nuclei are major challenges in segmenting individual objects. To address this, researchers adopted contour-aware information within networks. Kainz et al. (2017) proposed an early dual-path DNN framework using two LeNet-5-based DCNNs: Object-Net for gland-background classification and Separator-Net for inter-gland boundary detection. This approach achieved high accuracy in segmenting glands. Chen et al. (2017) proposed a multi-task framework, the Deep Contour-Aware Network (DCAN), to predict gland masks and contours simultaneously. This contour-based structure improved boundary accuracy and effectively separated touching glands with an object-level Dice score of 0.868 on the GlaS dataset. Graham et al. (2019a) proposed MILD-Net with MIL units to counter information loss due to max-pooling by re-introducing the original image (Fig. 14). It achieved an object-level Dice of 0.913 on the GlaS dataset, and an extended version, MILD-Net+, was also introduced for joint gland–lumen segmentation. With the same aim, Graham et al. (2019b) also proposed Rota-Net, which uses group equivariant convolutions (G-CNNs) with rotational symmetry to perform simultaneous gland and lumen segmentation. Upon evaluation on the GlaS dataset, it achieved an object-level Dice score of 0.919 and outperformed MILD-Net. Graham et al. (2023) also presented a single DL model using a shared ResNet-34 encoder with task-specific decoders for simultaneous segmentation of nuclei, glands, and lumina. This multi-task framework achieved improved segmentation performance on public datasets and external TCGA.

Other studies focused on hybrid and ensemble strategies to achieve more reliable segmentation results. For instance, Xu et al. (2017) developed a deep multichannel framework that fused outputs from separate foreground segmentation, edge detection, and object detection channels and yielded accurate gland segmentation masks. Qaiser et al. (2019) used Persistent Homology Profiles to capture atypical nuclear features, combining them with deep convolutional features to design an accurate tumor segmentation model. With this combination, it achieved the highest F1-score of 0.9243 on the GLas dataset. Similarly, Kim et al. (2021) combined Principal Component Analysis (PCA) and the Discrete Wavelet Transform (DWT) to design a U-Net++ based segmentation model, PCA-DWT (WWE), that segments normal and abnormal regions in a compressed domain. Khened et al. (2021) followed a divide-and-conquer strategy, combining multiple FCN architectures (U-Net with DenseNet-121, DeepLabV3+ with Xception, and U-Net with Inception-ResNet) to generate a final mask using averaging and achieved a Dice score of 0.78 on the DigestPath dataset. To address the critical issue of stain variability, Hou et al. (2019) proposed DAPNet, a domain-adaptive model that uses image- and feature-level adversarial training. DAPNet outperformed other standard baselines in cross-domain evaluation across both IoU and accuracy. Tang et al. (2018) followed the SegNet architecture to develop a model for gland segmentation in CRC histopathology images. It achieved a segmentation accuracy of 0.882 and a Hausdorff distance of 106.6471 on the Warwick-QU dataset. Binder et al. (2019) proposed two Dense-U-Net models that separately focus on gland and stroma segmentation. Unlike standard models that generally predict only the gland mask, this dual approach simultaneously predicts the object mask and its contour to effectively identify each individual gland.

CNNs excel at local feature extraction but struggle to capture global context. Therefore, recent studies have explored attention mechanisms and ViTs to capture global dependencies. For instance, Dabass et al. (2021) proposed an Attention-Guided Deep Atrous-Residual U-Net for gland segmentation. The novelty of this architecture lies in adding Atrous-Residual units to extract multi-level discriminative features and Attention units to capture global information. With extensive pre- and post-processing techniques, it achieved the highest Dice score of 0.945 on the CRAG dataset. To effectively model long-range dependencies, Zhao et al. (2023) proposed RGSB-UNet, integrated Residual Ghost Blocks and Switchable Normalization with a Bottleneck Transformer (BoT) within the U-Net for tumor segmentation. This hybrid CNN–Transformer design enabled stable model training with small batch sizes and improved global context modeling for better tumor segmentation. Nevertheless, Arslan et al. (2025a) argued that hybrid architectures remain constrained by the local receptive fields. For this, study (Arslan et al., 2025a) benchmarked the Swin Transformer over standard CNN architectures for multi-class tumor grade segmentation. They observed that the Transformer-based model achieved better performance with an average Dice score of 0.629. Zidan et al. (2023) then proposed SwinCup using a hierarchical Swin Transformer as the encoder and a Cascaded Upsampler (CUP) as the decoder. This architecture first captured a global understanding of the image and then used multi-scale features to accurately detect the tumor boundaries. SwinCup achieved an SOTA F1 score of 0.92 on the GlaS dataset but at the cost of heavy pre-training.

Transformers perform well and have achieved SOTA results across a variety of histopathology image analysis tasks. Despite this, their quadratic computational complexity limits their efficiency on large histopathology images. As a result, State Space Models (SSMs), such as the Mamba, have emerged to model long-range dependencies with linear complexity. For instance, Kareem et al. (2025) introduced InceptionMamba by redesigning the bottleneck block using an Inception-Mamba module (IMM) and a Feature Calibration Module (FCM), as shown in Fig. 12. By using depth-wise convolutions inside the IMM,

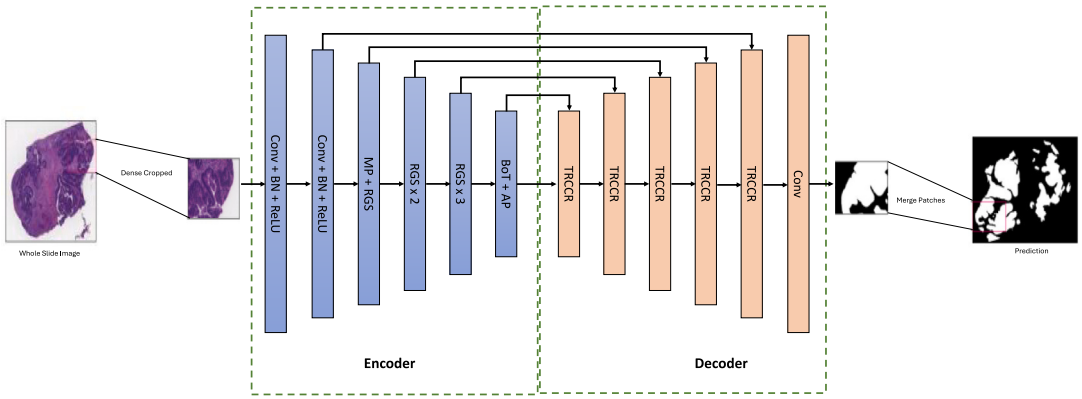


Fig. 13. Architecture of encoder- decoder with bottleneck transformer (RGSB-UNet) (Zhao et al., 2023).

the model became more efficient while capturing multi-scale features and long-range dependencies. With an F1-score of 91.88% on the GlaS dataset, InceptionMamba outperformed both CNN and transformer-based architectures. Recently, pure Vision Mamba models such as VM-UNet (Ruan et al., 2024) have been adopted to capture global context with reduced computational complexity. However, Kui et al. (2025) argued that these pure architectures often lack precise local feature extraction. For this, study (Kui et al., 2025) combined a CNN and Mamba architecture and designed GL-MambaNet for medical image segmentation. The addition of the Global–Local Feature Fusion Module (GLFFM) allowed the model to learn complementary features from both architectures. It showed exceptional performance on the GlaS dataset with a Dice score of 0.930 and outperformed various SOTA, including pure Vision Mamba baselines (VM-UNet).

Recent architectures like Transformers and Mamba achieve SOTA results, but they are data-hungry and require massive annotated datasets. Therefore, data-efficient approaches have emerged as a critical research direction to address the scarcity of annotations. Su et al. (2024) used a few-shot approach to design a DatasetGAN-based model that generated synthetic CRC histology images with masks using only a few labeled samples. Efficacy was assessed by training a UNet++ model that achieved reasonable tumor bud segmentation performance. In the same realm, Villanueva-Miranda et al. (2024) used the power of the foundational Segment Anything Model (SAM) using granular box prompts and found that fine-tuning on a small fraction of the CRAG dataset was sufficient to achieve superior segmentation accuracy.

Summary We evaluated a total of 31 studies on CRC histopathology tissue segmentation, of which 25 are DL-based, showing a clear dominance of DL methodologies in the field. Conventional methods, such as Otsu's thresholding combined with Watershed segmentation (Karthikeyan et al., 2024) and K-means with region growing (Rathore and Iftikhar, 2016), are simple, but their current scope is largely confined to serving as a prerequisite for classification tasks. Following the success of U-Net, DL-based methods have become the choice of researchers in tissue segmentation. Several enhancements, including the incorporation of the FML (Zhang et al., 2024a), EDM (Yuan et al., 2024), and MSEM (Luo et al., 2024), have improved gland segmentation performance. Touching and overlapping glands are more common in CRC histopathology. Studies have adopted distinct strategies to mitigate this. For instance, Kainz et al. (2017) proposed background identification and gland detection, Chen et al. (2017) preserved counter information by incorporating context-aware information, while MILD-NET (Graham et al., 2019a) achieved this by introducing MIL units to prevent feature loss. The inability of CNNs to handle global context has slowed their progress in recent studies. To capture global context, Dabass et al. (2021) proposed an Attention-Guided Deep Atrous-Residual U-Net for gland segmentation. Further, a hybrid CNN–transformer architecture, as shown in Fig. 13, was proposed by Zhao et al. (2023) to improve global context modeling for better tumor segmentation. However, Arslan et al. (2025a) argued that hybrid architectures remain constrained by the local receptive fields and benchmarked the Swin Transformer for multi-class tumor grade segmentation. Although transformers are good at modeling long-range dependencies, they simultaneously impose a higher computational burden that often leads to quadratic complexity. Modern architectures like Mamba have emerged in histopathology to model long-range dependencies with linear complexity. For instance, InceptionMamba (Kareem et al., 2025) and CNN–Mamba (Kui et al., 2025) outperformed both CNN and transformer-based models on the GlaS dataset. Finally, to address the bottleneck of large pixel-level datasets, innovative strategies like FSL-based DatasetGAN (Su et al., 2024) for synthetic image generation and the fine-tuned FM model SAM (Villanueva-Miranda et al., 2024) have achieved superior results with minimal samples. A brief overview of the methods, results, and objectives of DL approaches for CRC tissue segmentation is presented in Table 9.

4.4. Image prediction method

Molecular biomarker prediction is pivotal for early detection and prognostic assessment of CRC, providing valuable insights into mutational patterns that inform optimal therapeutic plans (Hamilton et al., 2014). However, manual molecular analysis is both time-consuming and expensive (Djuric et al., 2017). This field changed dramatically when recent studies demonstrated that phenotypic patterns visible in routine H&E-stained histology can reflect underlying genetic mutations (Fu et al., 2020). Image prediction thus addresses the critical clinical need to non-invasively detect molecular biomarkers directly from routine H&E-stained images. However, it is not easy for pathologists to visualize or capture genetic alterations in tissue images. Image prediction involves the use of ML and DL techniques to predict or detect the status of molecular biomarkers from routine histopathology images. The subsequent sections present applications of both conventional (ML) and DL-based approaches to CRC image prediction.

4.4.1. Conventional prediction methods

Conventional approaches are still valuable for predicting CRC biomarkers from histopathology images. Studies focused on HCF, such as textures and color features, demonstrate that visual patterns reflect underlying molecular status. For instance, Swanson et al. (2025) extracted GLCM-based texture features across three color spaces to train ML models for MSI classification in CRC tissues. The results showed that textural morphology can effectively predict MSI from H&E-stained slides. In contrast to using only texture features, Zhu et al. (2022) extracted morphological features alongside color and texture features and trained an RF model to predict MSI in colon tumors. The results

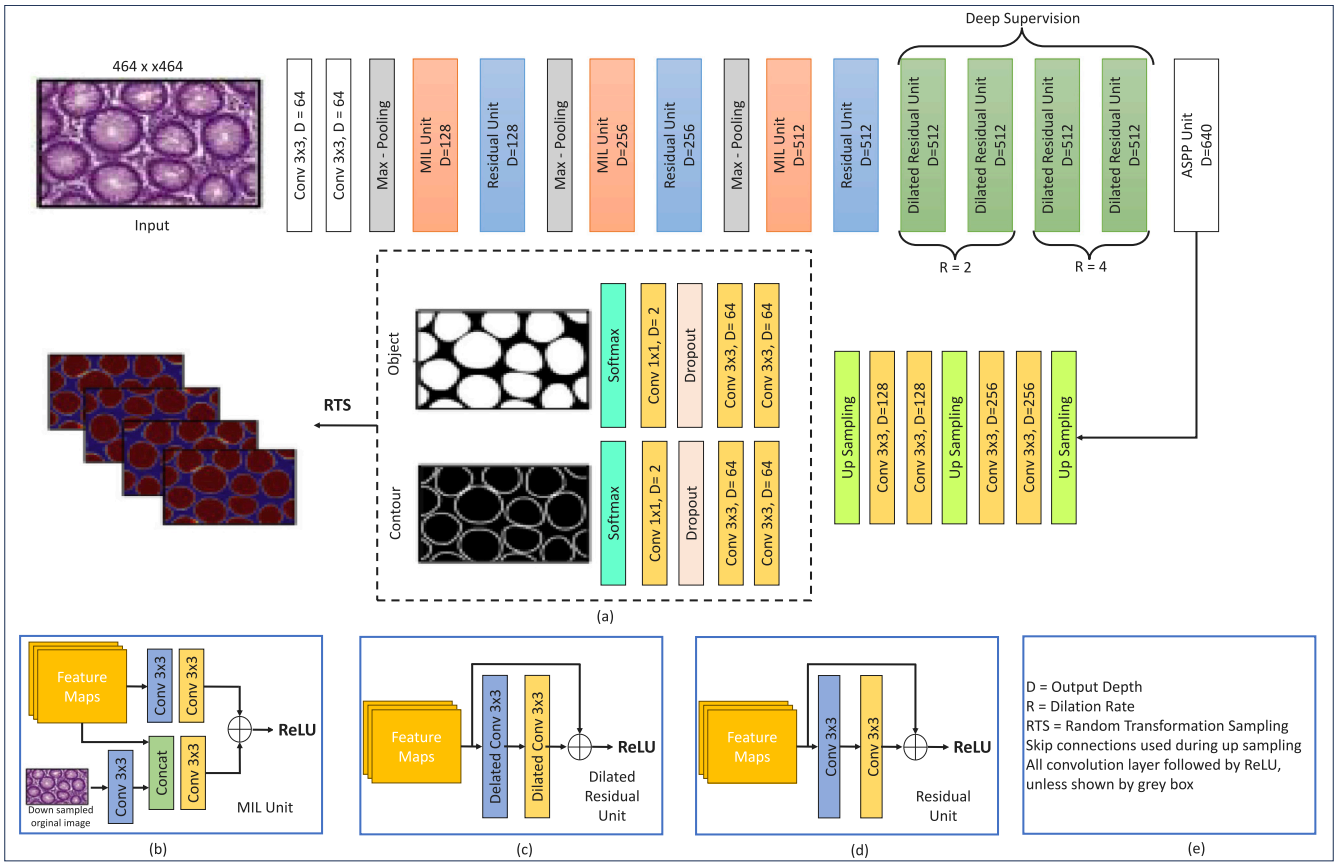


Fig. 14. MILD-Net architecture consisting of MIL units, residual and dilated residual blocks, random transformation sampling (RTS), and multi-branch outputs for object and contour detection (Graham et al., 2019a).

revealed that combining texture and color features was important for accurate MSI prediction. However, this ML-based approach did not outperform the DL model evaluated in the same study. The methodologies and performance outcomes of ML based genetic mutation prediction in CRC histopathology are summarized in Table 10.

4.4.2. Deep learning based prediction methods

The application of DL methods, specifically CNNs, in CRC histopathology image analysis has grown exponentially in the last decade. A seminal contribution in this direction was made by Kather et al. (2019b), who showed that DL models can predict MSI/MSS status directly from routine H&E-stained histology slides. The study used a two-stage ResNet-18 pipeline using TL for tumor detection and MSI prediction. The first network detected tumor regions, while the second classified MSI status using TCGA-CRC-DX dataset. Later, Kather et al. (2020) extended previous work by designing a unified framework using ShuffleNet to predict multiple genetic mutations and biomarkers from pan-cancer histology images. In CRC datasets, ShuffleNet achieved higher performance with an AUC of 0.89 and outperformed ResNet-18 in MSI prediction. In addition to MSI prediction, Konishi et al. (2023) developed a dual DL-based approach to predicting genetic mutations from tumor patches. On the TCGA CRC dataset, it achieved AUCs of 0.712 for BRAF and 0.721 for MSI mutations. Zhu et al. (2022) and Zhang et al. (2024b) both proposed end-to-end MSI prediction frameworks with integrated explainability. Study (Zhu et al., 2022) used ResNet-18 with Grad-CAM for image-level interpretability to highlight regions influencing MSI predictions, while the study (Zhang et al., 2024b) incorporated cellular analysis to provide biological interpretability of MSI-related patterns. Studies have also explored the application of CNNs to predict tumor mutational burden (TMB-H) using H&E-stained histopathology images. For this, Shimada et al. (2021) proposed a two-stage InceptionV3 pipeline that first classified CRC patches into neoplastic and non-neoplastic regions and then predicted TMB-H using H&E slides from the TCGA CRC and JP-CRC cohorts. It achieved an average AUC of 0.934 for slide-level TMB prediction. Rather than slide-level prediction, Liu et al. (2022) designed a tile-level TMB prediction model, DeepHE, based on a ResNet architecture on CRC slides. It achieved 83.0% accuracy at a TMB threshold of 10 and 85.0% accuracy at a threshold of 20.

CNNs perform well in patch-based biomarker prediction but struggle with WSI-based molecular tasks due to the lack of global contextual information. Studies are exploring MIL frameworks for effectively aggregating patch-level predictions to derive slide-level predictions. For instance, Cao et al. (2020) developed an MIL-based EPLA pipeline that used ResNet for patch-level likelihoods, followed by an ensemble-based aggregation using PALHI and BoW to improve slide-level MSI prediction. This ensemble-based aggregation achieved better performance than either of the individual aggregators. Using a similar approach, Li et al. (2024a) proposed an MIL-based clinical-pathological model to predict KRAS, NRAS, and BRAF mutations in left-sided CRC that used Inception-V3 for feature extraction and PALHI and BoW for patch aggregation. Wang et al. (2025a) enhanced this by adding an attention-based aggregation strategy into their MIL framework to predict MSI and TMB from CRC histopathology images. This intelligence aggregation strategy reduced the influence of irrelevant patches on decision-making and achieved 0.88 and 0.84 in accuracy for MSI and TMB with only 40% of the training data. Bilal et al. (2021) simplified the WSL process by introducing the IDaRS sampling technique, which selects informative tiles from a set of samples. This approach achieved SOTA AUROC of 0.90 for MSI prediction while also predicting multiple molecular markers and mutations. Saldanha et al. (2022) used Swarm Learning (SL) with a WSL framework to predict MSI and BRAF mutation status

Table 9
An overview and findings of studies employing deep learning techniques for colorectal cancer segmentation.

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Kui et al. (2025)	2025	GlaS, PH2, DSB2018	Data Augmentation	Global–local hybrid CNN–Mamba network (GL-MambaNet)	Dice 0.9309 (GlaS)	Introduces the use of Mamba for effective global dependency modeling in CRC gland segmentation tasks	Clinical validation on real-world CRC WSIs is not investigated
Kareem et al. (2025)	2025	GlaS	Data Augmentation	U-Net based encoder–decoder with ResNet backbone, Feature Calibration Module (FCM), and Inception Mamba Module (IMM)	Dice 0.918	Hybrid CNN–Mamba architecture for accurate gland segmentation	Limited validation on histopathology specific CRC datasets
Sengupta et al. (2025)	2025	EBHI-Seg	Data Augmentation	Comparative assessment of U-Net using pre-trained backbones	Dice 0.945 (EfficientNetB0U-Net)	Systematic comparison of multiple U-Net-based architectures for CRC segmentation	Evaluation is limited due to single dataset
Arslan et al. (2025a)	2025	CCTGS	Downsampling, Data Augmentation	Comparative evaluation of CNN and transformer-based encoder–decoder models for multi-class CRC tumor grade segmentation	mDSC 62.92% (Swin Transformer)	Introduces the first public multi-class CRC tumor grade segmentation dataset with pixel-level annotations	Benchmarking is restricted to a single-center dataset
Villanueva-Miranda et al. (2024)	2024	CRAG, GlaS	Data Augmentation	Fine-tuned Segment Anything Model using granular box prompts (GB-SAM)	Dice 0.900 (CRAG)	Adapts foundational model (SAM) to histopathology using granular box prompts	Few segmentation failures are observed in regions with dense and complex morphology
Zhang et al. (2024a)	2024	Self-built CCSS, Kvasir-SEG	Patch Extraction, Color Threshold	Symmetric Conical encoder–decoder (SC-Net) with MFEB and Feature Mapping Layer	mDice 86.11% (CCSS)	Ablation studies show that FML preserves low-level spatial details	Few false positives are observed around regions with similar morphology
Luo et al. (2024)	2024	GlaS, CRAG	Data Augmentation	U-Net based architecture with boundary fusion module (BFM) and a multi-scale enhancement module (MSEM)	F1 score 0.832 (CRAG)	Introduced boundary fusion and adaptive multi-scale feature enhancement improve segmentation malignant glands	Struggles to segment malignant glands in Test-B (GlaS) cases with blurred boundaries.
Su et al. (2024)	2024	86 CRC WSIs	Patch Cropping	Few-shot tumor bud segmentation using DatasetGAN	Dice 0.5293	Annotation-efficient tumor bud segmentation on routine CRC H&E	Domain gap persists between generated and real images

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Table 9 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Yuan et al. (2024)	2024	GlaS, CoCaHis	None	U-shaped network with depth-separable convolution and morphological methods (MMUNET)	Dice 92.76% (GlaS)	Morphology-aware feature fusion using EDM and EFM enhances boundary preservation	Demands high memory due to large convolution kernels and morphological operations
Graham et al. (2023)	2023	GlaS, CRAG, TCGA, Lizard	Patch Extraction, Data Augmentation	Multi-Task Learning FCN with a shared ResNet-34 encoder and task-specific decoders	PQ 0.640 (GlaS)	Single network enables simultaneous multi-task prediction	Higher training complexity due to multi-task setup
Zidan et al. (2023)	2023	GLAS, CRAG	Tissue Masking, Patch Extraction for pre-training datasets	Hierarchical Swin Transformer encoder with a Cascaded UPsampling decoder (SwinCup)	F1 score 0.92 (GLAS)	SSL-based pre-training improves segmentation accuracy and outperforms the SOTA by 23%	Despite heavy pre-training, the study evaluates performance only on benchmark datasets
Kassani et al. (2022)	2022	DigestPath	Patch Extraction	Comparative study using LinkNet, U-Net, and FPN backbones integrated with 17 multiple backbone	Dice 0.827 (LinkNet + DenseNet121)	Comparative study of 51 encoder-backbone combinations for CRC segmentation	Lack of generalization due to reliance on a single dataset
Rastogi et al. (2022)	2022	GlaS	Data Augmentation	Symmetric encoder–decoder with reduced filter depth for gland semantic segmentation	Dice 0.82	Lightweight U-Net architecture with post-processing enhancement for colorectal gland segmentation	Evaluation is confined to a single benchmark dataset
Dabass et al. (2021)	2021	GlaS, CRAG, HosC	Patch Extraction, Stain Normalization, Data Augmentation	U-Net with Attention units, Atrous-Residual units, and Transitional Atrous units	Dice 0.945 (CRAG)	Integrated a pathologist scoring system (0–10) to validate real-world diagnostic suitability	High computational demand due to complex Attention and Atrous-Residual units
Khened et al. (2021)	2021	DigestPath	Tissue Masking, Patch Extraction, Data Augmentation	Ensemble FCN with multiple backbone and a patch-based uncertainty analysis	Dice 0.782	Adapts uncertainty estimation for clinical reliability	High computational cost due to ensemble learning
Kim et al. (2021)	2021	390 WSIs of colorectal biopsy specimens	Patch Extraction	U-Net++ with PCA and DWT for CRC tumor segmentation	Dice 0.804	Introduced a novel segmentation approach in the compressed domain to prevent high-frequency information loss during resizing	Demands high computational resources

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Table 9 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Binder et al. (2019)	2019	GlaS	Stain Normalization	Dense-U-Net with two decoders	Dice 0.93	Introduced a Stroma-approach for segmentation for cross-organ generalization	Segmentation boundaries appear edgy due to the lack of expert stroma annotations
Qaiser et al. (2019)	2019	Warwick-UHCW, Warwick-Osaka	Stain Deconvolution, Patch Extraction, Data Augmentation	Hybrid tumor segmentation using persistent homology and CNN features	F1 score 0.9243 (Warwick-UHCW)	Introduce persistent homology profiles capture nuclear connectivity for tumor segmentation in CRC histology	Limited generalization due to sensitivity to empirically selected parameters
Graham et al. (2019b)	2019	GlaS	Data Augmentation	Rotation-equivariant fully convolutional network using group equivariant convolutions (Rota-Net)	F1 score 0.920	Simultaneous gland and lumen segmentation from CRC tissue images	Evaluation limited to GlaS dataset
Graham et al. (2019a)	2019	GlaS, CRAG, COMET-1, COMET-2	Patch Extraction, Data Augmentation	Minimal information loss dilated network (MILD-Net) with MIL units and ASPP	Dice 0.913 (GlaS)	Introduces MIL units to counter information loss due to max-pooling operation	Performance degrades in malignant regions due to ambiguous gland boundaries
Hou et al. (2019)	2019	GlaS, GlandVision	None	Dual Adaptive Pyramid Network (DAPNet) with dilated ResNet-18 backbone	ACC on Warwick-QU to GlandVision 0.88	Unsupervised cross-stain gland segmentation	Lower segmentation accuracy observed during adaptation from GlandVision to GlaS
Tang et al. (2018)	2018	GlaS	Data Augmentation	FCN based encoder–decoder network (Segnet) for gland segmentation	ACC 0.8636	Smaller network design	Observed lower segmentation accuracy on GlaS part A
Chen et al. (2017)	2017	GlaS	Data Augmentation	Multi-level contextual FCN with auxiliary supervision and a contour-aware multi-task learning framework	F1 score 0.887 (GlaS)	Additional supervision of contour information for effective separation of touching glands	Contour-based separation is sensitive in highly distorted malignant gland region
Kainz et al. (2017)	2017	GlaS	Color Deconvolution, CLAHE	Two-stage CNN with object and separator networks	Dice 0.75	Explicit separation of touching glands using a dedicated Separator-Net	No validation on independent datasets
Xu et al. (2017)	2017	GlaS	Data Augmentation	Deep Multichannel Neural Network (MCNN) for gland instance segmentation	Dice 0.908 (Test A)	Fusion of region, boundary, and location information from multiple CNN	Evaluation is limited to the GlaS challenge dataset

in CRC. SL allowed the model to be trained locally and synchronized through decentralized weight sharing. The framework used ResNet-18 for feature extraction, followed by a 4-layer fully connected network. This approach achieved AUROC values of 0.80 for MSI and 0.77 for BRAF and outperformed locally trained models while also maintaining data security. Managing the quality of patches is also critical as splitting

Table 10
An overview and findings of studies employing machine learning techniques for colorectal cancer prediction.

Author	Year	Datasets	Feature Extraction /Feature Selection	Classifier	Results	Key contributions	Limitations
Swanson et al. (2025)	2025	TCGA	Texture/	RF XGBoost	AUC 0.85 (XGBoost)	Interpretability using feature importance	Handcrafted texture features may not capture complex histological patterns
Zhu et al. (2022)	2022	TCGA-CC-DX, TCGA-CC-KR, TCGA-STAD	Color, Texture, Morphology/	RF	AUC 0.78 (CC-KR)	Feature-level interpretability using RF importance and interaction analysis	Depends on predefined tumor detection pipeline

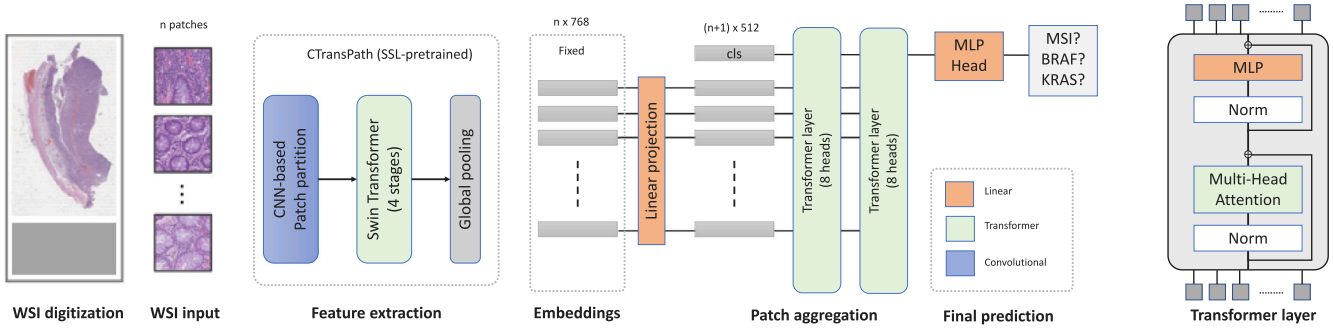


Fig. 15. Architecture of transformer model for biomarker prediction (Wagner et al., 2023).

WSI into patches can introduce noise and irrelevant information. To address this, Ke et al. (2020) designed a mislabel-aware module using a self-distillation model along with a patch auto-grading strategy within the MSI prediction pipeline. This multi-stage CNN framework achieved a maximum AUC of 0.879 on the TCGA-STAD dataset.

Undoubtedly, MIL-based CNN methods have shown strong potential for biomarker prediction in CRC histopathology images. However, CNNs struggle to capture the global contextual information essential for accurate biomarker prediction, and MIL frameworks typically treat patches independently, thereby failing to model long-range dependencies across the entire slide. Wagner et al. (2023) handled this issue by introducing transformers for both patch-level encoding and patch aggregation in biomarker prediction from CRC slides. The model, as shown in Fig. 15, achieved a sensitivity of approximately 0.9 on positive biopsy slides for MSI prediction and outperformed existing CNN and MIL-based SOTA methods. Singh et al. (2024) proposed KRASFormer, which improved the modeling of long-range dependencies and subtle textural patterns by using cross-covariance attention along the feature dimension for KRAS mutation prediction. This cross-covariance attention based KRASFormer outperformed existing CNNs and standard ViTs and yielded improved classification results. Transformer-based models have achieved SOTA results in biomarker prediction, but at the cost of high computational complexity. Cen et al. (2023) replaced expansive multihead self-attention modules with a simple sequential BiLSTM2D architecture and introduced DPSeq to predict key biomarkers in CRC tissue slides. DPSeq achieved excellent performance across multiple biomarker prediction tasks and outperformed multiple CNN and ViT baselines in both predictive accuracy and training time.

Architectural advances in the prognostic analysis of CRC using routine H&E-stained slides are rapidly evolving. Recent studies have found that integrating complementary modalities can further enhance prognostic performance. He et al. (2024) proposed a multimodal fusion model that integrated histopathological images with clinical data to improve MSI prediction accuracy. The model used a Multimodal Compact Bilinear (MCB) pooling module to fuse features from both data sources, where it achieved an AUC of 0.833 and outperformed unimodal methods. However, merely adding clinical metadata, such as stage information, may not capture the complex molecular drivers of MSI prediction. Tang et al. (2025) addressed this by integrating macroscopic radiological signatures with micro-level histopathological cues to predict MSI in CRC. The framework comprised two distinct CNNs for processing each modality, followed by an adaptive weighted concatenation. The key idea was to create a complementary feature set that improved prediction accuracy. On the external test set, this approach achieved an AUC of 0.905. Cheng et al. (2025) implemented a different approach by combining complementary staining slides to enhance performance and proposed a dual-modality transformer-based model that combined H&E and IHC slides for biomarker prediction in CRC tissue images. Integrating features from two staining methods enabled the framework to achieve an AUROC value near 0.97 for MSI prediction.

Summary We reviewed 21 CRC studies across MSI and MSS prediction, as well as other biomarkers such as KRAS, BRAF, and TMB prediction. Our analysis reveals that only 2 of these studies use conventional ML, while the remaining 19 use DL approaches. The conventional HCF-based method (Swanson et al., 2025; Zhu et al., 2022) demonstrated that texture and morphological features can capture important details for identifying MSI or MSS. Despite this, HCF-based methods typically fail to capture subtle and complex patterns within histology images. The field is gaining popularity following the findings of Kather et al. (2019b), who showed that DL models can predict MSI/MSS status directly from routine H&E-stained histology slides. Earlier studies (Konishi et al., 2023; Shimada et al., 2021) typically follow a two-stage pipeline where the first network detects tumor patches, followed by MSI/MSS classification. While these approaches achieved high accuracy, their understanding of the global context remains limited. For this, WSL-based techniques like MIL (Cao et al., 2020; Li et al.,

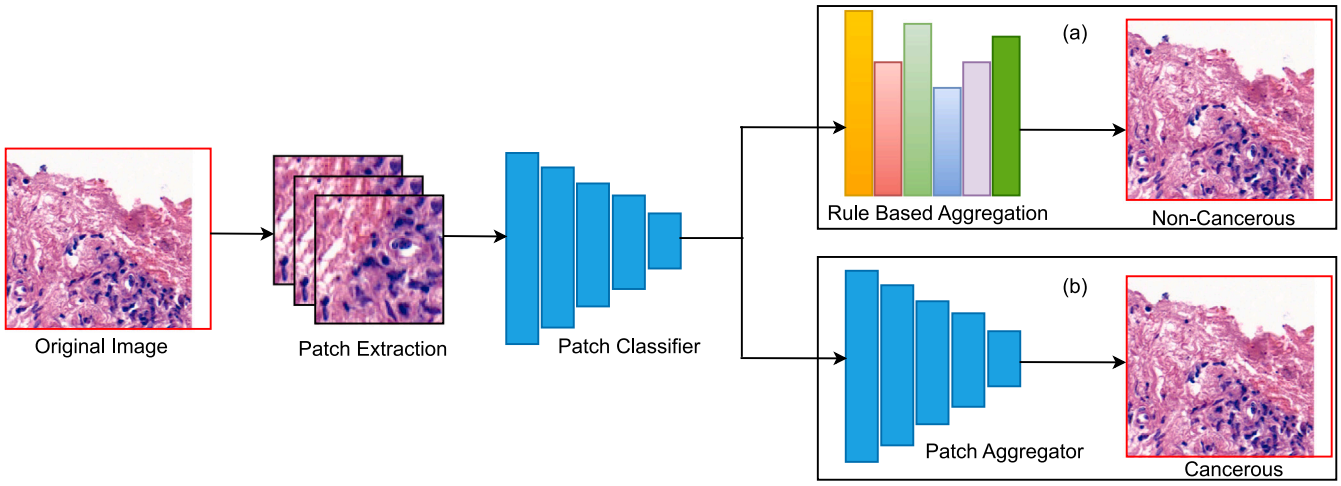


Fig. 16. Patch aggregation strategies for CRC slide-level classification. (a) Rule-based aggregation, where patch-level predictions are combined using predefined statistical rules. (b) Data-driven aggregation, where a learnable aggregator integrates patch-level information to produce final slide-level decisions.

2024a) have been widely explored for slide-level biomarker prediction. Although more advanced methods like Attention-MIL (Wang et al., 2025a) reduce the influence of irrelevant patches by assigning weights, they still treat patches as independent instances. In this regard, Wagner et al. (2023) proposed transformers for both patch and aggregation, and Singh et al. (2024) proposed KRASFormer to handle long-range dependencies. Further, DPSeq (Cen et al., 2023) replaced costly self-attention with BiLSTM, outperforming multiple CNN and ViT baselines in both time and accuracy. Beyond architectural enhancements, Multi-modal Fusion is drastically evolving this field. Histopathological images are combined with clinical metadata (He et al., 2024) and radiological signatures (Tang et al., 2025). More recently, Cheng et al. (2025) combined H&E and IHC slides using a dual-modality transformer and demonstrated that this integration enabled the model to learn complementary features, achieving higher accuracy in MSI prediction. The methods, contributions, and results of DL approaches for predicting genetic mutations in CRC histopathology are presented in Table 11.

5. Image post-processing

Post-processing techniques are a fundamental and concluding step in the CRC CPath pipeline. These are defined as a set of image processing methods applied immediately after the initial prediction generated by the ML or DL models, irrespective of the primary downstream task, such as classification, segmentation, detection, or prognosis. As histopathology images are complex and heterogeneous, the distinct output requirements of different downstream analysis tasks demand different post-processing methods to optimize outputs. This section outlines specific post-processing techniques used in the literature of CRC histopathology analysis.

5.1. Post-refinement in classification

A typical framework for CRC histopathology image classification begins with patch extraction and proceeds to patch-level classification. Based on the literature reviewed in Section 4.1, the process then involves an aggregation step to derive a final slide-level diagnosis. The patch aggregation method is summarized in the subsequent subsection.

5.1.1. Patch aggregation

One common practice in CRC histopathology is to classify the large WSIs into relevant classes. Since these WSIs are immense and incompatible with standard computational architectures, the common approach is to split the image into patches, classify them individually, and aggregate the results to obtain a final slide-level diagnosis. This step is critical because the more accurate the aggregation method is, the more accurate the final outcomes are. While modern end-to-end DL architectures, such as Transformers and MIL models, can learn patch aggregation within the network itself, this review focuses specifically on methods where aggregation is performed after the softmax layer. Studies in CRC histopathology classification typically adopt either rule-based aggregation methods or data-driven aggregation-based techniques for slide-level prediction, as illustrated in Fig. 16

Rule-based aggregation is commonly used in CRC slide-level classification, where patch-level predictions are combined through fixed statistical strategies such as majority voting and weighted averaging. This approach is shown in studies (Korbar et al., 2017) and Saeed et al. (2025). Korbar et al. initially classified CRC patches using ResNet-152 into five colorectal polyp classes and then used majority voting to assign slide-level labels based on the most frequent patch prediction. However, simple majority voting sometimes produced incorrect or unstable decisions when only a few misclassified patches dominated the vote. To make the results more robust, they additionally imposed rules based on the confidence scores of individual predictions and a minimum patch count threshold. Bilal et al. (2022) used the Average Top Half Aggregation Scheme, which calculates the average of patch-level scores above the median to derive the slide-level score. For the final slide label, they used a threshold selected through ROC curve analysis. Similarly, Saeed et al. (2025) used multiple patch aggregation methods, including Majority Voting and its weighted variant, as well as Probability Sum, to generate final slide labels. Upon analysis, they found that the Probability Sum method, which aggregates the scores of all contributing patches, was the most effective. Unlike score or voting based aggregation strategies, Kim et al. (2023) developed a deterministic approach to patch aggregation for multi-class classification in CRC. This deterministic approach used a structured hierarchy of percentage thresholds to intentionally prioritize high-risk classes in final slide predictions. Unlike slide-level classification, the study by Sirinukunwattana et al. (2016) introduces a specialized application of patch aggregation based on local neighborhood. They introduced the Neighboring Ensemble Predictor (NEP) that refines the classification of a single nucleus. Instead of relying on single-patch information, the NEP method considered spatial information from multiple neighboring patches and combined their probability scores using a weighted sum.

In contrast to rule-based aggregation methods, data-driven aggregation techniques use learning models that automatically determine the optimal way to fuse patch-level predictions into a final WSI classification. Unlike end-to-end aggregation, which integrates this step into the training of the feature extractor, this approach first computes patch results and then focuses on how they relate to one another. The key

Table 11
An overview and findings of studies employing deep learning techniques for colorectal cancer prediction.

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Cheng et al. (2025)	2025	CRC (MSI/MMRd), BRCA	Tissue Masking, Patch Extraction	Dual-modality transformer-based framework (DuoHistoNet)	AUC 0.97 (MSI)	Joint learning from H&E and IHC images	High computational cost due to dual-modality transformer and YOLO-based preprocessing
Wang et al. (2025a)	2025	TCGA	Patch Extraction	Cascaded two-stage MIL framework with attention-based aggregation for MSI and TMB prediction	ACC 0.88(MSI)	Interpretability using attention weights and Grad-CAM	Manual tumor region annotations
Tang et al. (2025)	2025	Multicenter CRC cohort, CECT + H&E pairs	Tissue Masking, Patch Extraction, Stain Normal-ization	Multimodal deep learning model using EfficientNet-b0 and ResNet-101 for MSI prediction	AUC 0.883	Multimodal MSI prediction model using radiology and histopathology	Requires manual tumor ROI annotation
Singh et al. (2024)	2024	TCGA-CRC-DX, Samples from NIB	Patch Extraction, Data Aug-mentation	Two-stage Vision Transformer framework	AUROC 0.691	Validated performance on an independent external cohort	Requires expert tumor ROI annotation
Zhang et al. (2024b)	2024	TCGA, Multicenter cohort	Patch Extraction, Stain Normal-ization	ResNet18-based patch-level CNN and slide-level aggregation	AUC 0.93 (multicenter)	Interpretable MSI prediction framework	Limited cross cohort generalizability
Li et al. (2024a)	2024	WSIs from 129 patients with left-sided CRC	Patch Extraction Stain Normal-ization, Data Augmenta-tion	CNNs with MIL-based WSI aggregation using PLHI and BoW	AUC 0.96	Pathology and clinical data integration	Lack of external validation
He et al. (2024)	2024	TCGA-CRC-DX	Patch Extraction, Stain Normal-ization	ResNet18 and RF based multimodal compact bilinear (MCB) fusion	AUC 0.833	Grad-CAM-based visualization for MSI-related regions	Single-center retrospective study with limited cohort size
Wagner et al. (2023)	2023	16 CRC cohorts	Tissue Masking, Patch Extraction, Stain Normal-ization	Transformer encoder with transformer-based patch aggregation	AUROC 0.97 (MSI)	Large-scale multi-cohort validation	Demands higher computational resources
Cen et al. (2023)	2023	TCGA-CRC-DX, MCO-CRC	Patch Extraction, Stain Normal-ization	DPSeq with BiLSTM2D-based sequencer for CRC biomarker prediction	AUROC 0.92 (TCGA-CRC-DX)	Replaced transformer self-attention with efficient BiLSTM2D	Performance depends on accurate tumor tile selection in the first stage
Konishi et al. (2023)	2023	TCGA	Patch Extraction, Stain Normal-ization	CNN-based feature extractor with genetic mutation classification models	AUC 0.721 (MSI)	Validated on large TCGA patient cohorts	Lower AUC observed for MSI and BRAF in CRC

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Table 11 (*continued*).

Author	Year	Datasets	Pre-processing	Methodology	Results	Key contributions	Limitations
Zhu et al. (2022)	2022	TCGA-CC-DX, TCGA-CC-KR, TCGA-STAD	Patch Extraction, Stain Normalization	ResNet-18-based CNN for MSI classification	AUC 0.84 (CC-KR)	Image-level interpretability using Grad-CAM heatmaps	Depends on predefined tumor detection pipeline
Liu et al. (2022)	2022	TCGA	Patch Extraction, Stain Normalization	DeepHE based on the ResNet50 for predicting TMB	ACC 0.850	Evaluated different CNN architectures for optimal TMB prediction	Manual tumor annotations
Saldanha et al. (2022)	2022	TCGA, Epi700, DACHS	Tissue Masking, Patch Extraction, Stain Normalization	Weakly supervised CNN with decentralized swarm learning	AUROC 0.83 (MSI)	Demonstration of swarm learning for CRC molecular prediction	Complex infrastructure required
Shimada et al. (2021)	2021	JP-CRC cohort, TCGA-CRC	Tissue Masking, Patch Extraction	Two-stage CNN pipeline using InceptionV3	AUC 0.934 (TMB-H)	External validation on an independent TCGA-CRC cohort	Performance is dependent on the accuracy of the first-stage detection
Bilal et al. (2021)	2021	TCGA-CRC-DX	Tissue Masking, Patch Extraction, Stain Normalization	Three-stage pipeline using ResNet with WSL (IDaRS)	AUROC 0.86(MSI)	Introduced IDaRS for iterative selection of the most predictive tiles from tumor regions	Lack of large-scale validation
Cao et al. (2020)	2020	TCGA, Asian-CRC	Patch Extraction	WSL framework based on ResNet-18 with PALHI and BoW pipeline for MSI prediction	AUC 0.884 (TCGA)	Introduced Ensemble Patch Likelihood Aggregation that combines PALHI and BoW predictions	Patch-based likelihoods may overlook the spatial relationships and global tissue architecture
Kather et al. (2020)	2020	TCGA, DACHS	Patch Extraction, Stain Normalization	ShuffleNet based tile classification with patient-level aggregation	AUC 0.89 (MSI)	Pan cancer prediction of genetic alterations directly from routine H&E histology	Requires expert tumor annotations
Kather et al. (2019b)	2019	TCGA, DACHS	Patch Extraction, Stain Normalization	ResNet-based MSI classification with patient-level aggregation	AUC 0.84 (DACHS)	Proved MSI can be predicted directly from H&E histology without genetic tests	Larger cohorts needed to learn rare morphological variants

is that the overall classification of WSIs is determined by the spatial distribution of the various classes rather than relying solely on the most frequently represented class. Studies in CRC histopathology classification typically explore a Recurrent Neural Network (RNN)-based strategy for data-driven aggregation to obtain a slide-level prediction. For instance, Iizuka et al. (2020) trained an RNN model for patch aggregation that captured the global tissue structural context and enabled accurate slide-level diagnosis. This study demonstrated that max pooling strategies are highly error-prone, as a single false-positive patch can lead to an incorrect slide-level prediction.

5.2. Post-detection refinement

Disease detection involves the identification and localization of objects ranging from macrostructures such as tumor regions to microstructures such as nuclei. Despite the advancement in automated detection methods, it is often the case that the raw output contains noise, artifacts, repetitions, and ambiguous structures. This is due to the complexity and heterogeneity of large histopathology images, which contain irregularly shaped, often touching objects that are difficult to localize and identify. Therefore, post-processing refinement methods are essential to remove these anomalies from the raw masks and refine the spatial precision. Any set of methods applied after the primary localization of potential objects (macro or micro) as determined by the network, and used to further refine, correct, and finalize the raw output data to provide the final result, is considered a post-processing technique.

5.2.1. Probability map refinement

It is a simple post-processing step that refines the raw probability map to obtain the final detection mask. Different studies have adopted various methodologies to refine the PMap. For instance, Sirinukunwattana et al. (2016) and Tofighi et al. (2019) both applied similar

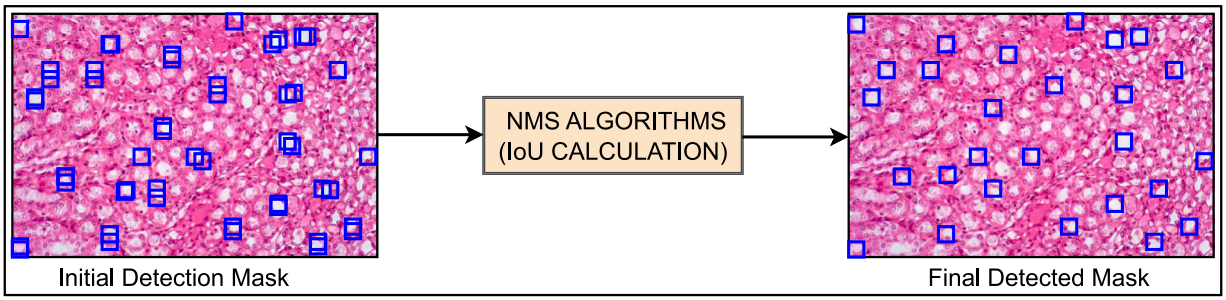


Fig. 17. Illustration of NMS-based post-processing as an example of nuclei detection, where overlapping detection candidates are filtered using IoU-based threshold to obtain the refined detected nuclei mask.

post-processing logic. They first computed local maxima using the PMap to detect nucleus centers. Finally, a threshold was applied, and only maxima whose values exceeded it were retained to avoid false positives. Similarly, Li et al. (2020) applied a simple threshold on the PMap generated by the detection network and retained only values above the threshold to obtain the final nuclei detection coordinates. In contrast to simple thresholding, Höfener et al. (2018) applied a median filter followed by Gaussian smoothing to the PMap to address noise and outliers. The final detection mask was generated by applying local maxima and thresholding. The study emphasized that such simple post-processing steps had a greater impact on nuclei detection performance than merely increasing the depth of the network. Xu et al. (2020) initially averaged the overlapping patch predictions to obtain the PMap and then applied a threshold to highlight the tumor regions within the WSI.

5.2.2. Non-maximum suppression (NMS) based refinement

In detection tasks, Non-Maximum Suppression (NMS) is an essential post-processing step that eliminates redundant bounding boxes in dense cellular regions, as shown in Fig. 17. With the same aim, Sun et al. (2021) set an intersection over union (IoU) threshold of 0.3 to identify redundant bounding boxes and applied NMS to retain only the highest-confidence detections. Beyond simple thresholding, NMS can be integrated into dual-channel fusion frameworks. For instance, Xie et al. (2024) first combined the coarse and fine detection branch PMaps using element-wise multiplication with prior positional information to obtain the final nuclei PMap and then applied NMS to produce clean, non-overlapping nuclei detections even in complex tissue contexts.

5.3. Post-segmentation refinement

Despite the use of complex architectures, segmentation of histological structures remains challenging due to several factors. First, it is often the case that large CRC histopathology images contain large swathes of normal tissue interspersed with smaller abnormal tissues. Second, morphological overlap among histological components in CRC slides also makes it difficult for the model to achieve precise segmentation. Therefore, in many cases, raw segmentation outputs often contain noise and irregular artifacts. To address these issues, studies have explored different post-processing strategies, ranging from simple morphological operations to advanced signal reconstruction. These methods are applied to the initial segmentation map and help refine and optimize the final segmentation output.

5.3.1. Morphological refinement

Given the structural complexity of histopathology images, morphological operations (MO) are frequently adopted as a post-processing step to clean and optimize raw segmentation masks. MOs such as erosion, dilation, opening, and closing provide a robust baseline for cleaning segmentation masks across a variety of CRC segmentation studies. For instance, Rastogi et al. (2022) used MO as a post-refinement step to optimize the raw predicted masks. Study (Rastogi et al., 2022) first applied a threshold of 0.7 to obtain binary masks and thereafter applied the MO methods to fill holes and remove noise objects smaller than 500 pixels. Similarly, Graham et al. (2019a), and Dabass et al. (2021) used a smaller threshold of 0.5 and applied an opening operation with a disk filter with radius value five to smooth glandular boundaries. For ease of clinical interpretation, the study (Dabass et al., 2021) also visualized the results by overlaying the predicted segmentation masks onto the original tissue images. In contrast, Kainz et al. (2017) used only blob-removal and hole-filling techniques to optimize binary masks due to the computational complexity of extensive morphological processing.

5.3.2. Instance separation

Overlapping and touching nuclei and glands often make it difficult for models to clearly delineate the boundaries. For this, Chen et al. (2017) addressed this by predicting gland objects and contour masks simultaneously. With the predicted probability maps, touching structures were separated by fusing the predicted object and contour masks. They further separated connected components by removing isolated regions and smoothing boundaries. At the end, a unique label was assigned to each connected component and represents an each individual gland. Similarly, Binder et al. (2019) used a single-input–multiple-output approach to predict both gland and contour probability maps. Then they multiplied the dilated contour masks with the gland predictions to split touching instances and applied dilation one more time to recover the original shape. At the end, a series of refinement operations, including threshold-based removal, hole filling, and a Conditional Random Field (CRF), were applied to sharpen boundaries and optimize the final segmentation. Xu et al. (2017) addressed the coalescence problem using three different channels: foreground segmentation, edge detection, and object detection channels, and combined them to generate a precise final mask.

5.3.3. Reconstruction and aggregation

Unlike mask refinement and spatial separation, some studies employ advanced reconstruction techniques to achieve final optimized segmentation masks. For instance, Kim et al. (2021) worked with the compressed domain, which decomposes histopathology images into wavelet subbands. Since the model predicts segmentation maps in compressed wavelet sub-bands, they used the Wavelet Weighted Ensemble (WWE) approach as a post-processing step. The study uses the Inverse Discrete Wavelet Transform (IDWT) as a reconstruction step to fuse multi-scale predictions and produce a final optimized mask that preserves fine details. To reduce variance and improve generalization, Khened et al. (2021) used an ensemble aggregation strategy. In this, the final segmentation mask was generated by averaging the posterior PMaps from multiple FCN architectures. The study also computes average prediction probabilities across overlapping regions to remove edge artifacts introduced by the patch-based segmentation approach, and generates a smooth tumor probability heatmap and uncertainty map for confidence analysis.

5.4. Post-prediction refinement

Unlike general tumor tissue classification, prognostic and biomarker analysis for CRC using routine histopathology are formulated as image-level tasks. A common computational approach therefore involves a multi-stage pipeline that starts with WSI tessellation, followed by patch-level feature extraction, and concludes with patch aggregation to achieve a unified slide-level prognostic score. This review assumes that any operation applied immediately after initial patch-level prediction is a post-prediction refinement. It includes techniques for patch aggregation to combine results for slide level diagnosis. These techniques are outlined in the subsequent subsections.

5.4.1. Patch aggregation

Patch aggregation is the most common post-processing technique for predicting CRC biomarkers from histopathology images. There are two primary reasons for this. First, biomarker prediction is essentially an image-level task. Second, there is an inherent computational bottleneck in current architectures. This essential but complementary demand entails that patch aggregation becomes mandatory to synthesize local patch-level predictions into a final, unified patient-level diagnosis. Aggregation strategies for CRC biomarker prediction from histopathology images can be broadly categorized into rule-based and data-driven methods (Fig. 16).

Rule-based aggregation is a simple yet effective deterministic technique for predicting CRC biomarkers. It uses a straightforward mathematical rule to combine individual patch-level scores into a final slide-level prediction. Commonly adopted approaches in CRC studies range from simple statistical pooling to threshold-based strategies. For instance, Kather et al. (2019b) introduced the MSIness score to obtain patient-level MSI prediction. The MSIness score was defined as the fraction of tumor tiles predicted as MSI-positive. Spatial heatmaps were also generated and smoothed using interpolation to illustrate the influence of different tissue regions on MSI status. Later, within a multi-biomarker prediction framework, Kather et al. (2020) derived slide-level predictions from the proportion of all tumor-predicted tiles. Instead of considering all tumor tiles, Cen et al. (2023) randomly sampled up to 500 tumor tiles per slide, applied simple mean pooling to derive final patient-level biomarker scores, and then applied a classification threshold to obtain MSI and BRAF predictions from CRC histology. However, random sampling of tumor tiles may sometimes overlook important diagnostic regions. Thus, Singh et al. (2024) designed a quality screening stage using ViT to filter out both non-tumor and low-quality tissue regions prior to slide-level inference. Finally, patient-level KRAS mutation scores were then obtained by simple averaging of tumor patch-level probabilities produced by the KRASFormer framework. In the multimodal framework, Tang et al. (2025) also computed simple averaging of tile-level probabilities to obtain WSI-level MSI predictions. Majority voting has also been widely adopted. Zhu et al. (2022) and Liu et al. (2022) applied the majority vote scheme for slide-level prediction. The study (Zhu et al., 2022) implemented two distinct strategies for slide-level MSI prediction based on the model. The feature-based RF model computed patient-level MSI scores by averaging tile-level predictions, whereas the ResNet model applied a 50% majority threshold to tile-level predictions to determine the final slide-level MSI status. While the feature-based model provided greater interpretability, the ResNet-based model achieved better patient-level performance. The study (Liu et al., 2022) also used a fixed-majority voting scheme and classified a patient as TMB-high when more than 50% of tumor tiles were predicted to be positive. Shimada et al. (2021) used an adaptive approach and applied an ROC-derived threshold to the proportion of positively predicted tiles to obtain slide-level TMB predictions. Bilal et al. (2021) compared different rule-based schemes within the IDaRS framework and found average tile scoring as the best-performing aggregation strategy for slide-level prediction of molecular pathway and mutation status in CRC histopathology.

Unlike rule-based aggregation strategies that aggregate patch-level predictions based on measures of central tendency, several studies on CRC biomarker prediction have explored data-driven methods that adaptively identify or combine tiles based on their relevance to the slide-level prediction. Cao et al. (2020) adopted a dual-aggregation pipeline within an MIL-based MSI prediction framework using BoW and PALHI. The BoW pipeline treated patch-level MSI likelihoods as visual words, whereas PALHI summarized them using histogram representations. The final patient-level MSI prediction was obtained by ensembling these features using XGBoost and NB classifiers. The study also used correlation analysis to improve clinical interpretability, demonstrating that this approach effectively aligned pathological signatures with genomic and transcriptomic features. Using a similar approach, Li et al. (2024a) developed a prediction framework using BoW and PALHI to identify KRAS, NRAS, and BRAF mutations in left-sided CRC. Nevertheless, this study employed XGBoost to fuse BoW and PALHI features into a final pathology signature for WSI-level genotype prediction. In spite of higher accuracy, this approach depends on patch-level likelihood scores and cannot explicitly model global contextual relationships across tissue regions. Wang et al. (2025a) addressed this by introducing an attention mechanism guided by Grad-CAM to learn patch importance scores and then employed integration modes such as AFS (All Feature Set) and MaxMinS (Maximum–Minimum Selection) to obtain the final WSI-level score. AFS achieved an AUC of 90.9% for MSI prediction, while MaxMinS achieved an average AUC of 88.18% for TMB status.

6. Current research landscape and synthesis

An analysis of the current research for CRC CPath is somewhat fragmented. A disproportionate number of studies focus on the analysis phase of CRC histopathology. A potential source of skepticism is dependence on publicly available datasets. Most publicly available CRC histopathology datasets are finely curated, pre-patched, and heavily pre-processed due to the unavailability and high cost of expert annotations. A majority of research, therefore, relies on these datasets. This has resulted in limited exploration of pre-processing and post-processing techniques. In pre-processing, challenges such as patch extraction, handling of stain variability, and tissue artifacts are often overlooked. Similarly, in post-processing, essential steps like patch aggregation for slide-level diagnosis and prognosis, output refinement, and optimization remain underexplored. In fact, this creates a bottleneck in a real-time clinical setting.

Ideally, a common practice to achieve a clinical level of trust is to train a model on a benchmark dataset and validate it on an external dataset. Although this generalization is essential, it is unfortunately confined to either internal cross-validation or external benchmark validation. Even advanced architectures, including transformer-based models, might struggle to generalize to real-time clinical data. Notwithstanding these limitations, our review also highlights several pioneering studies, such as Fu et al. (2025), Saeed et al. (2025), Pina et al. (2025), Zhang et al. (2024a), Dabass et al. (2021), Shimada et al. (2021), Singh et al. (2024), that successfully address various aspects of preprocessing, post-processing, and rigorous validation. In conclusion, it is anticipated that future research will shift its focus to cover the entire pipeline and to validate with external real-time data and pathologist supervision, rather than relying solely on isolated performance metrics on internal or curated datasets.

7. Open issues and future trends

As discussed throughout Section 4, the field of CPath has embarked on rapid advancements in CRC histopathology analysis. Both ML and DL approaches have been used, but DL approaches have shown promising results. Nevertheless, it is imperative to acknowledge that the integration of these models into real-world clinical workflows is

still at an incipient stage. This is due to skepticism arising at different stages of the analysis pipeline. Interestingly, it is not because of the failure of individual algorithmic power, but the absence of a unified computational framework. Therefore, we deliberately identified various pervasive challenges and delineated viable strategies to address them within the three-phase pipeline. Looking ahead, we propose several key areas for future research aimed at maturing these computational techniques for real-world clinical use.

7.1. Image pre-processing: Open issues and future directions

Image preprocessing is the first yet foundational step in CPath for CRC histopathology image analysis. This stage implicitly comprises a series of tasks with the same aim to prepare raw images so elegantly that downstream analysis can be performed effectively. However, this entails a variety of challenges, ranging from high-dimensional data management to stain inconsistency. The subsequent sections elucidate the open challenges existing in these steps, followed by potential solutions and future directions.

7.1.1. Stain inconsistency

Inter- and intra-dataset stain variability is an ongoing challenge in CRC histopathology analysis. The critical nature of this problem was initially highlighted in the study by Ciompi et al. (2017), who reported a significant impact of stain variability on the performance of CRC histopathology classification models. Thereafter, stain normalization has become a widely adopted strategy to harmonize these variations. Statistical methods such as Reinhard, Macenko, and Vahadane normalization are common in histopathology, but they are largely dependent on a reference template. In an effort to mitigate these issues, researchers have explored the use of a DL model to develop stain-agnostic models based on unpaired image-to-image translation. For instance, de Bel et al. (2021) used a CycleGAN with residual learning to increase the Dice coefficient by 4%, while Kausar et al. (2021) introduced SA-GAN, improving downstream classification by 6.9%. Nevertheless, training such GAN-based stain normalization models requires a copious reservoir of high-quality data. Moreover, these models are prone to introducing hallucination artifacts that compromise morphological fidelity and, as crucial technical pitfalls, break the chain of trust required for real-world clinical deployment.

Future Directions: Future research is shifting away from classical stain normalization toward stain-agnostic learning-based methods. In particular, residual learning-based strategies enable the model to learn only residual mappings rather than full image reconstruction. Beyond image translation, virtual staining methods such as VM-GAN (Wang et al., 2025b) and confusion-GAN (Li et al., 2024b) have emerged as a complementary direction that attempts to computationally generate standardized stain representations. In light of the task-specific nature of existing approaches and the limited availability of paired data in clinical settings, future work should explore foundation models for virtual staining. In this endeavor, a single model pre-trained on diverse stain transformation tasks can be easily fine-tuned for specific staining protocols.

7.1.2. Tissue artifacts

The introduction of artifacts into histopathology images is a common problem at different stages of the slide preparation pipeline. In the context of CRC histopathology, recent studies have been motivated by the findings of Wright et al. (2020), who found a negative impact of slide-level quality issues on model performance. In this endeavor, automated frameworks have been developed to detect and categorize artifacts in CRC histopathology. For instance, Hossain et al. (2023) focused on explicit segmentation of tissue artifacts, while Foucart et al. (2018) aimed to restrict analysis to artifact-free tissue regions. Nevertheless, excluding either artifact-containing regions or entire slides can reduce effective training sample size and entail a trade-off between data quality and model generalization. Artifact detection prior to analysis is a growing area of research in this domain. However, studies are stuck in a constant trade-off between removing artifacts and retaining valuable training data. This gap persists not only because of limited detection capabilities but also because of current research practices. The widespread use of pre-curated benchmark datasets and improvements in WSI scanners hide the true severity of artifacts. Nevertheless, exclusion of artifact detection from the analysis pipeline can impair the model's generalization across real clinical settings.

Future Directions: Future studies must focus on artifact-aware preprocessing methods that suppress irregularities without losing informative tissue. In this regard, GAN-based inpainting should be explored to restore artifact-free regions while maintaining morphological integrity. Adoption of FSL-based models can be a promising path to identify rare artifacts with minimal labeled data. Finally, integrating Real-time Quality Control (QC) directly into WSI scanners would allow artifacts to be identified during digitization, enabling immediate rescanning and ensuring only high-quality data enters the diagnostic pipeline.

7.1.3. Gigapixel image complexity

Histopathology images are multi-gigapixel in nature and contain a large proportion of redundant, non-informative, or background regions. Therefore, patch-based training is the method of choice in many studies. However, selecting informative patches is a strict requirement, as indiscriminate patch extraction not only wastes computational resources but also introduces learning biases. Grid and FPS sampling are frequently used in the CRC literature for patch selection. However, it is worth noting that, although FPS offers greater diversity than Grid, both techniques rest on a mathematical foundation and thereby ignore the biological content of the tissue. In this endeavor, recent research in CRC histopathology has shifted toward content-aware selection using Attention-inspired algorithms and Active Learning. Although these methods consider diagnostic relevance, selection bias may be introduced, whereby low-density yet clinically important features can be overlooked.

Future Directions: Future work should explore multi-magnification adaptive sampling, a form of hierarchical sampling. This allows models to dynamically adjust the level of detail, either based on magnification or sampling density, in specific regions to optimize efficiency and accuracy. Beyond intelligent patch sampling techniques, another important future direction is structure-aware sampling. This would allow the model to reduce edge effects and ensure that critical biological structures, such as glands or tumor nests, are not fragmented during patch extraction. Exploring multi-class semantic segmentation (Öztürk and Akdemir, 2019) could help differentiate between tissue regions with and without cells. This could enhance the selection of diagnostic relevant patches.

7.1.4. Data scarcity

Data scarcity is the major bottleneck in CRC CPath. High-quality, expert-annotated data is imperative for any data-driven analysis, particularly for modern DL architectures. Current countermeasures to handle data scarcity involve augmenting the data, either by transforming existing samples or by generating synthetic ones. Both strategies have been extensively explored in the realm of CRC histopathology analysis. Recent CRC literature has shifted towards GAN-based data augmentation. For instance, SAFRON by Deshpande et al. (2022) generates giant CRC tissues while preserving morphological features. Similarly, Wei et al. (2019) provide a viable solution to rare-class detection. It is important to point out that, although GANs offer an efficacious solution to mitigate the issue of data scarcity and class imbalance in CRC histopathology, adversarial training often induces mode collapse and synthetic bias, which restricts the representation of real-world clinical variability (Salimans et al., 2016).

Hallucination artifacts, unstable training, and a disproportionate focus on visual realism remain open issues in the medical domain, particularly in histopathology, as these factors adversely affect the entire diagnostic pipeline.

Future Directions: One viable future direction is to incorporate Diffusion Models for the generation of artificial CRC tissue. Unlike GANs, Diffusion Models use a maximum-likelihood framework to diversify training samples and closely mirror the data distribution. Several recent studies have successfully applied Diffusion models, such as Histodiffusion (Xu et al., 2024) and PathLDM (Yellapragada et al., 2024), in histopathology and achieved SOTA results, inspiring future research in the CRC domain to address persistent data scarcity and morphological diversity. Notably, PathLDM achieves high data efficiency by operating in a compressed latent space.

7.2. Image analysis: Open issues and future directions

Image analysis is the most crucial part of the CPath pipeline for CRC, where raw pixels are transformed into clinical insights through a series of computational objectives: classification, segmentation, and predictive modeling. Nevertheless, this phase is near saturation with current research, yet several open issues persist that affect both ML- and DL-based image analysis approaches.

7.2.1. Image classification

Patch-Centric Learning Paradigm and the Benchmark-Clinical Gap: Patch-based learning is a forced paradigm in CRC histopathology classification due to the inherent architectural and computational constraints of both ML pipelines and modern DL architectures. Large WSI repositories, such as TCGA, provide slide-level supervision, but patch-level annotations for classification are limited. Due to the ease of access to publicly labeled benchmarks, the vast majority of current research is based on isolated patches without any WSI context. Nonetheless, this approach simplifies computational load but introduces a relative gap between experimental success and clinical utility. Unfortunately, this represents a constraint rather than a limitation of any specific classification architecture. This phenomenon reflects an accuracy paradox: studies such as Chhillar and Singh (2024) and Babu et al. (2021) report high accuracy on patch-level data, while the slide-level diagnostic context required in clinical practice is outside the evaluation scope.

Limited Global Context: Understanding the spatial relationships between tissue components, for example, how malignant glands infiltrate the surrounding stroma, is de facto important for the clinical diagnosis of CRC. However, in an architectural context, learning spatial relationships is difficult due to large computational demand. Current ML-based studies, such as Tripathi et al. (2023), Junaid et al. (2025), and Sharkas and Attallah (2024), achieve optimal classification results in CRC tissue classification. However, it is imperative to understand that these methods are confined to patch-level classification. Popular DL models, such as CNNs, achieve SOTA performance in patch-based CRC tissue classification. Due to their limited receptive fields, the adaptability of CNNs has remained limited in the literature (Zeid et al., 2021). The current literature has diversified towards ConvNeXt architectures (Yengec-Tasdemir et al., 2023) and transformers (Tanveer et al., 2024; Fu et al., 2025) to model long-range dependencies, and has witnessed progress in the CRC domain. Nevertheless, training transformers required tons of labeled data at the cost of quadratic complexity. While efforts have been made to address this, such as using BiLSTM within ViTs (Choudhary et al., 2025) to capture context, these approaches have confined global understanding to the patch level only. In this context, the trade-off between computational feasibility and effective modeling of spatial tissue relationships remains an open challenge.

Clinical Interpretability Gap in Classification Decisions: Understanding the root cause of a classification decision is of utmost importance in clinical CRC histopathology. Although both ML and DL models have demonstrated strong performance in CRC classification, each suffers from distinct limitations in explainability. Interpretable ML models struggle with high-dimensional histopathology data and provide only statistical importance. On the other hand, DL models operate largely as black boxes. Recent research has witnessed the integration of Explainable AI (XAI) techniques in CRC histopathology classification as a viable countermeasure to bolster DL model transparency. Both attribution-based methods, such as Grad-CAM (Yengec-Tasdemir et al., 2023), LIME (Merabet et al., 2025), and Occlusion Sensitivity (Dabass et al., 2022), and non-attribution-based techniques, particularly attention-based visualization (Mir et al., 2025), have been explored in this domain. Despite this progress, the lack of a unified conceptual framework for systematically evaluating explainability methods leaves the trade-off between explanation granularity and human interpretability unresolved.

Annotation Bottleneck and Learning Bias: Data is indeed not the only requirement for both ML and DL models, it must be of high quality and rigorously curated to ensure reliable performance. In CRC histopathology classification, limited and inconsistent supervision directly affects the model's learning behavior. As a result, ML approaches are sensitive to annotation noise, while DL models can learn biased representations when trained under limited supervision. This restricts generalization beyond curated training distributions. Notably, coarsely labeled images are far more common than precise patch-level annotations in this domain. Due to the patch-centric learning paradigm, recent studies have adopted TL and WSL paradigms. Nonetheless, WSL itself requires an extensive volume of training data for robust generalization. In this direction, SSL and FSL are emerging as vital countermeasures in this domain that enable generalization under no or minimal supervision, respectively.

Future Directions: Future research endeavors in CRC classification should move beyond patch-level toward WSI-level classification using transformer-based architectures and explore SSMs to efficiently model long-range dependencies in high-resolution slides. Additionally, data-efficient methods like FSL and SSL should be explored to better identify rare tissue morphologies where annotated data is scarce. To achieve trust and faithfulness, future studies should conduct rigorous external validation across diverse cohorts. Finally, hybridizing the interpretability of HCFs with the representational power of deep learning will be advantageous in bridging the gap between performance and clinical interpretability.

7.2.2. Image detection

High False Positives CRC tissue morphology is complex and heterogeneous. This becomes more problematic when benign glands, inflammatory cells, and stromal components resemble malignant structures. Visual similarity affects both conventional HCF-based localization approaches and modern DL-based detection models, leading to frequent confusion between non-malignant and malignant objects. This phenomenon results in elevated false-positive rates and lower specificity (Ho et al., 2022; Dharshini et al., 2025; Sun et al., 2021). The core challenge in current research lies in the trade-off between specificity and sensitivity. While high sensitivity is prioritized in current practice, over-detection of objects in CRC histopathology remains an open challenge and increases unnecessary alerts and workload.

Dense and Overlapping Cellular Region: The biological structures in CRC histopathology, such as glands and nuclei, are typically crowded and overlapped, making individual object identification a major challenge for ML and DL models. Traditional superpixel-based approaches often lack the boundary precision needed to separate individual glands within a cluster. Whereas DL methods, such as SC-CNN (Sirinukunwattana et al., 2016) and POI-Net (Li et al., 2020), utilize spatial constraints but still struggle to identify individual nuclei in overlapping regions due to irregular shapes and clustered cellular arrangements. A

recent study by Javed et al. (2022) used multi-feature fusion to mitigate the overlapping and cluster nuclei detection and achieve a high F1-score on the CRCHistoPhenotypes dataset. However, the method still struggles with overlapping nuclei, resulting in false positives and false negatives. Under and over-detection directly affect the quantitative analysis required for accurate CRC grading, thus remaining a critical open technical bottleneck.

Clinical Interpretability Gap in Detection Decisions: Pathologists annotate tissue structures guided by diagnostic and biological relevance, such as using point-based centroids for nuclei (CRCHistoPhenotypes) and region-level masks for glandular structures (GlaS, CRAG). For clarity, this review uses nuclei detection as a representative case. Earlier nuclear detection methods relied on probability maps and local maxima localization to identify these centroids. Although this approach is naturally aligned with pathologists' point annotations, it can be sensitive to dense cellular regions (Sirinukunwattana et al., 2016; Pons et al., 2025). Conversely, modern DL detectors represent detected objects using bounding boxes. It is worth noting that, although it provides more explicit spatial representation, it introduces a localization ambiguity because box boundaries are estimated from single-pixel ground truth. The mismatch between model architectures and clinical annotation practices makes it hard for clinicians to interpret spatial reasoning, particularly in dense and overlapping regions.

Future Directions: Future studies should embrace advanced detectors, such as DEtection TRansformer (DETR), for CRC histopathology object detection. The key advantage over other detectors is that these models view object detection as a direct set prediction problem using bipartite matching and eliminate the need for NMS and anchor generation. Moreover, through its powerful global modeling using self-attention, DETR can build reasoning of neighboring spatial relations during detection. By incorporating this end-to-end transformer architecture into CRC histopathology, it could achieve more robust and generalizable detection results without the need for extensive post-processing. Another important future direction in this domain could be the development of uncertainty-aware detection models that can provide a confidence score for each prediction. This could not only assist in reviewing ambiguous detections but also help build trust for deployment in clinical settings.

7.2.3. Image segmentation

Annotation Exhaustion: A major problematic aspect shared by all supervised segmentation methods is their heavy reliance on annotations. Achieving fine, pixel-level annotations in CRC histopathology incurs heavy operational overhead. While traditional unsupervised methods like Otsu thresholding and K-means avoid this burden, their utility is now confined to auxiliary tasks like coarse tissue masking due to the complex morphology of CRC tissues. In contrast, a range of DL models, particularly but not limited to U-Net (Sengupta et al., 2025), Attention U-Net (Dabass et al., 2021), Hybrid U-Nets (Zhao et al., 2023), and Swin Transformers (Arslan et al., 2025a), have shown SOTA performance but are often restricted to carefully curated benchmark datasets. Yet, training complex DL models to achieve strong generalization capabilities simultaneously demands large volumes of fine-grained pixel-level annotations, unfortunately imposing a trade-off between segmentation granularity and annotation exhaustion. As a consequence, it constrains the development of large annotated datasets and can introduce subjective biases.

Poor Generalization to Complex Tissue Morphology: Due to constraints on acquiring CRC histopathology segmentation datasets, most SOTA segmentation models are trained on well-annotated benchmark datasets (GlaS, CRAG, and EBHI-Seg) that generally model well-defined glandular structures. Nevertheless, the practical applicability of these results in real-world clinical settings remains uncertain due to subtle morphological variability present in clinical datasets. Notably, studies (Qaiser et al., 2019; Hou et al., 2019) have also noted that their models can easily overfit to fine-grained patterns and exhibit noticeable performance drops under domain mismatch. Complex tissue morphology and domain-mismatch issues are still open challenges that hinder clinical trust in real CRC segmentation pipelines.

Boundary Ambiguity: Preserving anatomical integrity in CRC histop athology segmentation is often difficult to achieve due to ambiguous boundaries, particularly in malignant glands and overlapping nuclei. The majority of recent studies primarily emphasize achieving high Dice scores and therefore rely on standard loss functions, typically a hybrid of cross-entropy and Dice loss. Nevertheless, such loss functions focus on region overlap while ignoring boundary precision and topological consistency, thereby failing to delineate touching objects and over-segmentation. A recent study by Chen et al. (2017) mitigated this issue by introducing an explicit contour-aware loss alongside standard loss functions to provide boundary supervision in a gland segmentation framework. However, it is important to note that the mere incorporation of contour information may not be sufficient to capture the complete spectrum of boundary supervision in all cases (Graham et al., 2019a). Several inspiring studies (Chen et al., 2017; Xu et al., 2017) further addressed this issue by explicitly incorporating specialized modules within their frameworks to enhance boundary supervision. Collectively, due to Dice-driven formulation and ambiguous tumor boundaries, preserving anatomically reliable boundaries across diverse morphological and domain conditions is yet unresolved.

Clinical Interpretability Gap in Boundary Delineation: Achieving clinically meaningful interpretability with current methods is still a critical challenge due to the lack of a clear rationale behind model predictions. The current method of choice in many CRC histopathology segmentation methods uses heatmaps and uncertainty estimation maps to provide post-hoc insights. Nevertheless, it is worth noting that such visualizations provide only coarse localization and fail to explain delineation decisions, particularly at fragmented and hazy tissue boundaries. Although these techniques visually assist pathologists in identifying ROIs, their mathematical foundations cannot convey the corresponding biological meaning in a verbal or diagnostic setting. Addressing this foundational mismatch endures as an open hurdle for the CPath community.

Future Directions: Data-efficient approaches, like FSL and SSL, should be integrated into the tissue segmentation pipeline. This could reduce annotation dependency and preserve fine-grained structural details under limited supervision. Additionally, beyond dice-centric model training, future studies should explore boundary-aware, topology-preserving, and structure-guided loss functions to reduce boundary ambiguity and distortion, particularly in malignant and overlapping regions. Last but not least, pathological FMs like SAM have shown impressive zero-shot and few-shot segmentation capabilities. Finally yet importantly, pathological FMs like SAM have shown impressive zero-shot and few-shot segmentation capabilities. Exploring these large-scale models in CRC histopathology segmentation could be highly auspicious for future research.

7.2.4. Image prediction

Spatial Context and the Tumor Microenvironment (TME): Understanding the tumor microenvironment (TME) is a de facto requirement for predicting CRC biomarkers directly from routine histopathology images. However, this is an image-level task that requires modeling long-range dependencies, which fundamentally conflicts with the current patch-centric analysis paradigm. Starting with two-stage CNN-based pipelines (Kather et al., 2019b), most studies in this domain have relied on MIL with feature pooling (Cao et al., 2020) and attention-based pooling (Wang et al., 2025a). Yet, while CNN-based and early MIL approaches are confined to modeling the long-range dependencies required for TME modeling, attention-based MIL treats patches independently, thereby limiting its contextual modeling capacity. The current

SOTA has witnessed a shift toward Transformer-based WSL (Wagner et al., 2023) and ViT architectures (Singh et al., 2024), which are known for capturing long-range dependencies. It is imperative to note that training transformers with WSIs demands extensive data and incurs quadratic complexity. The trade-off between contextual modeling and computational cost is still an open challenge in this domain.

Clinical Interpretability Gap in Prediction Decisions: Clinical interpretability is a more persistent issue in prognostic modeling than in diagnostic CRC histopathology. In diagnostic workflows, a pathologist can visually validate a model's decision by cross-examining identified lesions, whereas prognostic modeling is more formidable, as the lack of direct visual correlates makes the internal decision-making process opaque. Recent studies (Swanson et al., 2025; Zhu et al., 2022) have shown that HCFs can detect interpretable patterns in MSI and MSS, but their scope is limited by complex and invisible biomarker signatures. The current method of choice is dominated by DL models that achieve superior performance but operate as a black box. Recent studies mitigate this interpretability gap by employing techniques such as Grad-CAM heatmaps and Attention-MIL (Zhu et al., 2022; Wang et al., 2025a). Nevertheless, their faithfulness and clinical trust in prognostic modeling remain an open challenge, as it is currently not possible to validate solely by visual inspection (Van der Laak et al., 2021).

Weak Supervision: A majority of recent CRC biomarker prediction is formulated as an image-level task and is predominantly addressed using WSL frameworks. This can reduces the need for manual annotations but entails its own set of challenges. One major issue is that biomarker signals are not uniformly present within the heterogeneous tumor regions (Kather et al., 2019b). Current methods, ranging from MIL with max pooling to attention-based MIL and IDaRS, are widely adopted in CRC biomarker prediction pipelines. While traditional MIL selects a single tile to reach the final MSI or MSS decision and is highly sensitive to noise, attention-based MIL and IDaRS mitigate this by weighting or ranking top tiles (Wang et al., 2025a; Bilal et al., 2021). However, these approaches still lack explicit spatial supervision. More recent literature has adopted transformer-based architectures (Singh et al., 2024) that effectively handle long-range dependencies, but these architectures still operate under weak supervision and leave the problem unresolved.

Future Direction: Modeling TME is pivotal for predicting survival outcomes from routine CRC histopathology images. Therefore, future studies should explore advanced end-to-end architectures, like transformers and, in particular, SSMs like VisionMamba, which can learn long-range dependencies with improved computational efficiency. Multimodal integration, although still an evolving area in CRC histopathology, is propitious for future research. Similar to multimodal integration, multi-view latent feature stacking (KARAGÖZ et al., 2024) offers a viable solution that aggregates features from different tissue levels and magnifications to better capture the spatial context of the tumor microenvironment in CRC. Hybridization of HCFs with modern DL, together with XAI techniques, could further advance interpretability and foster clinical trust.

7.3. Image post-processing: Open issues and future directions

Post-processing in the CRC histopathology image analysis pipeline is the final yet crucial step where raw model outputs are optimized and aggregated into a final task-specific decision. This section discusses the open challenges in post-processing and explores future directions for unified output optimization.

7.3.1. Post-classification refinement

Patch aggregation is post-processing strategies used to enhance classification robustness in the CRC literature. Studies in CRC adopt two aggregation strategies: rule-based heuristics and data-driven models for slide-level classification. While rule-based methods, ranging from simple majority voting to score-based thresholding (Korbar et al., 2017; Kim et al., 2023; Bilal et al., 2022), improve explainability, they are sensitive to real-world artifacts, as a few misclassified patches can disproportionately alter the final slide-level diagnosis (Saeed et al., 2025). In contrast, data-driven methods (Iizuka et al., 2020) offer to capture global context for slide-level diagnosis. Yet, their black-box nature limits clinical validation and interpretability. Therefore, combining global context with interpretability can provide a clear direction for future research.

Future Directions: A viable countermeasure is to explore end-to-end architectures, such as Transformers and MIL-based methods for implicit patch aggregation. Additionally, hybrid frameworks such as CLAM (Van der Laak et al., 2021) can be used. This integrates global context with clustering-based constraints to better align patch aggregation with histological structures.

7.3.2. Post-detection refinement

DL-based detection models, particularly for nuclei detection in CRC tissue, either detect the centroid or the bounding box of target nuclei. Nevertheless, raw outputs often contain redundant detections and noise due to the dense nuclei in colorectal tissue. For centroid-based detection tasks, PMap refinement is the most common strategy, where local maxima are computed to locate precise nucleus centers, and then a threshold is used to filter out low-confidence candidates (Sirinukunwattana et al., 2016). For modern detectors, Non-Maximum Suppression (NMS) is the most common method that aims to remove duplicate boxes using either the feature fusion techniques (Xie et al., 2024) or IoU threshold (Sun et al., 2021). Despite their popularity, it is imperative to understand that both PMap thresholding and NMS are heuristic approaches and require empirical tuning. Due to this manual parameter selection, these methods are highly sensitive to noise and frequently produce false-positive detections in dense nuclear regions.

Future Directions: Future research should develop context-aware detection frameworks that use global tissue information to model the entire tissue architecture. Graph neural networks are highly embraced in this direction, where nuclei are modeled as graph nodes with explicit spatial neighborhood relationships. Additionally, integrating uncertainty estimation into detection modules can reduce false-positive predictions by providing a confidence score for each detection.

7.3.3. Post-segmentation refinement

Based on the reviewed literature, DL-based segmentation models typically employ three primary post-processing strategies to refine and optimize masks. Morphological refinement is the most common approach for boundary smoothing and noise removal using operations like opening and hole-filling (Rastogi et al., 2022; Dabass et al., 2021). Another important refinement is instance separation, which aims to separate touching glands and nuclei using contour-aware fusion and a CCA to assign unique labels to each object (Chen et al., 2017). Finally, reconstruction and optimization approaches are used to assemble patches into slide masks using averaging of prediction probabilities (Khened et al., 2021). Nevertheless, the reliance on heuristic post-processing opens new avenues for future research.

Future Directions: Foundational models such as SAM represent an emerging approach in CRC histopathology segmentation. These models are pretrained on very large-scale datasets and exhibit stronger capabilities for handling complex morphology and overlapping objects using prompt-based guidance and minimal fine-tuning (Villanueva-Miranda et al., 2024). However, patch-to-slide aggregation within these large-scale frameworks is still an active area of development.

7.3.4. Post-prediction refinement

Patch aggregation is predominantly used as a post-processing technique in the literature of CRC image prediction. Since biomarker status is a patient-level characteristic, aggregating local patch-level insights into a unified slide-level diagnosis is essential. Current strategies can be broadly categorized into rule-based and data-driven methods. Similar to classification post-refinement, commonly used rule-based strategies are mean pooling, majority voting, and ROC-based thresholding. Although these methods are deterministic and interpretable at the decision level, they assign equal importance to all patches, thereby overlooking patch specific relevance. Data-driven aggregation uses specific models to learn and prioritize discriminative regions. While these methods (Cao et al., 2020; Li et al., 2024a) assign patch-level importance scores, they treat each patch independently, thereby lacking global tissue modeling.

Future Directions: Future research endeavors in CRC classification should explore patch-level aggregation using GNNs and Transformer-based architectures that can capture the long-range spatial dependencies between disparate tissue regions. In addition, a crucial countermeasure is to incorporate XAI techniques to generate visual heatmaps that justify slide-level predictions and enhance clinical trust.

8. Conclusion

Following remarkable progress in DL, the CPath has become a cornerstone for CRC diagnosis and prognosis. CRC CPath inherently follows a three-phase pipeline comprising pre-processing, analysis, and post-processing. However, the majority of existing studies reveal a strong inclination towards the analysis phase only. This imbalance has resulted in a fragmented research landscape. Consequently, the clinical acceptance of CRC CPath is still a bottleneck. This review demonstrates that an effective CRC diagnosis cannot be achieved through the analysis phase only. Instead, a seamless integration of pre-processing and post-processing stages is essential for real-world deployment. This review concludes that a unified three-phase pipeline is not just an optional enhancement but a fundamental requirement for achieving clinical adoption in CRC diagnosis.

CRediT authorship contribution statement

Sanjeev Kumar: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Pramod Kumar Mishra:** Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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